Chapter 4

NGIBBS: FAST PHASE EQUILIBRIA AND MATERIAL PROPERTY CALCULATIONS FOR GENERAL MULTICOMPONENT SILICATE SYSTEMS VIA DEEP LEARNING

Thermodynamic models that compute equilibrium phase assemblages by free energy minimization, such as MELTS and HeFESTo, are often the computational bottleneck of the workflows that use them. The problems that would benefit most from these models include compositional inversions of seismic data, geodynamic models of a chemically heterogeneous mantle, and physical models of realistic magmatic systems that are heterogeneous in composition and temperature over space and time. These problems require large numbers of equilibria over a composition space that lookup tables cannot cover. Recently, machine-learning surrogate models have been built to address this need, but those that have been built to date each emulate a narrowly defined slice of their parent model. This chapter presents nGibbs*, a set of general deep-learning emulators that reproduce MELTS 1.0.2 and 1.2.0 and the HeFESTo models of Earth's and Mars' mantles across broad ranges of pressure, temperature and bulk composition. Each emulator predicts the full equilibrium assemblage, the abundance of every phase and the endmember composition of every solution phase. Like the parent models* nGibbs *emulates, phase abundances vary continuously across phase boundaries, and mass is conserved. Physical properties are computed from the predicted assemblage with the parent models' own equations of state. On a validation dataset generated from bulk compositions distinct from the training data, the emulators predict the presence of the major silicate and oxide phases with precision and recall of 0.93 or better and conserve every element to a median relative error of about* 5×10^{-6}*%. Along a depleted-mantle isentrope, emulated densities and seismic velocities differ from HeFESTo by 0.02% on average. The HeFESTo emulators run four orders of magnitude faster than*

HeFESTo on an ordinary CPU, and the MELTS emulators two orders of magnitude faster than alphaMELTS. The emulators are differentiable, run on CPUs or GPUs, and are interchangeable with alphaMELTS in existing PetThermoTools workflows. These fast, stable, and compositionally general models make composition a practical free parameter in seismic inversions and coupled geodynamic models.

Author contributions. This chapter is being prepared for publication with co-author P. D. Asimow. **B.J.T.:** conceptualization, methodology, software (nGibbs), investigation (training-data generation with HeFESTo and MELTS), formal analysis, data curation, visualization, writing – original draft. **P.D.A.:** conceptualization, supervision, formal analysis, funding acquisition, writing – review and editing. Generative AI was used as described in the front matter.

4.1 Introduction

Motivation

Free energy minimizers translate a bulk composition and two intensive variables into a stable phase assemblage, the abundance and composition of every phase, and the physical properties that follow from them. The MELTS family of models (Ghiorso and Sack, 1995; Ghiorso et al., 2002; Gualda et al., 2012; Ghiorso and Gualda, 2015) serves this role for silicate liquids and crustal to uppermost-mantle conditions, and HeFESTo (Stixrude and Lithgow-Bertelloni, 2005; Stixrude and Lithgow-Bertelloni, 2011; Stixrude and Lithgow-Bertelloni, 2024) serves it for the solid mantle from the surface to the core–mantle boundary. Perple_X (Connolly, 2005) and MAGEMin (Riel et al., 2022) utilize the database and formulations of Holland and Powell (2011), Holland et al. (2018), and Holland and Powell (1998), which apply from the surface to the core–mantle boundary and support liquids. Each equilibrium requires the solution of a constrained, nonlinear optimization problem, and within a larger workflow that solution is often the slowest step. A single MAGEMin minimization takes on the order of 10^{-1} s depending on the water content of the system (Riel et al., 2022; Candioti et al., 2026), whereas MELTS is an order of magnitude faster (10^{-2} s per assemblage), and HeFESTo is an order of magnitude slower (seconds per assemblage). Efficient coupling to a geodynamic model requires less than 1 ms per point (Kerswell

et al., 2024; Candioti et al., 2026). Minimizers are also largely serial, while geodynamic solvers are built for distributed or shared-memory parallelism, so the thermodynamic calculation dominates the run time of a coupled model (Q. Yuan et al., 2026; Candioti et al., 2026). Robustness is a second cost. A MELTS calculation that fails to converge can halt a batch of calculations or require the library to be reinitialized, which has limited the use of Monte Carlo methods with these models (M. L. M. Gleeson et al., 2025).

The common workaround is to precompute a lookup table of properties over pressure and temperature for one or a few bulk compositions and to interpolate within it during the geodynamic calculation (Connolly, 2005; Zunino et al., 2011; Rummel et al., 2020; Q. Yuan et al., 2026). This works well when composition is fixed but becomes impractical when composition is itself a variable, because tracking compositional evolution requires interpolation in a space of ten or more dimensions (Rummel et al., 2020; Candioti et al., 2026). The problems for which thermodynamic coupling matters most are exactly those in which composition varies: open-ended searches over composition space, inversions that treat composition as an unknown, and forward models in which every parcel of rock carries its own chemical history. These problems need a forward model that is fast, robust, and as general in composition as the thermodynamic model it replaces. Several such problems, described below, motivate this chapter.

Compositional inversion of seismic and geophysical data. Gibbs energy minimization inside a stochastic sampler allows geophysical observations to be inverted directly for mantle composition and thermal state, rather than for seismic velocity alone (Khan and Connolly, 2008). Khan and Connolly (2008) applied this approach to the Love number, tidal dissipation, mean density and moment of inertia of Mars in the system CaO–FeO–MgO–Al_2O_3–SiO_2. Khan et al. (2021) later inverted InSight body-wave travel times for the upper mantle structure of Mars, but assumed a single homogeneous bulk composition derived from the martian meteorites, so that the sampled parameters were primarily thermal. Each forward model in such an inversion is a column of equilibria, and a sampler must evaluate many such columns, so the cost of minimization limits how many compositional dimensions can be explored. On Earth, the seismic character of the mantle transition zone is sensitive to composition at a level that such inversions could exploit. The post-spinel transition in the

Mg–Fe binary spans only ∼0.01 GPa, or about 250 m, at mantle compositions and temperatures, which accounts for the observed sharpness of the 660-km discontinuity without requiring chemically distinct upper and lower mantles (Ishii et al., 2019). The dissolution of aluminous garnet into bridgmanite, in contrast, is expected to produce steep but gradual gradients in velocity and density between about 660 and 750 km (Wood and Rubie, 1996). Surrogate-based sensitivity analysis suggests that the wadsleyite–ringwoodite transition and the akimotoite-out reaction near 660 km both shift with mantle fertility (Hartmeier and Lanari, 2026). Whether lateral variations in discontinuity depth and sharpness reflect temperature, composition, or both is therefore a question that a fast, composition-aware forward model could address directly.

For example, the depth, sharpness and velocity contrast of the transition-zone discontinuities depend on both temperature and composition. The binary post-spinel transition is narrow enough to explain the sharpness of the 660-km discontinuity (Ishii et al., 2019), garnet dissolution into bridgmanite broadens the velocity and density gradients beneath it (Wood and Rubie, 1996), and the wadsleyite–ringwoodite and akimotoite–bridgmanite transitions shift with mantle fertility (Hartmeier and Lanari, 2026). Ishii et al. (2019) also proposed that latent heat can thicken the post-spinel transition by up to ∼7 km where vertical flow is faster than thermal diffusion, so that maps of discontinuity sharpness could locate upwellings and downwellings.

Composition of the lowermost mantle. The large low-velocity provinces (LLVPs) and ultralow-velocity zones (ULVZs) at the base of the mantle are commonly interpreted as compositionally distinct material. Proposed origins vary wildly, including a crystallizing basal magma ocean (Labrosse et al., 2007), subducted banded iron formations (Dobson and Brodholt, 2005), and iron-rich mantle material from the Moon-forming impactor that is 2.0–3.5% denser than the ambient mantle (Q. Yuan et al., 2023). Jackson and Thomas (2021) found that assemblages containing iron-rich (Mg,Fe)O match the seismic velocity reductions and density increases of many ULVZ case studies, and that the range of observations allows multiple distinct but coincident origins. Testing these hypotheses requires searching for compositions, including unusual ones unknown on the Earth's surface, whose equilibrium assemblages reproduce observed velocity reductions at core–mantle boundary conditions. That search is open-ended in composition space.

A chemically heterogeneous mantle. The Sr–Nd–Pb isotopic record of oceanic island basalts is described by four principal mantle end-members (DMM, HIMU, EMI and EMII) with distinct compositions and proposed origins, from recycled sediment to intra-mantle metasomatism (Hart, 1988; Stracke et al., 2005). The major element and lithological consequences of this heterogeneity are also observable. A mechanical mixture of basalt and harzburgite has a different phase assemblage and seismic velocity from pyrolite of the same bulk composition, and such a mixture matches transition-zone seismic observations better than pyrolite (Xu et al., 2008; Stixrude and Lithgow-Bertelloni, 2012). Geodynamic models that compute mineral physics self-consistently, however, often represent this heterogeneity with a single variable, the basalt fraction in a basalt–harzburgite mixture (e.g., Nakagawa et al., 2010; Q. Yuan et al., 2026). That parameterization cannot represent the full range of diverse compositions implied by the geochemical record, nor the fate of subducted material that falls off the basalt–harzburgite join, such as sediments, altered oceanic crust, or banded iron formations. A flexible approach to modeling this heterogeneity is needed to capture the full complexity of the mantle's composition and dynamics to be able to explain the observed seismic anomalies and plumes (Keller et al., 2023; Dobson and Brodholt, 2005).

Magma ocean solidification. Fractional crystallization of a magma ocean is expected to produce a gravitationally unstable cumulate stratigraphy that overturns (Elkins-Tanton et al., 2003; Elkins-Tanton et al., 2011). One-dimensional overturn models assume that the cumulates reorganize on every scale into a monotonic density profile, and Elkins-Tanton et al. (2003) note that viscosity is likely to limit that reorganization and leave a laterally heterogeneous mantle. Miyazaki and Korenaga (2022) proposed that Rayleigh–Taylor instabilities with length scales below 100 km leave iron-rich blobs embedded in a high-Mg# pyroxenite matrix, and that a wet mantle of this kind permits plate tectonics and removes atmospheric CO_2 more than ten times faster than a homogeneous pyrolitic mantle. Testing this hypothesis in a forward geodynamic model requires an equation of state for every parcel as its composition and depth change, which is the case that lookup tables handle worst.

Igneous processes at smaller scales. Recent works have made impressive progress developing physical models for the mechanical behavior of open-system

magma mushes (Bergantz et al., 2015). Such models aim to describe thermally and chemically heterogeneous and evolving material whose physical properties are a direct consequence of composition and temperature as described by thermodynamics, but are necessarily simplified for computational efficiency. Similarly, melt migration within a magma mush is a complex process that is well studied, but produces geochemical signatures which are poorly understood (O'Hara and Mathews, 1981; Lesher and Walker, 1988; Maier et al., 2020; Yao and Mungall, 2021). Thermodynamic models of reactive porous flow may elucidate these signatures, but current models treat each melt–mush reaction as a single step, run until the percolating melt and the mush reach thermal equilibrium, whereas natural samples record a protracted, open-system process (Boulanger and France, 2023) that may be better approximated by spatiotemporally-resolved, local-equilibrium thermodynamics that is significantly more computationally expensive.

Previous Work

Machine-learning surrogates have been developed to address the cost of phase equilibrium calculations, and those built to date are each scoped to a particular application. Kerswell et al. (2024) trained models that predict bulk physical properties, such as density and seismic velocity, of peridotite. Candioti et al. (2026) validated the Holland et al. (2018) igneous database in MAGEMin against crystallization experiments on arc magmas and then trained a multi-task neural network on ∼980,000 MAGEMin equilibria. Their training data span 0.1–10 kbar, 650–1300 °C, and 100 synthetic compositions. The network predicts the presence, fraction and density of liquid and fluid, the liquid composition, and the system density. Solid properties are computed afterward by difference from the system, liquid and fluid properties. Individual minerals are not predicted. The authors report that it runs about 3,000 times faster than MAGEMin on 64 CPU threads, framing their results as a proof of concept. They recommend retraining on application-specific data, and argue that a single surrogate spanning porous, mush and suspension flow regimes is impractical (Candioti et al., 2026).

Hartmeier and Lanari (2026) present a particularly impressive example of a machine-learning surrogate. They trained a combined classifier and regressor on two million MAGEMin equilibria computed with the Stixrude and Lithgow-Bertelloni (2022) database. It predicts the fractions of 20 phases

and the compositions of the solid solutions in the system Na_2O–CaO–FeO–MgO–Al_2O_3–SiO_2 over 1–40 GPa and 500–2500 °C. Conservation of mass and compositional closure are softly enforced, entering the loss function as penalties. They achieve relative errors around 1% for phase fractions and 0.5% for solid solution compositions. The model's classifier delivers impressive precision and recall, mostly 95% and up. Hartmeier and Lanari (2026) also showed that the surrogate's analytical gradients identify which inputs control each phase boundary, a use of differentiability that conventional minimizers do not offer.

The IGyTheM model (Integrating Geodynamics with Thermodynamics using Machine Learning; Q. Yuan et al., 2025) couples a HeFESTo-trained surrogate to the Citcom mantle convection code, so that thermodynamically derived density enters the momentum equation. Its training database covers 0–140 GPa and 273–4273 K. It represents the mantle as a mechanical mixture of basalt and harzburgite, so composition enters as a single mixing variable.

These surrogates show that neural networks can replace minimizers at a fraction of the cost. Each, however, emulates a narrow slice of its parent model, defined by a compositional subspace (arc magmas, or the basalt–harzburgite join), a pressure range, a volatile treatment, or a limited choice of outputs. None of them can be used in place of its parent model for an arbitrary task that the parent model could perform.

Properties and Innovations of nGibbs

nGibbs is designed as a general emulator: one that can stand in for its parent thermodynamic model for essentially any calculation that model would be used for, within the parent model's own range of validity. Its released emulators reproduce HeFESTo for Earth (0–140 GPa) and Mars (0–25.5 GPa), and the closed-system isothermal forms of MELTS 1.0.2 and 1.2.0, with and without chromium. Emulators for open-system (buffered oxygen fugacity) MELTS calculations have been trained but are not yet released. Similarly to Hartmeier and Lanari (2026), each emulator predicts the quantities that define the equilibrium assemblage: the phase abundance and chemistry. Building upon Q. Yuan et al. (2026), *nGibbs* computes physical properties which are the thermodynamic consequence of an equilibrium assemblage.

Four properties distinguish *nGibbs* from the surrogates above. First, its training and validation compositions are drawn from whole-rock databases of igneous

and, for MELTS 1.2.0, sedimentary rocks, that are perturbed to cover exotic compositions that are rare or unknown at Earth's surface (§4.2). Second, conservation of mass is enforced at inference rather than only encouraged during training: an iterative, constrained correction adapted from Asimow and Ghiorso (1998) adjusts the predicted assemblage until it reproduces the input bulk composition exactly, without allowing any elemental abundance to become negative (§4.2). Third, phase abundances are continuous across phase boundaries, because the gate and the magnitude of each phase are the same variable (§4.2). Fourth, through PetThermoTools (M. L. M. Gleeson et al., 2025), they are called through the same interface as alphaMELTS, so that existing workflows can seamlessly switch between the emulator and the parent model for benchmarking (§4.4).

4.2 Methods

Network architecture

Motivation: the discontinuity problem

Before the introduction of the continuous-saturation framework described below, the early *nGibbs* architecture predicted phase presence with a binary classifier and phase abundance with a distinct regressor, multiplying the two so that the predicted abundance is the regressed magnitude gated by a thresholded presence probability. This gated model was simple and easy to train, but it has a structural defect. Because the gate and the magnitude were distinct functions, nothing tied the magnitude to zero as the gate closes. Therefore, when the presence probability of a phase crosses its threshold, the predicted abundance jumps discontinuously from a finite value to exactly zero, which is (for most phases) an unexpected behavior for a mixed system.

Natural systems of interest to geologists and planetary scientists are typically complex, chemically-mixed systems. These systems, and the thermodynamic models that describe them, behave continuously. In addition to the continuous quality of phase abundance, thermodynamic models like MELTS and HeFESTo also track each phase's chemical affinity, which is a continuous quality that measures the free energy cost of introducing a metastable phase to the system. Even when the phase is absent, the chemical affinity is a meaningful, continuous quantity that smoothly goes to zero as the phase approaches saturation. Likewise, at a phase-out boundary, the abundance of the departing

phase goes to zero smoothly, linearly in reaction progress. A gated model is therefore structurally wrong precisely at the phase boundaries that constitute the most interesting content of a phase diagram.

The architecture used for all released models removes this defect by making the gate and the magnitude the same variable, analogously to a prediction of chemical affinity. Some emulators predict a scaled chemical affinity, while others simply activate their continuous molar output with a ReLU function. The models trained with a scaled affinity perform slightly better, but require a more complex data processing pipeline that cannot be applied to datasets where the data are not structured in paths that hold a state variable constant. Because the HeFESTo data were generated along mock-isentropic pressure-temperature paths (and not true isentropes), the HeFESTo emulators were trained with the simpler, ReLU-activated architecture, hereafter referred to as the affinity-naive architecture. The MELTS emulators, which were trained on data generated by isobaric cooling, were trained with scaled affinities with an architecture hereafter referred to as affinity-aware.

Variables

Here a table is provided to name the key variables used in the equations below that describe the training and function of nGibbs. These equations rely additionally on general letter codes to describe the dimensionality of the data that are used internally within the nGibbs codebase and are borrowed here for clarity: P phases, VP compositionally variable phases, C components (endmembers of all phases), VC components of the compositionally variable phases, E elements and O oxides. Two further sizes are used here: F, the number of physical non-compositional input features (e.g. pressure and temperature, with or without fO_2), and B, the number of rows in a mini-batch (the number of assemblages processed in a single pass during training). Every quantity in Table 4.1 is per row; a batch stacks B rows. For emulator outputs, the hatted symbol is the prediction and the plain symbol is the ground truth taken directly from the thermodynamic model (e.g. MELTS or HeFESTo).

Inputs

Each emulator takes pressure and one further intensive variable (temperature for the isothermal models, specific entropy for the isentropic models) and a bulk

Prediction	Truth	Dim.	Meaning
Inputs			
	u	F	Physical state (pressure, temperature, ...), min–max normalized to $[0, 1]$
$\hat{b}$	b	E	Bulk composition in element moles, normalized to sum to 1. $\hat{b}$ is the bulk reconstruction $\hat{b} = \hat{m}A$
Emulator outputs			
$\hat{n}$	n	P	Phase abundance (moles); 0 for an absent phase
$\hat{x}$	x	VC	Endmember mole fractions of each variable phase (sum to 1 within a phase: $\sum x_i = s_i\|1 - b\|$)
$\hat{m}$	m	C	Extensive endmember components/species in moles; $\hat{m} = (\hat{x}V + b) \circ (\hat{n}M)$
$\hat{s}$	s	P	Phase presence, 1 if present and 0 if absent; $s = [n > 0]$
Quantities used only in training			
g	–	P	Phase affinity/abundance: the raw output of each phase's mole head; $\hat{n} = \max(g, 0)$
$\tilde{n}$	–	P	Smoothed, positive, non-zero abundance used to evaluate mole loss; $\tilde{n}(g, \Upsilon)$: Eq. (4.41)
–	$\bar{s}$	VC	Phase Presence s projected to phases' endmembers (chemistry mask) $\bar{s} = s\,M\,V^\top \in \{0, 1\}$
Training constants			
Υ_0		P	Reference abundance scale per phase (5th percentile of present abundances)
Υ		P	Annealed smoothing scale; $\Upsilon = a\,\Upsilon_0$
a		1	Annealing factor, decreased linearly over an episode
w		P	Phase weights
ω		VC	Endmember weights; $\omega = w\,M\,V^\top$
$\lambda_s, \lambda_x, \lambda_n, \lambda_b$		1	Weights for the binary phase presence, chemistry, molar, and bulk loss terms
Fixed stoichiometric/projection matrices			
A		C×E	Element moles per component mole
M		P×C	Which components belong to which phase; $M \in \{0, 1\}$
V		VC×C	Indexing matrix that places the VC components into the full component list; $M \in \{0, 1\}$
f		C	Which components belong to fixed-composition (pure) phases; $b \in \{0, 1\}$

Table 4.1: Key to the symbols. "–" marks a quantity with no prediction or no ground-truth counterpart.

composition as inputs. The emulators are trained on and expect the same units for pressure, temperature, and entropy that the real thermodynamic MELTS and HeFESTo use: kelvin and GPa for HeFESTo, and degrees Celsius and bars for MELTS. Both expect specific entropy in units of J/g/K. nGibbs receives as input normalized cation atomic abundances, following $\mathbf{b} \in \mathbb{R}^E$ with $\sum_e b_e = 1$. Compositions are supplied by the user either as oxide weight percents or as element molar abundances for any model; oxide inputs are automatically converted internally to the normalized cation abundances expected by nGibbs.

Entropy is offered as a useful alternative to temperature because isentropic paths are the natural coordinate for the convecting mantle (Davies, 1977; Brown and Shankland, 1981; Asimow et al., 1997; Asimow, 2001; Ganguly, 2005; Tirone, 2016). Supplying S directly removes an inner iteration for T at every pressure step, which is the dominant cost of constructing an adiabat by conventional means.

The full non-compositional feature vector (i.e. pressure, temperature, entropy, $f\mathrm{O}_2$) is min–max normalized to $[0, 1]$. The minima and maxima used for these normalizations are fit on the training data, and are carried by the emulator and automatically applied during inference. This avoids presenting the network with inputs on a different scale than the one it was trained on, leading to errors that would be difficult to diagnose.

The neural networks driving *nGibbs* expect that the input composition vector sum to a total of one. During inference, this is explicitly enforced by a renormalization so that, similarly to MELTS and HeFESTo, the user can supply a composition that sums to values other than one for convenience. The outputs of *nGibbs*, however, are not rescaled and will always map to a cation sum of one regardless of user input.

Compositional basis

Three compositional bases are used. *Components* (m) are the endmembers of the reference code's solution models (C in total across all phases); *oxides* (O) are accepted as an input because it is conventional to measure and report rock compositions as oxides; and *elements* (E) are the conserved quantities that *nGibbs* conserves. Ferric and ferrous iron are distinguished as oxides but not as elements for models with oxygen buffering, for which $O = E + 1$, but closed system models distinguish ferric and ferrous iron in the element basis as

well so that they are conserved: $O = E$. Conversion between bases is exact stoichiometry, expressed as fixed matrices that relate components to oxides and oxides to elements. The product of the two, A, is the most commonly used stoichiometry matrix because it relates model outputs to the bulk elemental composition it conserves. These matrices are model constants, not fitted parameters. The phases and components of the MELTS and HeFESTo emulators, with the formula each component carries in the component-to-oxide matrix, are listed in Tables 4.2 and 4.3.

Table 4.2: Phases and components of the MELTS emulators, with formulae from the component-to-oxide stoichiometry used by *nGibbs*. Liquid components are oxides.

Phase	Component	Formula
olivine	forsterite	Mg_2SiO_4
	fayalite	Fe_2SiO_4
	monticellite	$CaMgSiO_4$
orthopyroxene, clinopyroxene	diopside	$CaMgSi_2O_6$
	clinoenstatite	$Mg_2Si_2O_6$
	hedenbergite	$CaFeSi_2O_6$
	alumino-buffonite	$CaMg_{0.5}AlTi_{0.5}SiO_6$
	buffonite	$CaMg_{0.5}Fe^{3+}Ti_{0.5}SiO_6$
	essenite	$CaFe^{3+}AlSiO_6$
	jadeite	$NaAlSi_2O_6$
spinel	chromite‡	$FeCr_2O_4$
	hercynite	$FeAl_2O_4$
	magnetite	$Fe^{2+}Fe_2^{3+}O_4$
	spinel	$MgAl_2O_4$
	ulvospinel	Fe_2TiO_4
plagioclase, K-feldspar	albite	$NaAlSi_3O_8$
	anorthite	$CaAl_2Si_2O_8$
	sanidine	$KAlSi_3O_8$
garnet	almandine	$Fe_3Al_2Si_3O_{12}$
	grossular	$Ca_3Al_2Si_3O_{12}$
	pyrope	$Mg_3Al_2Si_3O_{12}$
leucite	leucite	$KAlSi_2O_6$
	analcime	$NaAlSi_2O_6{\cdot}H_2O$
	na-leucite	$NaAlSi_2O_6$
biotite	annite	$KFe_3AlSi_3O_{10}(OH)_2$
	phlogopite	$KMg_3AlSi_3O_{10}(OH)_2$
rhm-oxide	geikielite	$MgTiO_3$
	hematite	$Fe_2^{3+}O_3$
	ilmenite	$FeTiO_3$
	corundum	Al_2O_3
apatite	apatite	$Ca_5P_3O_{12}(OH)$
whitlockite	whitlockite	$Ca_3P_2O_8$
quartz	quartz	SiO_2
tridymite	tridymite	SiO_2
fluid	H_2O	H_2O
	CO_2†	CO_2
liquid	oxides	SiO_2, TiO_2, Al_2O_3, FeO MgO, CaO, Na_2O, K_2O P_2O_5, H_2O, Cr_2O_3‡, Fe_2O_3 CO_2†
alloy-solid	Fe-metal	Fe
muscovite†	muscovite	$KAl_3Si_3O_{10}(OH)_2$
hornblende†	pargasite	$NaCa_2Mg_4Al_3Si_6O_{22}(OH)_2$
	ferropargasite	$NaCa_2Fe_4Al_3Si_6O_{22}(OH)_2$
	magnesiohastingsite	$NaCa_2Mg_4Fe^{3+}Al_2Si_6O_{22}(OH)_2$
graphite†	graphite	C
calcite†	calcite	$CaCO_3$
nepheline†	na-nepheline	$Na_4Al_4Si_4O_{16}$
	k-nepheline	$K_4Al_4Si_4O_{16}$
	vc-nepheline	$Na_3Al_3Si_5O_{16}$
	ca-nepheline	$Na_2CaAl_4Si_4O_{16}$

† MELTS 1.2.0 emulators only. ‡ Cr variants only.

Table 4.3: Phases and components of the HeFESTo emulators, with formulae from the component-to-oxide stoichiometry used by *nGibbs*.

Phase	Component	Formula
plagioclase	anorthite	$CaAl_2Si_2O_8$
	albite	$NaAlSi_3O_8$
spinel	spinel	$MgAl_2O_4$
	hercynite	$FeAl_2O_4$
	magnetite	$Fe^{2+}Fe_2^{3+}O_4$
	picro-chromite	$MgCr_2O_4$
olivine	forsterite	Mg_2SiO_4
	fayalite	Fe_2SiO_4
wadsleyite	mg-wadsleyite	Mg_2SiO_4
	fe-wadsleyite	Fe_2SiO_4
ringwoodite	mg-ringwoodite	Mg_2SiO_4
	fe-ringwoodite	Fe_2SiO_4
orthopyroxene	enstatite	$Mg_2Si_2O_6$
	ferrosilite	$Fe_2Si_2O_6$
	mg-tschermaks	$MgAl_2SiO_6$
	ortho-diopside	$CaMgSi_2O_6$
clinopyroxene	diopside	$CaMgSi_2O_6$
	hedenbergite	$CaFeSi_2O_6$
	clinoenstatite	$Mg_2Si_2O_6$
	ca-tschermaks	$CaAl_2SiO_6$
	jadeite	$NaAlSi_2O_6$
	acmite	$NaFe^{3+}Si_2O_6$
HP clinopyroxene	hp-clinoenstatite	$Mg_2Si_2O_6$
	hp-clinoferrosilite	$Fe_2Si_2O_6$
Ca-perovskite	ca-perovskite	$CaSiO_3$
akimotoite	mg-akimotoite	$MgSiO_3$
	fe-akimotoite	$FeSiO_3$
	corundum	Al_2O_3
	hematite	$Fe_2^{3+}O_3$
	eskolaite	Cr_2O_3
garnet	pyrope	$Mg_3Al_2Si_3O_{12}$
	almandine	$Fe_3Al_2Si_3O_{12}$
	grossular	$Ca_3Al_2Si_3O_{12}$
	mg-majorite	$Mg_4Si_4O_{12}$
	na-majorite	$Na_2MgSi_5O_{12}$
	andradite	$Ca_3Fe_2^{3+}Si_3O_{12}$
	knorringite	$Mg_3Cr_2Si_3O_{12}$
quartz	quartz	SiO_2
coesite	coesite	SiO_2
stishovite	stishovite	SiO_2
bridgmanite	mg-bridgmanite	$MgSiO_3$
	fe-bridgmanite	$FeSiO_3$
	al-bridgmanite	Al_2O_3
	ferric-bridgmanite	$Fe_2^{3+}O_3$
	ferric-bridgmanite-ls	$Fe_2^{3+}O_3$
	ferric-al-bridgmanite	$Fe^{3+}AlO_3$
	cr-bridgmanite	Cr_2O_3
post-perovskite[E]	mg-post-perovskite	$MgSiO_3$
	fe-post-perovskite	$FeSiO_3$
	al-post-perovskite	Al_2O_3
	ferric-post-perovskite	$Fe_2^{3+}O_3$
	cr-post-perovskite	Cr_2O_3
ferropericlase	periclase	Mg_4O_4
	wustite	Fe_4O_4
	wustite-ls	Fe_4O_4
	alpha-naalo2	$Na_2Al_2O_4$
	magnetite	$Fe^{2+}Fe_2^{3+}O_4$
Ca-ferrite	mg-ca-ferrite	$MgAl_2O_4$
	fe-ca-ferrite	$FeAl_2O_4$
	na-ca-ferrite	$NaAlSiO_4$
	high-pressure-magnetite	$Fe^{2+}Fe_2^{3+}O_4$
	cr-ca-ferrite	$MgCr_2O_4$
kyanite	kyanite	Al_2SiO_5
nepheline	nepheline	$NaAlSiO_4$
α-iron	alpha-iron	Fe
γ-iron	gamma-iron	Fe
ε-iron	epsilon-iron	Fe
α-PbO_2-SiO_2[E]	alpha-pbo-type-sio2	SiO_2
NAL phase[M]	mg-nal-phase	$NaMg_2Al_5SiO_{12}$
	fe-nal-phase	$NaFe_2Al_5SiO_{12}$
	na-nal-phase	$Na_3Al_3Si_3O_{12}$

[E] Earth model only. [M] Mars model only.

nGibbs models closed systems that conserve oxygen content, as well as open systems where oxygen fugacity is imposed. Iron is the major redox-active element in *nGibbs*. Every element besides iron is tracked at a single, fixed oxidation state (with a limited exception for carbon) and therefore the oxygen content of the system is fully specified by the ferric-to-total iron ratio. The only other redox-active element is carbon in MELTS 1.2 emulators, wherein carbon exists as an oxide in melt, fluid, and calcite, and as reduced carbon in graphite. Graphite is represented as $C(FeO)_4(Fe_2O_3)_{-2}$ in oxide space in order to conserve oxygen, which introduces strong coupling between Fe^{2+}/Fe^{3+} and graphite for closed-system MELTS 1.2.0 models. This coupling vanishes for open system models where oxygen is not conserved. Similarly, native iron phases are represented as $(FeO)_3(Fe_2O_3)_{-1}$ in oxide space in *nGibbs*.

nGibbs accepts one of two inputs that define the redox state: an explicit, distinct definition of both FeO and Fe_2O_3, which the closed-system models conserve; or an oxygen fugacity (fO_2) in log unit offset from the fayalite-magnetite-quartz buffer (ΔFMQ), which the open-system models impose. Oxygen buffering is available in neither HeFESTo nor its *nGibbs* emulator. In these open models, only cation mass is conserved and the oxygen content of the system is a consequence of the predicted phase assemblage, except for the liquid, for which total iron is speciated following the procedure of Kress and Carmichael (1991).

Forward pass

The network sees the input vector state variables u and bulk composition b, represented as the vector $[u, b]$ and:

1. a shared encoder produces a central working representation of the system's state;
2. one small network per phase gives the signed abundance/affinity g;
3. one softmax-activated network per chemically-variable phase gives that phase's endmember fractions $\hat{x}$.

This workflow is illustrated for a single phase in Fig. 4.1.

During inference, the predicted abundance is the rectified affinity,

$$\hat{n} = \max(g, 0). \tag{4.1}$$

Phases and chemical endmembers that contain an element that is absent from b are forced to be zero.

Component moles and the reconstructed bulk composition follow from stoichiometry:

$$\hat{m} = (\hat{x}V + b) \circ (\hat{n}M), \tag{4.2}$$

$$\hat{b} = \hat{m}A. \tag{4.3}$$

Here $\circ$ is the element-wise product, so $\hat{m}$ is each component's phase abundance multiplied by its fraction in that phase. Conservation of mass (adjusting $\hat{m}$ until $\hat{b} = b$) is enforced only at inference (§4.2).

Thermodynamic justification for softmax activated phase composition prediction

The softmax activation function produces a positive, nonzero output that sums to 1, and is most often applied in machine learning for classification problems in tandem with cross-entropy loss, as its outputs are often interpreted as a probability or likelihood distribution over categorical classes (Holm et al., 2023; Pearce et al., 2021; Li et al., 2019; Yang et al., 2018). For class i of ϕ categorical classes, the softmax function is $e^{z_i} / \sum_{j \in \phi} e^{z_j}$ where z_i is the logit (linear dense layer output) for class i.

The softmax function is used here for a bounded regression problem, to predict the mole fractions of chemical endmembers in a phase, which are also positive, nonzero, and sum to 1. Originally, we chose the softmax function because of this quality, and early tests showed it outperformed alternative activations (such as a renormalization of ReLU outputs) for predicting chemical endmember fraction. Although this application of the softmax is unusual, it is justified in that its functional form mirrors that of the thermodynamic problem of predicting chemical endmember fractions in a phase at equilibrium.

For a general solution, the chemical potential of endmember i in phase ϕ is $\mu_i = \mu_i^\circ(P,T) + RT \ln a_i$, with activity $a_i = \gamma_i x_i$ (e.g., Anderson, 2005; Ganguly, 2008). The activity coefficient $\gamma_i \equiv a_i / x_i$ collects the departure from ideal molecular mixing, including excess (interaction) terms and multi-site configurational effects. At equilibrium, μ_i equals the apparent chemical

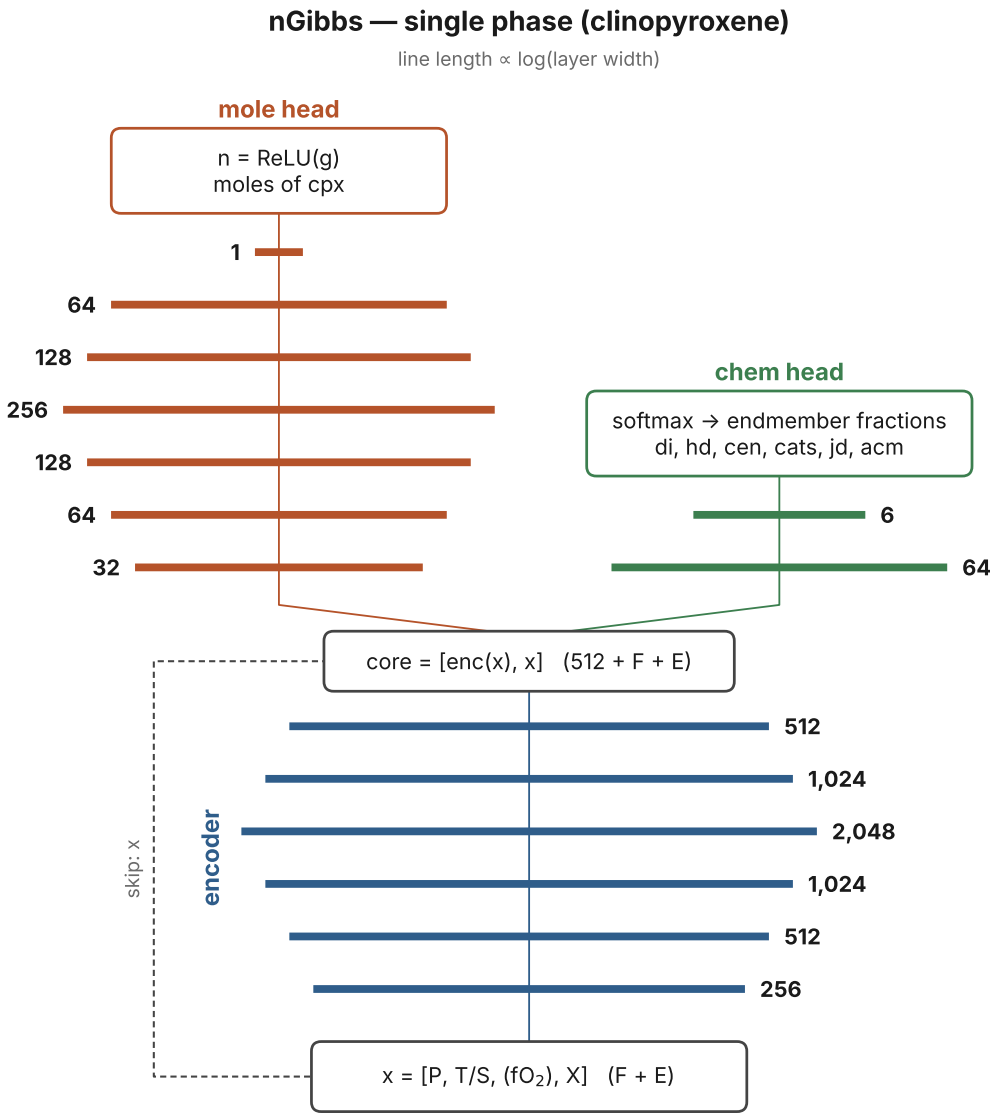


Figure 4.1: Schematic of a forward pass through the neural network geometry, drawn for a single phase (e.g. clinopyroxene). Every phase has its own mole head, which outputs the signed abundance/affinity g, and every phase with variable chemistry also has a chem head, which outputs the softmax-activated endmember fractions $\hat{x}$. The encoder, in contrast, is not specific to any phase but encodes a single system-wide shared state from $[u, b]$ that is passed to all of the heads. These are somewhat large models that carry about 30 MB of weights each. A single trained model thus has a larger storage footprint than all of alphaMELTS or HeFESTo, but remains much faster to run.

potential μ_i^* that the coexisting assemblage imposes on that endmember, so

$$\ln x_i \;=\; z_i \;\equiv\; \frac{\mu_i^* - \mu_i^\circ(P,T) - RT\ln\gamma_i}{RT}. \tag{4.4}$$

The standard state chemical potential $\mu_i^\circ(P,T)$ and the non-ideal contribution ($RT\ln\gamma_i$) may be gathered together in an effective reference chemical potential $\tilde{\mu}_i^\circ(P,T) \equiv \mu_i^\circ(P,T) + RT\ln\gamma_i$.

$$\ln x_i \;=\; z_i \;\equiv\; \frac{\mu_i^* - \tilde{\mu}_i^\circ(P,T)}{RT}. \tag{4.5}$$

Imposing the constraint that the sum of the chemical end members is 1 ($\sum_{i\in\phi} x_i = 1$), it follows that

$$x_i \;=\; \frac{x_i}{\sum_{j\in\phi} x_j} \;=\; \frac{e^{z_i}}{\sum_{j\in\phi} e^{z_j}} \;=\; \mathrm{softmax}(\boldsymbol{z})_i. \tag{4.6}$$

so the softmax is the exact functional form describing the equilibrium composition as a function of the chemical potential with respect to a reference state, normalized by RT. The network does not need to track the non-ideal terms because it predicts the logits $\hat{z}_i$ that satisfy it directly, which absorbs the activity model into the learned mapping from $(P, T,$ bulk composition$)$ to $\boldsymbol{z}$. In the ideal limit ($\gamma_i = 1$), the logit output z represents a unitless chemical potential of each endmember relative to the reference state.

A non-negative, closure-preserving reparameterization of the pyroxene and spinel components of the MELTS models

The softmax function always produces outputs that are strictly positive and sum to one, so the network cannot emit a non-physical composition (e.g. negative atoms) for phases composed of components that are themselves legal compositions, regardless of its unconstrained raw output. For spinels and pyroxenes, however, MELTS predicts legal compositions that are outside the simplex defined by the end members (i.e., some components can be negative). A softmax activation therefore cannot be applied to these components directly, but these compositions can be remapped to another basis that is compatible with a softmax activation: components that are non-negative and sum to one.

Even for spinels and pyroxenes, which can contain negative components, there exists a limit lying outside the simplex that expresses legal compositions,

Table 4.4: Site occupancies of the seven MELTS pyroxene components (Sack and Ghiorso, 1994).

Component	M2	M1	T, T
diopside (x_0)	Ca	Mg	Si, Si
clinoenstatite (x_1)	Mg	Mg	Si, Si
hedenbergite (x_2)	Ca	Fe^{2+}	Si, Si
alumino-buffonite (x_3)	Ca	$\frac{1}{2}$Mg + $\frac{1}{2}$Ti	Al, Si
buffonite (x_4)	Ca	$\frac{1}{2}$Mg + $\frac{1}{2}$Ti	Fe^{3+}, Si
essenite (x_5)	Ca	Fe^{3+}	Al, Si
jadeite (x_6)	Na	Al	Si, Si

implying that there exists some compositional basis that can remap legal compositions to lie within the simplex. We were able to analytically derive a solution for pyroxene, but spinel was instead solved numerically. All other phases' compositional bases in MELTS and HeFESTo are already strictly non-negative.

Pyroxene

We write the mole fractions of the seven MELTS pyroxene components (diopside, clinoenstatite, hedenbergite, alumino-buffonite, buffonite, essenite, jadeite) as $x_0, \ldots, x_6$. Pyroxene has the general formula M2 M1 T_2O_6, with one large octahedral site (M2), one small octahedral site (M1), and two tetrahedral sites (T); the site occupancies of the seven components follow Sack and Ghiorso (1994) and are given in Table 4.4. The atoms-per-formula-unit totals implied by that table are

$$\begin{aligned} \mathrm{Ca} &= x_0 + x_2 + x_3 + x_4 + x_5, & \mathrm{Na} &= x_6, \\ \mathrm{Mg} &= x_0 + 2x_1 + \tfrac{1}{2}(x_3 + x_4), & \mathrm{Al} &= x_3 + x_5 + x_6, \\ \mathrm{Fe}^{3+} &= x_4 + x_5, & \mathrm{Ti} &= \tfrac{1}{2}(x_3 + x_4). \end{aligned} \tag{4.7}$$

Seven components subject to closure leave six independent directions. These separate naturally into three groups. Three elements are carried by exactly one component each and can be read off directly: Fe^{2+} (hedenbergite), Na (jadeite), and Mg on the M2 site specifically (clinoenstatite, the Mg–Ca exchange vector distinct from the M1 magnesium already carried by diopside). Three further

elements—Fe^{3+}, Al and Ti—are each split across two or three components, so no single component fraction stands in for them; the corresponding coordinates are obtained by requiring their pairwise sums to equal the element totals of Eq. (4.7),

$$r_1 + r_3 = \mathrm{Al}, \qquad r_2 + r_3 = \mathrm{Fe}^{3+}, \qquad r_1 + r_2 = 2\,\mathrm{Ti}, \tag{4.8}$$

which is a 3×3 system with the unique solution given below. The remaining two quantities, Ca and Si, are never tracked directly; they are pinned by requiring each crystallographic site to be exactly full—a second, local layer of closure nested inside the overall constraint $\sum_i x_i = 1$. Collecting these,

$$r_0 = x_2, \quad r_1 = x_3 + \tfrac{1}{2}x_6, \quad r_2 = x_4 - \tfrac{1}{2}x_6, \quad r_3 = x_5 + \tfrac{1}{2}x_6, \quad r_4 = x_6, \quad r_5 = x_1. \tag{4.9}$$

Each r_i is bounded for any physically legal pyroxene, but not on $[0, 1]$. Rescaling and recombining gives six quantities that are individually bounded on $[0, 1]$:

$$\begin{aligned} u_0 &= \tfrac{1}{2}r_0, \quad u_1 = \tfrac{1}{2}(r_1 + r_2), \quad u_2 = \tfrac{1}{2}(r_1 + r_3), \\ u_3 &= \tfrac{1}{2}(r_2 + r_3), \quad u_4 = \tfrac{1}{2}r_4, \quad u_5 = \tfrac{1}{2}r_4 + r_5. \end{aligned} \tag{4.10}$$

Unit-sum closure of the full transformed vector is a separate matter and does *not* follow from site positivity. Because $\sum_i x_i = 1$, the constant $u_C \equiv 1 = \sum_i x_i$ is itself a linear function of the mole fractions and can be appended as a seventh coordinate. This makes the map square (and hence invertible), and gives the diopside reference endmember, for which $u_0, \ldots, u_5$ all vanish, a non-zero value in the transformed space. The resulting vector $(u_C, u_0, \ldots, u_5)$ is non-negative for any legal pyroxene but does not sum to one. Closure was imposed by evaluating the linear map of Eqs. (4.9)–(4.10) at each of the seven pure endmembers and dividing each resulting row by its own row sum, yielding the fixed matrix U_{px} of Table 4.5, in which every row is individually non-negative and sums to unity.

Because closure was imposed by endmember normalization rather than derived from the site-occupancy argument, its validity for a general (non-endmember) composition is not guaranteed *a priori*, and was instead established empirically: applying U_{px} to the full set of MELTS-computed pyroxene compositions in the training data yields vectors that are non-negative and sum to unity, and $\mathsf{U}_{\mathrm{px}}^{-1}$ recovers the original mole fractions within numeric precision.

Table 4.5: The pyroxene reparameterization matrix U_{px}. Rows are the seven MELTS pyroxene components; columns are the transformed coordinates. Every row is non-negative and sums to unity. The same matrix is applied to both the orthopyroxene and clinopyroxene solutions.

Component	u_C	u_0	u_1	u_2	u_3	u_4	u_5
diopside	1	0	0	0	0	0	0
clinoenstatite	1/2	0	0	0	0	0	1/2
hedenbergite	2/3	1/3	0	0	0	0	0
alumino-buffonite	1/2	0	1/4	1/4	0	0	0
buffonite	1/2	0	1/4	0	1/4	0	0
essenite	1/2	0	0	1/4	1/4	0	0
jadeite	2/5	0	0	1/5	0	1/5	1/5

Spinel

We were unable to reduce the spinel components (chromite, hercynite, magnetite, spinel *sensu stricto*, ulvöspinel) to a comparably clean set of site-occupancy inequalities. No analogous analytic transform was therefore pursued. Instead a masked, invertible linear transform U_{sp} was fit numerically.

The matrix was parameterized as lower triangular, which guarantees invertibility for any non-zero diagonal. Chromite uniquely expresses Cr contents and is therefore always non-negative. It was mapped onto its own dedicated output coordinate and excluded from the fit so that chromium-free models would be able to use the same transform. The remaining entries were optimized by gradient descent (Adam; PyTorch) against approximately 3×10^6 MELTS spinel compositions drawn from the training data, minimizing a two-term objective: a mean-squared term driving each transformed composition's row sum toward unity, and a leaky-ReLU penalty on negative transformed entries driving them toward zero. The learning rate was annealed by a factor of ten every 2,500 of 10,000 steps; the diagonal was reset to unity whenever it decayed toward zero, preserving invertibility; and coefficients falling below 0.05 in magnitude were periodically hard-thresholded to zero to keep the fitted matrix sparse. The solution satisfyingly converged to one that can be expressed in terms of relatively simple fractions.

The resulting matrix and its analytic inverse were empirically validated in the same way as the pyroxene transform: by histogramming the transformed compositions and their row sums, checking their extrema, and comparing round-

Table 4.6: The spinel reparameterization matrix U_{sp}. Rows are the four (non-chromite) MELTS spinel components; columns are the transformed coordinates. Every row is non-negative and sums to unity.

Component	u_C	u_0	u_1	u_2
Hercynite	1	0	0	0
Magnetite	2/5	3/5	0	0
Spinel	19/20	0	1/20	0
Ulvöspinel	1/5	0	0	4/5

tripped compositions against the originals across the full training set. Original compositions are recovered within numeric precision and the transformed components of all MELTS spinels are confined almost entirely to $[0, 1]$.

Legality constraint for pyroxene and spinel compositions

The transforms derived above in §4.2 guarantee that every legal composition lands in the simplex ($\hat{x} \geq 0$, $\sum \hat{x} = 1$), but not the reverse: the legal compositions fill only a portion of the simplex, and other regions of the simplex are forbidden because they non-physically map to negative atoms. Because the softmax can return any point in the simplex, it is possible for the predicted output to be illegal without the addition of further constraints. To address this, we apply a post-processing step to the softmax outputs of pyroxene and spinel to force them to be legal compositions. This correction is commonly applied to orthopyroxene (6-7% of predicted OPX), for which CaO is low generally and is easily pushed negative. For spinel and clinopyroxene, these corrections are more rare, affecting less than 2% of predicted spinel and 0.1% of predicted clinopyroxene.

Correcting a single negative oxide

A single negative oxide can be safely corrected with trivial arithmetic.

The first step in correcting a single negative oxide is to separate the positive and negative oxide contributions from each component (S_+ and S_- respectively). For oxide i, let q_i be the column of oxide stoichiometry for the transformed components as given in Table 4.7.

$$S^+ = \sum_{q_i>0} q_i \, \hat{x}_i, \qquad S^- = \sum_{q_i<0} |q_i| \, \hat{x}_i, \qquad \text{oxide moles} = S^+ - S^- \tag{4.11}$$

Component	Al_2O_3	CaO	FeO	Fe_2O_3	MgO	Na_2O	SiO_2	TiO_2	Cr_2O_3
Pyroxene									
u_C	0	1	0	0	1	0	2	0	–
u_0	0	1	3	0	−2	0	2	0	–
u_1	0	1	0	0	1	0	0	2	–
u_2	2	1	0	0	−1	0	0	0	–
u_3	0	1	0	2	−1	0	0	0	–
u_4	0.5	−2	0	0	−4	2.5	4	0	–
u_5	0	−1	0	0	3	0	2	0	–
Spinel									
Chr	0	–	1	0	0	–	–	0	1
u_0	1	–	1	0	0	–	–	0	0
u_1	$-\frac{2}{3}$	–	1	$\frac{5}{3}$	0	–	–	0	0
u_2	1	–	−19	0	20	–	–	0	0
u_3	$-\frac{1}{4}$	–	$\frac{9}{4}$	0	0	–	–	$\frac{5}{4}$	0

Table 4.7: Transformed components and their stoichiometries. These are the rows of the inverted transform matrices $\mathsf{U}_{\text{px}}^{-1}$ and $\mathsf{U}_{\text{sp}}^{-1}$. Mapped into an oxide space, these transformed components contain negative atoms and therefore a softmax output in this space is capable of producing illegal negative atoms.

The oxide is negative if $S^- > S^+$. We employ a simple fix that aims to preserve as much of the original prediction as possible. We find two scaling factors to scale the positive and negative contributing components to balance the oxide. We multiply every component fraction on the positive side by one factor a and every fraction on the negative side by one factor b. If we set the oxide to zero and enforce the closure constraint, we formulate a system of two equations that fix the two unknowns a and b:

$$\text{oxide becomes exactly zero:} \quad aS^+ = bS^-, \tag{4.12}$$

$$\text{fractions still sum to one:} \quad aP + bN = 1, \tag{4.13}$$

where P and N are the plain sums of the component fractions on each side.

$$P = \sum_{\hat{x}_i > 0} \hat{x}_i, \qquad N = \sum_{\hat{x}_i < 0} \hat{x}_i, \qquad P + N = 1 \tag{4.14}$$

From Eq. (4.12), $b = aS_+/S_-$; substituting into Eq. (4.13) gives

$$a = \frac{S^-}{S^-P + S^+N}, \qquad b = \frac{S^+}{S^-P + S^+N}. \tag{4.15}$$

Pyroxene: CaO < 0

From Table 4.7,

$$S^+ = \hat{x}_C + \hat{x}_0 + \hat{x}_1 + \hat{x}_2 + \hat{x}_3, \qquad S^- = 2\hat{x}_4 + \hat{x}_5. \tag{4.16}$$

All positive coefficients are 1, so $P = S^+$ and $N = \hat{x}_4 + \hat{x}_5$, and Eq. (4.15) simplifies to:

$$b = \frac{1}{3\hat{x}_4 + 2\hat{x}_5}, \qquad a = \frac{S^-}{S^+\,(3\hat{x}_4 + 2\hat{x}_5)}. \tag{4.17}$$

The CaO correction is applied to clinopyroxene only.

Pyroxene: MgO < 0

$$S^+ = \hat{x}_C{+}\hat{x}_1{+}3\hat{x}_5, \quad P = \hat{x}_C{+}\hat{x}_1{+}\hat{x}_5; \qquad S^- = 2\hat{x}_0{+}\hat{x}_2{+}\hat{x}_3{+}4\hat{x}_4, \quad N = \hat{x}_0{+}\hat{x}_2{+}\hat{x}_3{+}\hat{x}_4, \tag{4.18}$$

a, b then can be calculated using Eq. (4.15). The MgO correction is applied to clinopyroxene only.

Spinel: FeO < 0 and $Al_2O_3 < 0$

Spinel has two oxides that can go negative, sometimes simultaneously, substantially complicating the problem. From Table 4.7,

$$\mathrm{FeO} = \hat{x}_{\mathsf{Chr}} + \hat{x}_0 + \hat{x}_1 + \tfrac{9}{4}\hat{x}_3 - 19\,\hat{x}_2, \tag{4.19}$$

$$\mathrm{Al_2O_3} = \hat{x}_0 + \hat{x}_2 - \tfrac{2}{3}\hat{x}_1 - \tfrac{1}{4}\hat{x}_3. \tag{4.20}$$

In models without Cr, $\hat{x}_{\mathsf{Chr}} = 0$.

Three pools. Because the two constraints share components, the components are grouped by the signs they carry in the two equations, and each group gets its own factor:

Pool	Components	FeO	Al_2O_3	Factor
$A = \hat{x}_{\mathsf{Chr}} + \hat{x}_0$	Chr, u_0	+	+	a
$B = \hat{x}_1 + \hat{x}_3$	u_1, u_3	+	−	b
$C = \hat{x}_2$	u_2	−	+	c

In all cases, the closure constraint is enforced:

$$aA + bB + cC = 1, \tag{4.21}$$

If only one oxide is negative, then only the violated oxide is set to zero, and the pool it does not involve held fixed. Then, the second equation for each scenario is:

- *FeO only* ($b = 1$): Eq. (4.21) and $aA + \hat{x}_1 + \frac{9}{4}\hat{x}_3 = 19\,cC$;
- *Al_2O_3 only* ($c = 1$): Eq. (4.21) and $a\hat{x}_0 + C = b\left(\frac{2}{3}\hat{x}_1 + \frac{1}{4}\hat{x}_3\right)$.

Both violated. If both oxides are negative, or if fixing one would push the other below zero, all three factors are solved together with three constraints: closure, $\mathrm{FeO} = 0$, and $\mathrm{Al_2O_3} = 0$,

$$\begin{pmatrix} A & B & C \\ A & \hat{x}_1 + \frac{9}{4}\hat{x}_3 & -19C \\ \hat{x}_0 & -\left(\frac{2}{3}\hat{x}_1 + \frac{1}{4}\hat{x}_3\right) & C \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{4.22}$$

Figure 4.2 compares spinel compositions predicted by the MELTS 1.0.2 isothermal emulator with those computed by MELTS, before and after this correction.

Conservation of mass

The assemblage predicted by the emulator is not strictly constrained to reproduce the bulk composition; it is only encouraged to with a loss term during training (§4.2). Therefore, a post-processing step is necessary to ensure that the final assemblage is consistent with the requested bulk composition. *nGibbs* uses a conservation-of-mass correction adapted from Asimow and Ghiorso (1998), who find the smallest relative change to $\hat{m}$ that makes $\hat{b} = b$. The implementation in *nGibbs* differs from their procedure most strongly in two ways: (1) the minimization is solved in parallel for thousands of assemblages at once on a GPU, and (2) the correction is done iteratively, with constraints enforced at each step so that no elemental abundance becomes negative.

The section makes use of variables defined in Table 4.1 as well as those unique to this section in Table 4.8. Index forms are given alongside the matrix forms throughout, with c running over components and e, e' over elements.

The neural network returns the component moles $\hat{m}$ (the endmembers of every phase, in the transformed basis of Section 4.2). The element moles they imply

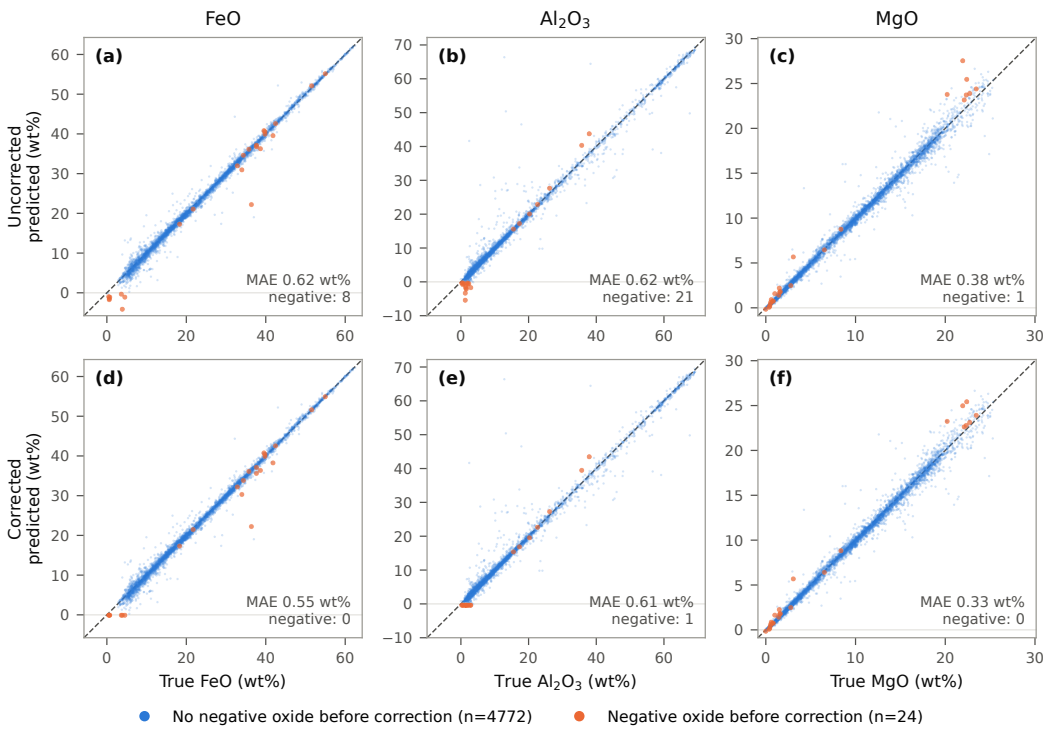


Figure 4.2: Spinel compositions predicted by the closed-system MELTS 1.0.2 isothermal emulator (Cr variant) against MELTS for the 15,000 test assemblage subset. The top row (a–c) is the raw prediction of the emulator, and the bottom row (d–f) is the same data with the legality correction of §4.2. Orange points had at least one negative oxide in spinel before correction and are colored the same in both rows. Dashed lines are 1:1; each panel gives the mean absolute error (MAE) and the number of predicted spinels with that oxide below zero.

Symbol	Dim.	Meaning
δb	E	Bulk residual in element moles, $\delta b = \Gamma\, b - \hat{m}A$
$\delta\hat{m}$	C	Correction to the component moles
ζ	C	Relative correction, $\zeta_c = \delta\hat{m}_c/\hat{m}_c$
D	C×E	Abundance-weighted stoichiometry, $D = \mathrm{diag}(\hat{m})\, A$, i.e. $D_{ce} = \hat{m}_c A_{ce}$
ν	E	Lagrange multipliers, one per element
κ	1	Regularization added to the diagonal of G (10^{-12})

Table 4.8: Symbols added for conservation of mass. Dimensions as in Table 4.1.

are $\hat{m}A$ ($[1 \times \mathrm{C}][\mathrm{C} \times \mathrm{E}] \to [1 \times \mathrm{E}]$). The reconstructed bulk is normalized to sum to one, then its residual is given by

$$\delta b = b - \hat{m}A \quad . \tag{4.23}$$

Conservation of mass seeks a correction $\delta\hat{m}$ to the component moles such that

$$\delta\hat{m}\, A = \delta b, \qquad \text{i.e.} \qquad \sum_c \delta\hat{m}_c\, A_{ce} = \delta b_e \quad \text{for each element } e. \tag{4.24}$$

This is E equations (about ten) in C unknowns (several tens), so the system is underconstrained and has infinitely many solutions. Because the network's prediction is the best available estimate of the assemblage, it is most reasonable to choose the solution that changes it least. We define this as the solution that minimizes the sum of squared relative changes to the component moles. For convenience, we define the relative change ζ_c for each component c as

$$\zeta_c = \frac{\delta\hat{m}_c}{\hat{m}_c}, \tag{4.25}$$

We minimize the sum of squared relative changes,

$$\tfrac{1}{2}\,\zeta\,\zeta^\top = \tfrac{1}{2}\sum_c \left(\frac{\delta\hat{m}_c}{\hat{m}_c}\right)^2. \tag{4.26}$$

The factor $\frac{1}{2}$ is arbitrarily added to elegantly cancel the 2 produced by the differentiation below. We introduce a new variable: D, the component abundance-weighted stoichiometry matrix, to relate ζ to the bulk residual δb.

$$\zeta\, D = \delta b, \qquad D = \operatorname{diag}(\hat{m})\, A \quad (\mathrm{C} \times \mathrm{E}), \tag{4.27}$$

We therefore formulate the problem as a minimization of Eq. (4.26) subject to the constraints of Eq. (4.27). In other words: our solution is the smallest relative change to the component moles necessary to satisfy conservation of mass.

Solving with Lagrange multipliers

Lagrange multipliers reformulate this problem into a set of ordinary equations. One new unknown, the multiplier ν_e, is introduced for each element. Each constraint is written as an expression that must equal zero. The multipliers ν_e relate the derivatives of the objective function to the constraints. Requiring every derivative of the result to vanish then expresses two conditions at once: the constraints hold, and no change that respects them can lower the objective $\frac{1}{2}\,\zeta\,\zeta^\top$. This is the Lagrangian,

$$\mathcal{L}(\zeta, \nu) = \tfrac{1}{2}\,\zeta\,\zeta^\top - \Big(\zeta D - \delta b\Big)\,\nu^\top, \tag{4.28}$$

with ν the row of E multipliers. Now, differentiating with respect to the relative correction ζ gives

$$\frac{\partial \mathcal{L}}{\partial \zeta} = \zeta - \nu\, D^{\top} = 0 \qquad \Longrightarrow \qquad \zeta = \nu\, D^{\top} \quad [1 \times \mathrm{E}][\mathrm{E} \times \mathrm{C}] \to [1 \times \mathrm{C}]. \tag{4.29}$$

This reveals that the C relative corrections are not independent: they are all determined by just E numbers, the Lagrange multipliers. The problem has been reduced from C unknowns to E, and now the system is well-constrained.

Substituting Eq. (4.29) into the $\zeta\, D = \delta b$ constraint gives

$$\nu\, D^{\top} D = \delta b, \tag{4.30}$$

and now the only unknowns are the E multipliers ν, which can be solved with an LU (lower-upper) decomposition as

$$\nu^{\top} = SOLVE(D^{\top} D, \delta b^{\top}). \tag{4.31}$$

With ν known, Eq. (4.29) relates the relative correction ζ to the multipliers.

$$\zeta = \nu A^{\top} \operatorname{diag}(\hat{m}) \tag{4.32}$$

The absolute correction $\delta\hat{m}$ is then obtained by multiplying the relative correction by the component moles (Eq. (4.25)).

$$\delta\hat{m} = \zeta \operatorname{diag}(\hat{m}) = \nu A^{\top} \operatorname{diag}(\hat{m})^2 \quad [1 \times \mathrm{E}][\mathrm{E} \times \mathrm{C}][\mathrm{C} \times \mathrm{C}] \to [1 \times \mathrm{C}], \tag{4.33}$$

Why the Lagrange form is used. Asimow and Ghiorso (1998) obtain the same solution by SVD; however, their problem is distinct from this one in that they correct one assemblage at a time, whereas *nGibbs* applies this operation to tens of thousands of assemblages in parallel. The Lagrange form is used here because batched SVD handles these matrices poorly on a GPU. In an informal test on 15,000 assemblages from the HeFESTo Earth validation dataset (B = 15,000, E = 8, C = 68), `torch.linalg.pinv` took 10.1 s on the GPU (implying that the calculation was run sequentially, perhaps because the matrices were too large) against 0.29 s on the CPU (single precision), whereas the Lagrange solve took 25 ms on the GPU (double precision; 48 ms on the

CPU). The Lagrange form, however, requires double precision floating point arithmetic to recover an accurate solution, whereas the SVD is robust to single precision arithmetic because the $\nu\left(D^\top D\right)$ solve of the Lagrange formulation squares the condition number. This cost is marginal, however, beside the speedup of the Lagrange form on both CPU and GPU, and it elegantly avoids the need to bus data between CPU and GPU when *nGibbs* is run on a GPU.

Regularization

In practice, a small number κ is added to every diagonal entry of $\left(D^\top D\right)$,

$$\nu\left(D^\top D + \kappa I\right) = \delta b, \qquad \kappa = 10^{-12}, \tag{4.34}$$

where I is the E×E identity matrix. This is because $D^\top D$ has all-zero columns and rows for any bulk composition lacking any particular element and is therefore singular. For a row lacking an element, its equation becomes $\kappa\,\nu_e = 0$ for the regularized case and ν is zero because $\delta b_e = 0$ for an absent element. Without regularization, the equation for this row would be $0\nu_e = 0$, for which ν_e is undefined.

On the HeFESTo Earth Model validation dataset, the smallest non-zero diagonal entries of $D^\top D$ are $\sim 3 \times 10^{-7}$ (median over rows; Fig. 4.3), so κ must be far smaller than that. A κ as small as 10^{-12} requires double precision floating point values, but this is already required by the solve of the Lagrange multipliers and therefore poses no additional complication.

Seeding phases for unexplained elements

The conservation of mass correction can only move an element through components that contain it and are present ($\hat{m}_c > 0$). If the predicted assemblage contains no carrier of an element that is in the bulk, that element's residual cannot be removed: the regularized solve leaves it in place (Section 4.2).

Before the mass conservation correction is applied, *nGibbs* first identifies unexplained elements. Element e is unexplained if $b_e > 0$ and no component with $\hat{m}_c > 0$ contains it. For each of these unexplained elements, the phase that is predicted by *nGibbs* to be closest to saturation out of the set of phases that contain the unexplained element is forced to saturation. Its abundance, naturally, is set to the amount that exactly supplies the unexplained element.

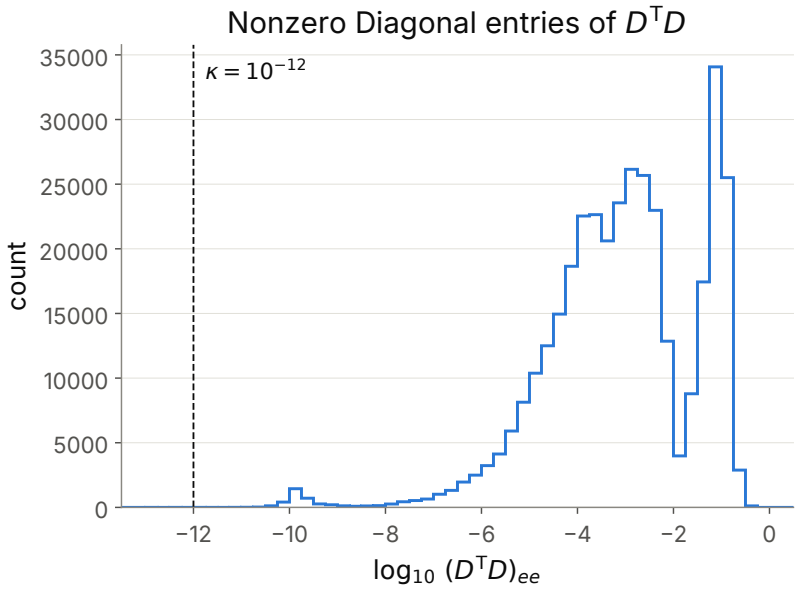


Figure 4.3: Histogram of the non-zero diagonal entries of $D^\top D$ formed by the conservation of mass correction (three iterations) on 15,000 assemblages of the isentropic (NPS) HeFESTo Earth validation dataset. The dashed line marks $\kappa = 10^{-12}$, a regularization term that is small enough to not affect the solution.

We first calculate a_e, the moles of element e in one mole of a given phase p. This is

$$a_e = M_{pc}\, diag(\hat{x}V + f)_{cc}\, A_{ce} \quad [\mathrm{P} \times \mathrm{C}][\mathrm{C} \times \mathrm{C}][\mathrm{C} \times \mathrm{E}] \to [\mathrm{P} \times \mathrm{E}] \tag{4.35}$$

The phase is added with the abundance $\hat{n}_p$ that supplies the missing element in its proportion. For unexplained element $\mathcal{U}$,

$$\hat{n}_p = a_e\, b_e \quad \text{for } e \in \mathcal{U} \tag{4.36}$$

The phase's components get $\hat{m}_c = \hat{n}_p\,(\hat{x}V + f)_c$ for every c in $M_{pc} = 1$ for phase $\mathcal{U}$.

The sequence loops until all elements for each row are explained. $\hat{m}$ is then renormalized so that $\sum_c \hat{m}_c = 1$. Rows that need no seeding are unaffected.

The iterative correction

The mass conservation correction of Section 4.2 places no bound on $\delta\hat{m}_c$, so $\delta\hat{m}_c$ is able to force $\hat{m}$ to become negative. A negative $\hat{m}_c$ is impossible; it means that a phase would have negative atoms, which is non-physical. Generally, this is a

rare occurrence because the high magnitude relative change $d\hat{m}_c/\hat{m}_c < -1$ is highly disfavored by the objective function of §4.2 and therefore is indicative of a poor prediction. This is the "overstepping" failure Asimow and Ghiorso (1998) note. Setting such a component to zero restores a physically sensible assemblage but breaks the conservation of mass again, because the moles it would have supplied are now missing. The step is therefore repeated (three times by default) in order to spread the remaining residual over the components that are still present. For each repetition, $\hat{m}$ is updated as: $\hat{m} \leftarrow \max\left(\hat{m} + \zeta\,\mathrm{diag}(\hat{m}),\ 0\right)$

MELTS emulators in *nGibbs* have another complexity for pyroxenes and spinel: the physically allowed range of components is only a region within the broader accessible range of the transformed component predictions, unlike all other phases. Thus, *nGibbs* additionally pushes these components to lie within their legal simplexes after each iteration following the method described in §4.2.

Figure 4.4 compares the bulk composition reconstructed from the raw network output and from the mass-conserving assemblage with the requested bulk composition for the HeFESTo Earth isentropic emulator.

Data generation

Bulk composition sampling

Bulk compositions are sampled from whole-rock analyses in the GEOROC database in 2022 and the EarthChem PetDB database in April 2025. Only samples with total weight percent within 2% of 100 were kept, then all samples were renormalized to a total = 100%. Evolved compositions are more common in the GEOROC database in particular, so evolved compositions were down-sampled to be symmetric with respect to mafic and ultramafic compositions. The prefiltered dataset included all GEOROC entries, whereas only mafic and ultramafic compositions were pulled from PetDB. A document containing a supersample of citations for the studies that measured these compositions can be found on the nGibbs GitHub page.

For the HeFESTo models, one fifth of the training data are drawn from the population of compositions between 5 and 20 wt% MgO, three tenths of the data are drawn from the population of compositions with greater than 20 wt% MgO, and the remaining half is a mixture of these two mafic and ultramafic populations in random proportions. Roughly 80% of these base compositions were substantially randomly altered so that *nGibbs* would perform well in

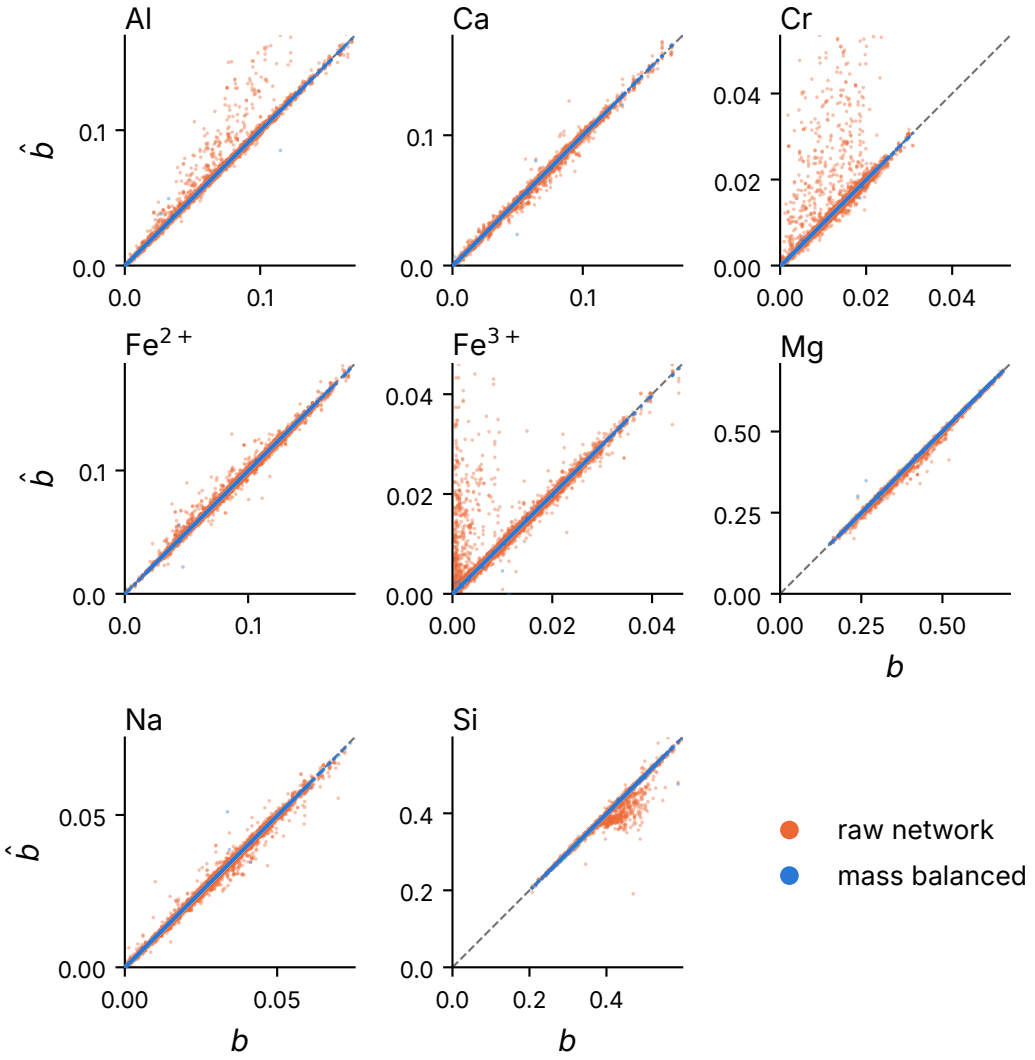


Figure 4.4: Reconstructed ($\hat{b} = \hat{m}A$) against requested (b) bulk composition in element mole fraction for 15,000 assemblages of the isentropic HeFESTo Earth validation dataset. Orange points are the raw network output without the mass conservation correction and the blue points are the results after correction (§4.2). Dashed lines are 1:1. Fe^{2+} and Fe^{3+} are conserved as separate elements by the closed-system models.

a wider variety of parameter spaces, including rock compositions that are uncommon or unknown on Earth's surface. For the Earth model, one quarter of the compositions are not perturbed. For the MELTS emulators, half of the training data is generated from any composition in the compiled database, one fifth of the data are pulled from ultramafic compositions with greater than 25 wt% MgO, and three tenths of the data are pulled from compositions with greater than 5 wt% MgO. These bulk compositions are perturbed by Gaussian noise with a standard deviation of 5%, a substantially more modest perturbation than that applied to the HeFESTo emulator training data. For the MELTS 1.0.2 emulators, half of the isobaric cooling runs are performed with fractional crystallization and half are performed with batch crystallization, whereas only half of the igneous compositions are performed with fractional crystallization for the MELTS 1.2.0 emulators.

For volatile-focused MELTS 1.2.0 emulators in particular, 35% of the training and validation datasets are drawn from batch crystallization of sedimentary compositions within the Gard database (Gard et al., 2019). Of these sedimentary bulk compositions, about one quarter are drawn from carbonate rich compositions, with CO_2 contents greater than 10 wt%, whereas the remaining sedimentary compositions contain less than 10 wt% CO_2. The remaining 65% of the training data were drawn from igneous whole rock data from GEOROC and PetDB, similarly to the other models.

In order to train emulators that perform well in a variety of redox conditions, each sampled composition is randomly assigned an oxygen fugacity from -5 to +5 $\log_{10}\Delta$FMQ for MELTS training data, or -7 to +3 $\log_{10}\Delta$FMQ for HeFESTo training data. Open system models simply carry this oxygen fugacity as a feature, whereas closed models translate this oxygen fugacity into a molar $^{+3}Fe/^{+2}Fe$ ratio using an empirical parameterization. This is not meant to be explicitly correct; it is meant to generate training data with an appropriate distribution of molar $^{+3}Fe/^{+2}Fe$ ratios.

$$^{+3}Fe/^{+2}Fe \approx 10^{(0.2\times\log_{10}\Delta\mathrm{FMQ})-1} \tag{4.37}$$

Manganese and nickel are excluded from the compositional basis of all released models. Manganese and nickel are not handled by HeFESTo, and we have observed that MELTS solutions that carry MnO and NiO frequently stabilize

phases with compositions rarely observed in nature, and there is little value in emulating the regions of a model that are not themselves trusted. Excluding these oxides also removes two dimensions from the component basis at small cost to the applications the emulators target.

State-variable sampling

For the MELTS 1.0.2 and 1.2.0 emulator families, each sampled composition is isobarically cooled from 1800°C in 1°C increments at a random pressure between 1 and 12000 bars. These computed assemblages have well-defined entropies and therefore are acceptable for use in training isentropic models.

HeFESTo Earth and Mars models are each trained around a core of (approximately) adiabatic simulations at reasonable potential temperatures (1200–1650°C for Earth; 500–1750°C for Mars). These adiabats are supplemented with cooler conditions due to a conductive lid regime near the surface, which is assumed to be between 0-250 km thick for Earth and 100-700 km thick for Mars. Similarly, warmer conditions are modeled at the approach to the core-mantle boundary (CMB) for the Earth model, which assume a mantle temperature between 3200–4700°C at the CMB and a 100–300 km wide conductive layer.

Although HeFESTo natively calculates isentropes, such a calculation requires an iteration of T at every step to retrieve the imposed entropy, which roughly doubles the computation time. Therefore, adiabatic PT paths were instead approximated to reduce computational expense in the generation of the training dataset. To accomplish this, a second-order linear regression of temperature as a function of specific entropy and pressure was fit to a dataset of dozens of HeFESTo isentropes at representative compositions and potential temperatures. A similar linear regression was fit for specific entropy as a function of temperature at P = 0 GPa on the same dataset to translate potential temperatures to specific entropy. Similarly to the iron speciation parameterization described above, this approach is not intended to be explicitly correct; it is intended to control the distribution of the training dataset in pressure-temperature-entropy space without unnecessary computational expense. These regressions are used to impose a roughly adiabatic pressure-temperature path for training data. These paths are perturbed with small random noise to prevent the emulators from learning relationships between composition and entropy.

Execution

MELTS simulations are orchestrated with Python scripts preparing, running, and parsing MELTS simulations using alphaMELTS versions 2.3.1 and 2.3.2 (P. M. Smith and Asimow, 2005) run with GNU parallel (Tange, 2021). Batch crystallization was run with alphaMELTS 2.3.1 and fractional crystallization was run with alphaMELTS 2.3.2. In future work, we will rebuild the MELTS datasets to consistently use alphaMELTS 2.3.2, but the results are unlikely to significantly differ. Separate datasets are generated for the closed-system emulators, and for the open-system MELTS 1.2.0 emulator. All MELTS calculations were carried out on a 20-core Intel Core i7-14700H CPU @ 3.40GHz with 32 GB of RAM. Additionally, each simulation was run with parameters adjusted to prohibit multiple crystallization of any phase, including clinopyroxenes and spinel, for which different endmembers are well-known to coexist in nature (Lindsley, 1983; Stosch and Seck, 1980). It is important to note in this regard that orthopyroxene and clinopyroxene, as well as alkali-feldspar and plagioclase, are treated as separate phases in MELTS and therefore can coexist in nGibbs as well. Future work will explore the division of clinopyroxenes into distinct low-Ca and high-Ca phases in nGibbs. Additionally, some phases that are modeled by MELTS were deliberately suppressed: sphene, melilite, aegirine, aenigmatite, cummingtonite, orthoamphibole, cristobalite, kalsilite, corundum, rutile, perovskite, ortho-oxide and sillimanite. Both of these deliberate omissions were made for simplicity and to avoid learning MELTS idiosyncrasies for phases that are not well-modeled.

Some HeFESTo simulations were run on the same machine as the MELTS runs, but the majority were run on the Caltech High Performance Computing cluster as SLURM array jobs.

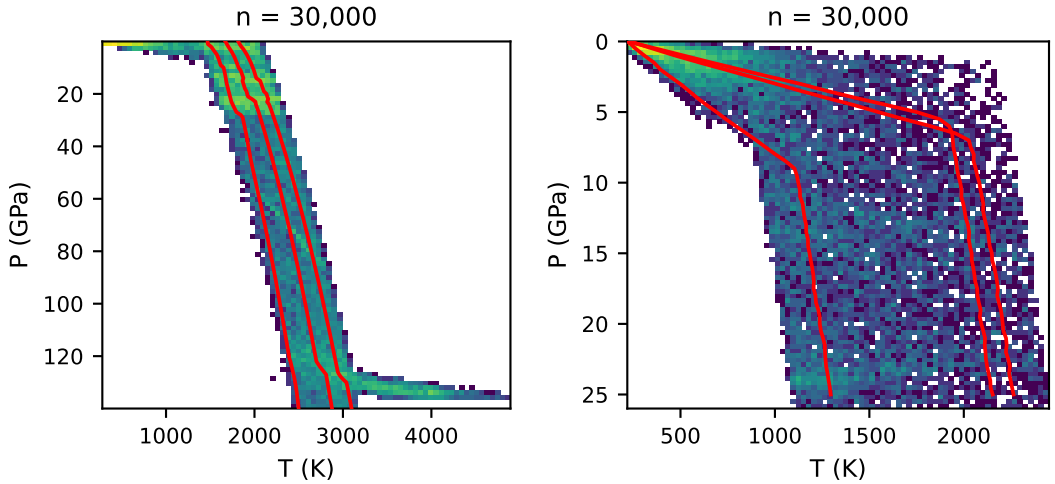
After pressure transects are calculated with HeFESTo, phase changes are identified and iteratively resampled to improve the model resolution around phase boundaries. A random subset of these phase changes is chosen (up to 3x the original simulation count), and the HeFESTo states bounding the phase transition are used to build a P-T grid sampling with higher resolution. The pressure interval is divided into five steps, which is evaluated at at least one isotherm, but more isotherms are added for every full 25 K between the bounding conditions of the phase. Along an isentrope both P and T change between the bracketing states, so this first resampling step produces a two-

dimensional PT search of the region in which the boundary must lie. Each of these isotherms again bounds the phase boundary, but now at a fixed temperature. A second iteration of resampling is performed at 0.01 GPa resolution, between the states that bound the phase boundary. Again, as many as 3x the original number of simulations are called in this step.

A fraction of simulations fail to converge or terminate without usable output. These are detected and discarded at import. Attrition differs sharply between codes and compositional domains: approximately 0.1% for HeFESTo, approximately 15% for MELTS on igneous bulk compositions, and approximately 40% for MELTS 1.2 on sedimentary bulk compositions.

Coverage

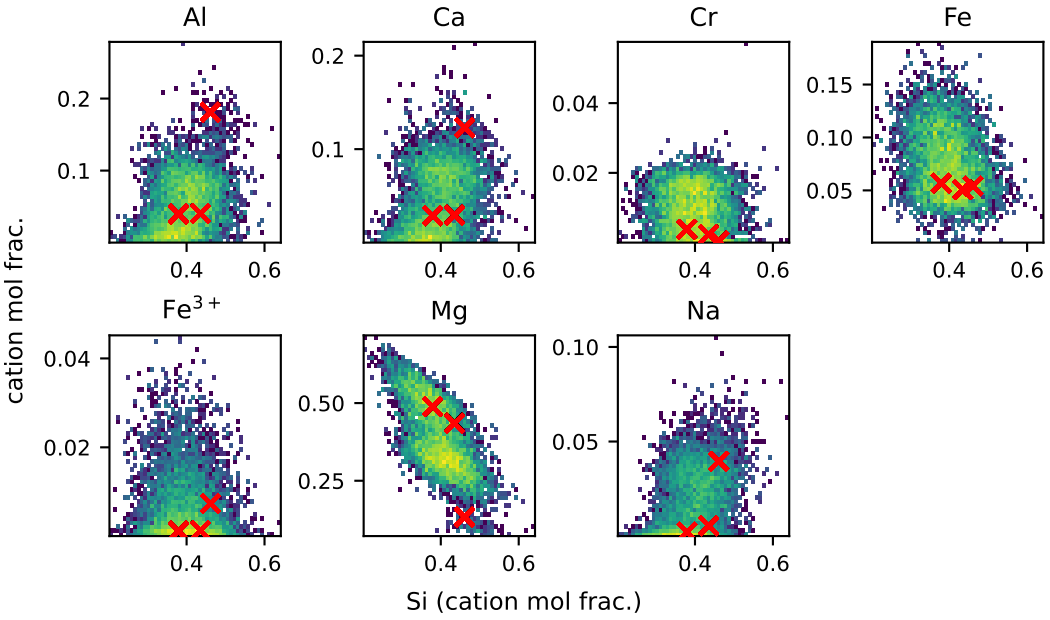
The training domain of each model's pressure and temperature is shown for the HeFESTo models in Fig. 4.5, and in composition space for all models in Figs. 4.6–4.8.



(a) HeFESTo Earth (b) HeFESTo Mars

Figure 4.5: Pressure–temperature distribution of the validation datasets of the HeFESTo Earth (a) and Mars (b) emulators. Red lines mark the deployment-test reference composition paths used in Figs. 4.15–4.21.

(a) HeFESTo Earth



(b) HeFESTo Mars

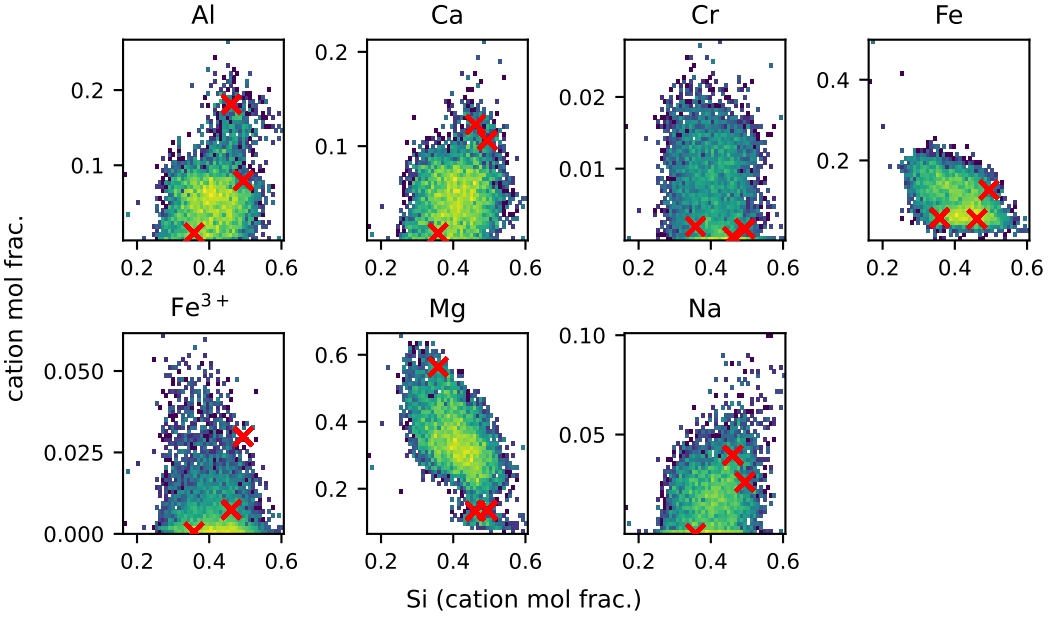


Figure 4.6: Harker diagrams of the validation datasets (as a proxy for the training data) of the HeFESTo Earth (a) and Mars (b) emulators in cation mole fraction, as two-dimensional histograms with a logarithmic color scale. Red crosses mark the deployment-test reference compositions.

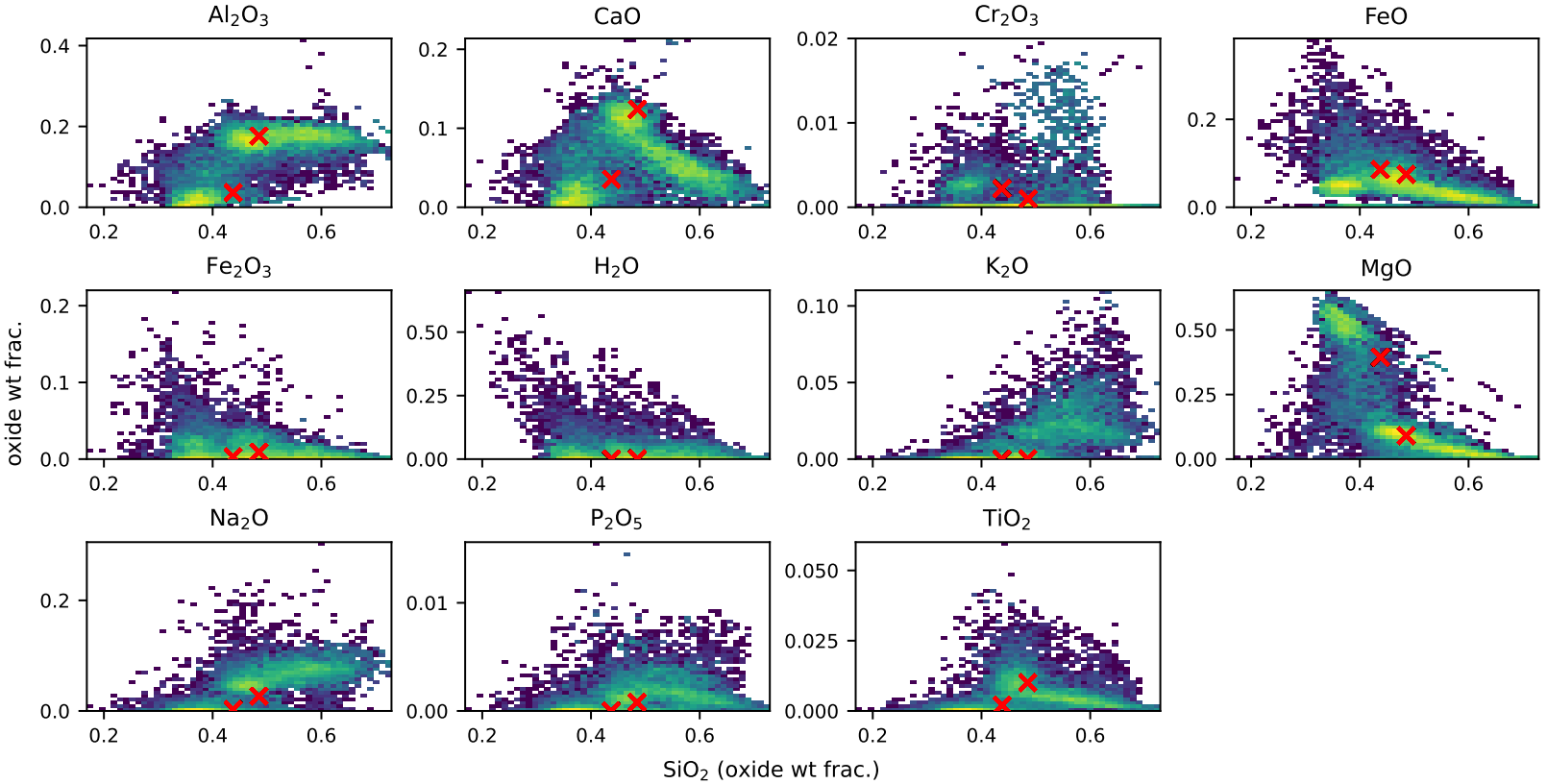


Figure 4.7: Harker diagrams of the validation datasets (as a proxy for the training data) of the closed-system MELTS 1.0.2 emulators (Cr variant) in oxide weight fraction, as two-dimensional histograms with a logarithmic color scale. Red crosses mark the deployment-test reference compositions.

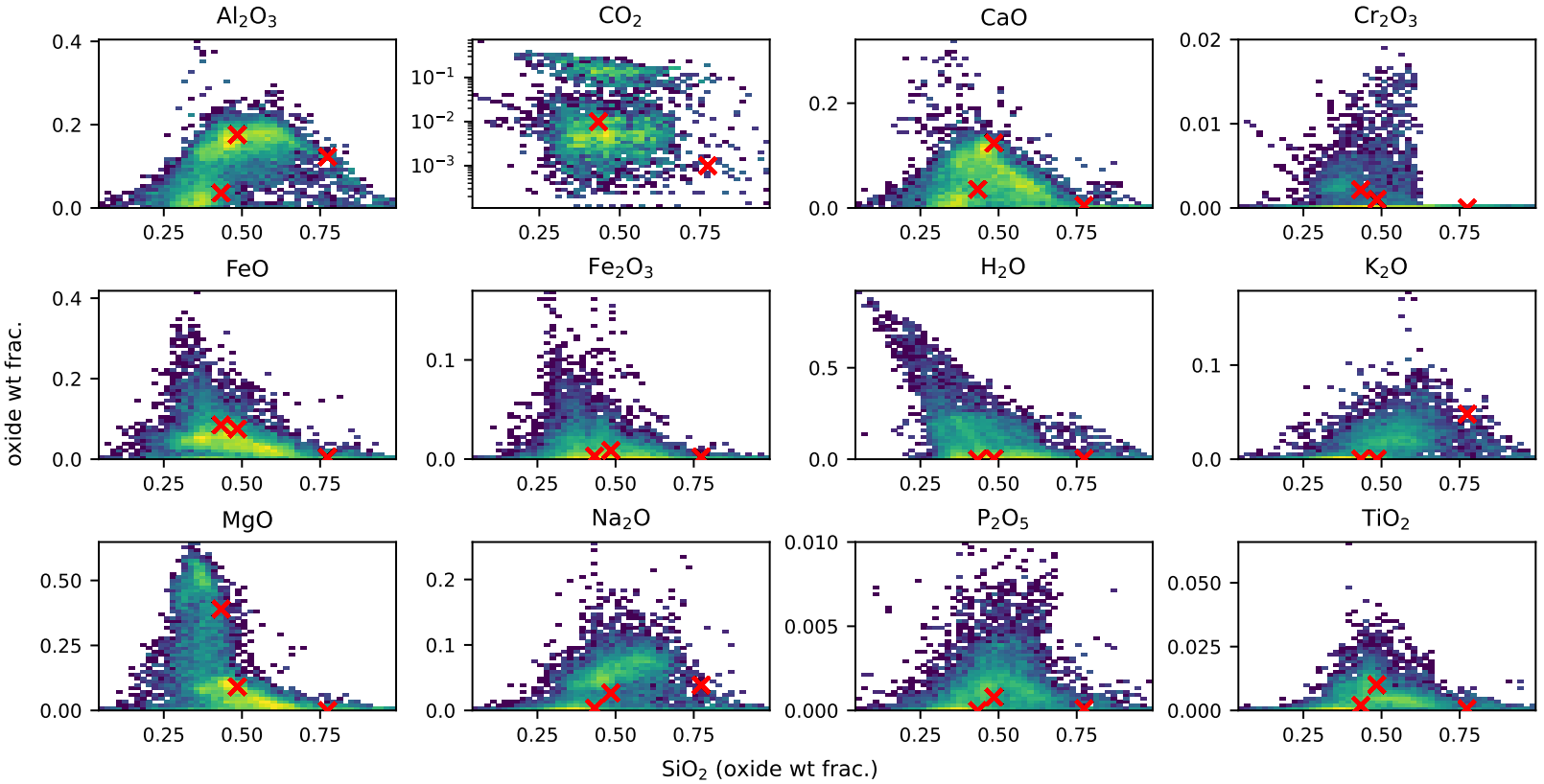


Figure 4.8: As Fig. 4.7, for the closed-system MELTS 1.2.0 emulators (Cr variant), whose training data include sedimentary bulk compositions.

Data preparation

Four operations are applied to prepare MELTS and HeFESTo outputs for training.

Consistency filtering

Rows are removed where the reference code's output is internally inconsistent, e.g. where the composition reconstructed from the reported assemblage disagrees with the stated bulk composition beyond tolerance.

Compositional filtering

A series of filters is applied to remove rows containing phases with unusual compositions generally not seen in nature. These compositions are considered to be MELTS idiosyncrasies that needn't be passed on to nGibbs. These filters include:

1. K-feldspar: at least 6 wt% K_2O and at most 4 wt% CaO.
2. Clinopyroxene: at least 6 wt% CaO, at most 9 wt% TiO_2, and at most 12 wt% Al_2O_3.
3. Orthopyroxene: at most 5 wt% TiO_2, 10 wt% Al_2O_3, and 6 wt% CaO.
4. Silicate melt: at least 30 wt% SiO_2, at most 13 wt% TiO_2, and at most 10 wt% P_2O_5.
5. Plagioclase: the sanidine component may not exceed 0.2 (mole fraction).

Rare phase removal

Phases occurring in less than 1 assemblage per thousand are considered to be too rare in the training set to be effectively learned and are removed from the component basis entirely to simplify the model. This operates on entire rows; assemblages with rare phases are discarded for containing a rare phase at any low modal proportion. It is presumed that this operation is minimally damaging, because very few assemblages are affected by design.

Shuffling

The training and validation datasets are shuffled to remove any correlation between the order of the rows and the order in which they were generated. Conventional dataloaders in PyTorch can automatically shuffle data during training, but this option is not available for *nGibbs*, because its datasets are too large to fit in memory and instead were streamed in static batches of 1,000,000 rows each from disk (SK hynix Platinum P41, 2 TB). The order in which these batches are streamed is shuffled every epoch, but the contents of each bundle are fixed.

Partitioning

To monitor model overfitting during training, a validation dataset distinct from the training set is used. Training and validation datasets are generated with the same pipelines, but are separated *by unique bulk composition*. Bulk compositions were partitioned between training and validation datasets and are treated independently as separate datasets throughout the processing pipeline. This ensures that the validation dataset tracks the same fundamental structure as the training dataset while avoiding data leakage. Successive states along one isobaric or isentropic path are strongly autocorrelated, so a row-wise random split would place very similar data in both the training and validation datasets. The resulting error would measure interpolation between adjacent states of a rock the model had already seen, rather than generalization to a rock it had not, and would be optimistic by a wide and unquantifiable margin. Splitting by composition, before the reference thermodynamic models were applied, makes the validation dataset's error track overfitting to the training dataset more faithfully. This overfitting metric is instrumental to develop an emulator that can be effectively applied to new data.

Compositions were assigned in the ratio 80:20 between training and validation. The validation partition is used every epoch for model selection, to tune adaptive regularization, and for early stopping. Unless otherwise stated, all performance metrics presented here in Section 4.3 are for the validation datasets; the emulator's performance on the training datasets is not reported.

Training

The objective

For each batch the trainer minimizes

$$L = \lambda_s L_s + \lambda_x L_x + \lambda_n L_n + \lambda_b L_b \tag{4.38}$$

covering phase presence (L_s), chemistry (L_x), phase abundance (L_n) and bulk (L_b). There is no phase presence term in the affinity-aware model (§4.2). The weighting factors λ are tuned to balance the contributions of each term to the total loss, preferentially weighting the chemistry and abundance terms. When training some models, the loss from present examples of important rare phases was additionally up-weighted by a per-phase factor to encourage the model to learn phases that otherwise would contribute little to the total loss.

Chemistry and bulk losses use the symmetric relative squared error:

$$\rho(\hat{y}, y) = \frac{(\hat{y} - y)^2}{|\hat{y}| + |y|}, \tag{4.39}$$

with the denominator floored at 10^{-6}. The abundance loss uses the same function in the affinity-naive model, but plain mean squared error (MSE: $\rho(\hat{y}, y) = (\hat{y} - y)^2$) is used in the affinity-aware model (§4.2). The symmetric relative squared error proved effective in training the model to predict both small and large values accurately, for positive targets that span orders of magnitude. MSE is used for the affinity-aware model abundance heads because the network maps to negative labels (affinities) as well as positive labels which behave continuously across zero: a property that MSE preserves but the symmetric relative squared error does not.

Chemistry loss. The chemistry loss L_x is only evaluated for phases that are present in the ground truth: *nGibbs* is not trained to predict metastable phase chemistry. The loss is zeroed for phases without ground truth compositions, but not for phases that are present in the ground truth and not the predicted assemblage: this loss is evaluated everywhere a ground truth composition is available. Because this masking behavior strongly changes the absolute value of the loss as a function of the number of present phases, the loss is normalized by the number of present phases in each batch.

Bulk loss. The reconstructed bulk (Eq. 4.3) is compared with the input bulk over the elements present in it:

$$L_b = \sum_{e:\, b_e>0} \rho(\hat{b}_e, b_e), \tag{4.40}$$

This measures how far the raw predictions are from satisfying conservation of mass, which reduces the correction that has to be applied at inference (§4.2). The backpropagation graph for this loss is complex and covers the entire network, thus it can cause exploding gradients when applied to untrained networks. Here it is only applied late in training to further improve the model's performance after it already produces strong predictions.

Abundance loss. For the abundance loss L_n, every phase is supervised, including absent ones. *nGibbs* emulators can be trained to predict chemical affinities for absent phases (affinity-aware) or it can be trained to predict phase abundances and presence without this information (affinity-naïve). In both cases, they produce an unbounded, unactivated output $\hat{g}$, that corresponds to a phase abundance when present and an effective chemical affinity when negative, tracking how close the phase is to saturation. The affinity-aware models learn this behavior with direct supervision: they are trained with a scaled affinity value, whereas the affinity-naive models are not provided with any negative labels, and instead are trained to treat this negative space as a confidence in the prediction of phase absence, rewarding confidently correct predictions of phase absence, and likewise penalizing confidently incorrect predictions of phase absence.

Affinity-naïve abundance loss. The labels for the affinity-naïve model are the literal phase abundances (n), which are zero for absent phases, never negative. Therefore, the loss is not computed on the predicted abundance $\hat{n} = \max(g, 0)$ itself but on a smoothed version of it ($\tilde{n}$), which is $\hat{g}$ activated with a softplus function (Fig. 4.9).

$$\tilde{n} = \Upsilon \,\mathrm{softplus}(g/\Upsilon) = \Upsilon \ln\left(1 + e^{g/\Upsilon}\right) \tag{4.41}$$

The softplus function is continuous for all real inputs and only approaches zero asymptotically. It is effectively a smoothed version of the ReLU function, where gradients are alive throughout and the transition from positive to zero

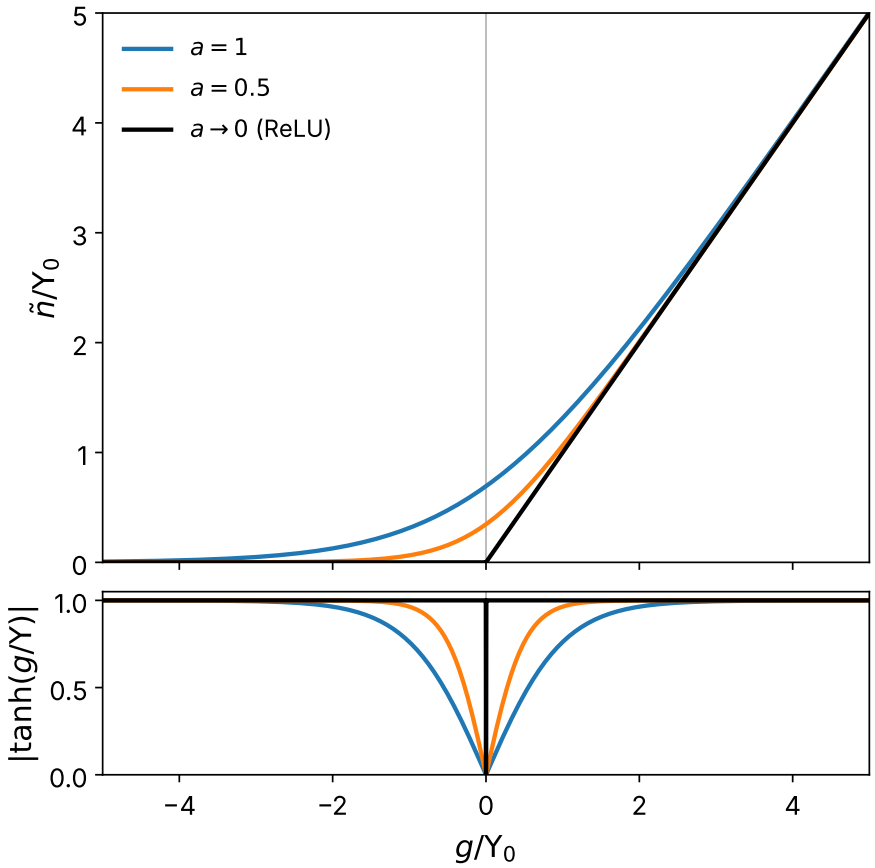


Figure 4.9: The softplus function (Eq. 4.41) generalized as a function of g/Υ_0 for different values of the scheduled annealing parameter a. The softplus function is a smooth approximation of the ReLU function, which is used to activate the predicted phase abundances $\hat{g}$ in the affinity-naïve model. As training proceeds, the softplus function collapses to the ReLU function, while allowing the model to learn a smooth transition from zero to positive values in the earlier stages of training. The smoothing scale Υ controls the strength of the smoothing, with larger values producing a smoother transition from zero to positive values. The lower plot is $\tanh(g/\Upsilon)$, which is the weight of the BCE loss on binary phase presence/absence. This binary loss is similarly scaled by the smoothing scale Υ so that the loss is weak near saturation to avoid teaching the model discontinuous behavior across 0.

values is smooth instead of kinked, following the thermodynamic logic that a phase generally grows from zero abundance at saturation and gradually decays to zero as it becomes undersaturated. The utility of this function applied to this problem is clear given its limits:

$$\lim_{g\to-\infty}\tilde{n}=0, \qquad \tilde{n}\approx g \text{ for } g/\Upsilon>3. \tag{4.42}$$

The cost of the softplus activation in this context is that the smoothed $\hat{n}$ systematically overestimates the true abundance of phases near zero, and therefore the strength of the smoothing needs to be reasonably restricted for the model to learn phase abundances well. The input to the softplus is thus scaled by a smoothing scale $\Upsilon = a\Upsilon_0$, where Υ_0 is the 5th percentile of that phase's abundance where the phase is present and a is a scaling factor within [0,1]. This limits the range of values for which this smoothing error is high to only the smallest abundances that are approaching desaturation. The smoothing is further weakened as training progresses by annealing the a parameter from 1 to ≈ 0, and the softplus function anneals into the ReLU function ($\tilde{n}\approx\max(g,0)$), matching *nGibbs*'s behavior during inference.

Presence loss L_s for the affinity-naïve model. As a means to further incentivize the model to learn the correct assemblages, the affinity g is also used as the logit of a binary cross-entropy against s:

$$L_s=\frac{\sum_j c_j\Big[-s_j\ln\text{sigmoid}\Big(5g_j/\Upsilon_{0,j}\Big)-(1-s_j)\ln\Big(1-\text{sigmoid}\Big(5g_j/\Upsilon_{0,j}\Big)\Big)\Big]}{\sum_j c_j},$$
$$c_j=\Big|\tanh(g_j/\Upsilon)\Big|, \tag{4.43}$$

where the bracketed term is the binary cross-entropy (BCE) of the logit $5g_j/\Upsilon_{0,j}$ against binary label s_j. The logit is scaled by the fixed Υ_0, not the annealed Υ, so the classifier's confidence means the same thing throughout training: it reaches sigmoid(5) $\approx$ 0.99 at $g=\Upsilon_0$. The weight for this binary loss $|\tanh(g/\Upsilon)|$ uses the annealed Υ. It is 0 at the boundary and close to 1 a few Υ away. The gradient is carried by the abundance loss near the boundary, where an almost-saturated phase is thermodynamically similar to a just-saturated phase, a subtlety that the BCE loss cannot capture. The weight c depends on g, a model output. Thus, we normalize the BCE loss by the sum of the weights

$\sum_j^P c_j$ so that the model is not encouraged to make near-zero predictions for g simply to minimize the BCE loss.

Affinity-aware abundance loss

The affinity-aware model is trained to predict an activated abundance y, a transformed version of the signed abundance g that remaps negative values but preserves positive values. Unlike an affinity-naïve model, there exists a ground truth g that is used to supervise the affinity-aware model even for phases that are absent.

Estimating affinities (g) from dn/dT of isobaric transects. Affinities are positive for absent phases and 0 for present phases. They are expressed in units of energy, not molar abundances. This unit mismatch produces an undesirable discontinuity at zero if unaccounted for, reducing the information content of the affinities within the context of the molar abundances that we seek to predict well. If the affinities are scaled well, they smoothly transition into molar abundances at saturation, reinforcing physical dn/dt values even near zero, where other approaches might learn spurious patterns near the boundary in response to the objective function changing character across the boundary. This smoothness constraint is simply that g be continuous at exactly 0, which can be written:

$$\lim_{g\to 0^-}\left(\frac{dg}{dT}\right)_P = \lim_{g\to 0^+}\left(\frac{dg}{dT}\right)_P = \lim_{n\to 0^+}\left(\frac{dn}{dT}\right)_P. \tag{4.44}$$

Affinities are therefore rescaled into a "virtual abundance" unit with a constant unique to each phase and each transect. For an affinity α in an arbitrary unit:

$$\begin{cases} g = n, & n > 0 \\ g = \alpha \frac{lim_{n\to 0^+}(dn/dT)_P}{lim_{\alpha\to 0^+}(d\alpha/dT)_P}, & n = 0. \end{cases} \tag{4.45}$$

$(dn/dT)_P$ and $(d\alpha/dT)_P$ always have opposite signs, so $g \leq 0$ while $n = 0$ because the rescaling factor is always negative. This operation erases the memory of the units of the affinity and instead privileges the slope of the molar abundance (dn/dT) at saturation. Indeed, the affinity is locally linear near zero, as is the molar abundance. The rescaled affinity is therefore well-approximated

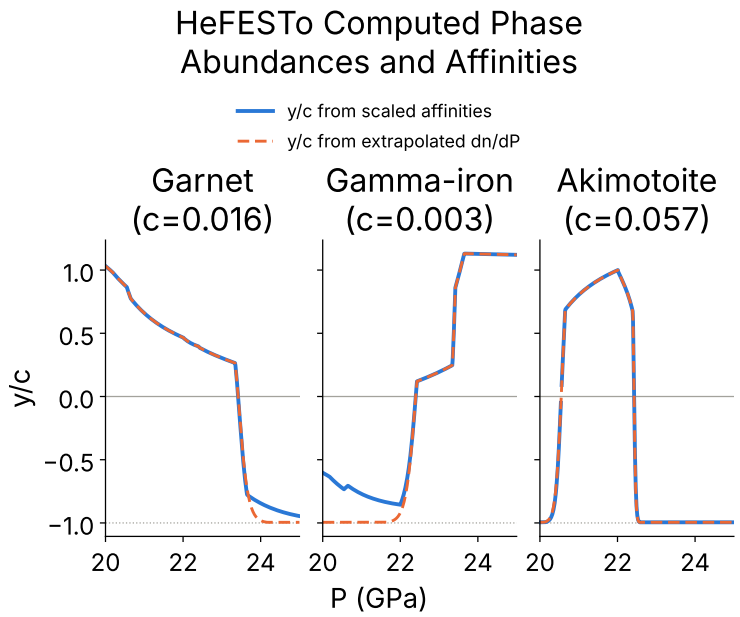


Figure 4.10: Activated and scaled abundance labels y/c for garnet, γ-iron and akimotoite along a HeFESTo isentrope of harzburgite (Stixrude and Lithgow-Bertelloni, 2024, after Workman and Hart, 2005) from 20 to 25 GPa in 10 MPa steps. Blue: scaled abundance and affinity labels built from HeFESTo's own phase affinities, rescaled by the ratio of $|dn/dP|$ to $|dA/dP|$ at saturation (Eq. (4.45), in pressure rather than temperature). Orange dashed: labels built without affinities, by extrapolating dn/dP through zero (Eq. (4.47)). Both are floored at $g = -3c$ and activated with $y = c\tanh(g/c)$ below zero. The scale c, given above each panel, is the median $|dn/dP|$ at saturation times 0.25 GPa, bounded between one fifth of the phase's maximum abundance and that maximum. The extrapolated labels are nearly identical to the affinity-based labels near the saturation boundary, but sometimes diverge further away. The far-field divergences are relatively unimportant because they have little bearing on the predicted assemblage, unlike the near-field behavior.

by a linear extrapolation of the abundance into negative values. Figure 4.10 compares labels built both ways along a HeFESTo transect.

$$\alpha(T)\frac{lim_{n\to 0^+}(dn/dT)_P}{lim_{\alpha\to 0^+}(d\alpha/dT)_P} \approx (lim_{n\to 0^+}(dn/dT)_P) \times (T - T_{sat}) \tag{4.46}$$

With this in mind, there is little benefit to collecting real affinity values from the reference thermodynamic models. We evaluate the ground truth g for each phase as:

$$\begin{cases} g = n, & n > 0 \\ g = (lim_{n\to 0^+}(dn/dT)_P) \times (T - T_{sat}) & n = 0. \end{cases} \tag{4.47}$$

The MELTS training data were generated as isobaric cooling runs, either as batch crystallization (constant bulk composition) or fractional crystallization (solids are removed from the system at every step). Each run is split into temperature transects, and the slope dn/dT is estimated at n=0 with differentials of n and T. Let n_i refer to a phase's molar abundance just after saturation, i.e. the first nonzero value of n going either up or down in T. Then, $n_{i-1} = 0$, and

$$\text{batch:} \quad (lim_{n\to 0^+}(dn/dT)_P) \approx \frac{n_{i+1} - n_i}{T_{i+1} - T_i}, \tag{4.48}$$

$$\text{FxTal:} \quad (lim_{n\to 0^+}(dn/dT)_P) \approx \frac{n_i}{T_i - T_{i-1}}, . \tag{4.49}$$

It is incorrect to use this approach to estimate the affinities for fractional crystallization *sensu stricto*, because the composition (and therefore $(dn/dT)_P$) is not constant as T decreases. However, we found that this approximation is good for small $|T - T_{\text{sat}}|$, and the accuracy of the affinity is less important further from the saturation boundary. Therefore, we accept this rough approximation so that we are able to weight the training datasets towards liquidus assemblages, which are often of interest to the community.

Far from the saturation boundary, the first-order approximation of g is less accurate, and accurate predictions of g for absent phases do little to improve the quality of the predicted assemblage. Therefore, we activate the raw affinity g with a hyperbolic tangent function to compress the negative range of g to limit the influence of strongly undersaturated phases on the abundance loss. Because phases have very different abundances in common rocks (e.g. $n_{\text{clinopyroxene}} \gg n_{\text{apatite}}$), we need to define a scale factor c for each phase which corresponds to some characteristic abundance for a phase. We choose to define c as the typical (median) abundance that a phase accumulates over a temperature interval $\Delta = 25$ K after saturation:

$$c = \text{median}(|\,(lim_{n\to 0^+}(dn/dT)_P) \times \Delta|), \qquad \Delta = 25 \text{ K}. \tag{4.50}$$

Intuitively, the temperature window Δ represents the window of temperature approaching saturation for which we would like to accurately predict affinities. Phases beyond Δ °C from saturation are considered arbitrarily far and so we place them on the plateau of the tanh function. Any confident prediction that such a phase is absent will contribute little to the loss. We use the median rather than the mean slope ($(lim_{n\to 0^+}(dn/dT)_P)$) because it is log-normally distributed.

We define an activated label ($y(g, c)$) that is used during training:

$$y(g) = \begin{cases} g, & g \geq 0, \\ c\tanh(g/c) + \varepsilon g, & g < 0, \end{cases} \qquad \varepsilon = 0.02 \tag{4.51}$$

where epsilon is a small proportionality constant used to invigorate the loss gradient when the predicted $\hat{g}/c$ is lower than -3, for which the tanh term is nearly flat ($\tanh(-3) \approx -1$). This allows the network to learn well at the beginning of training, when predicted $|\hat{g}|$ can be large, inaccurate, and unstable. Additionally, ground truth g is floored at $-3c$ to prevent systematic predictions of very low g values where the gradient is weak (but the network has no such constraint, which would kill the gradient below -3c). Note that g, y, and n are identical for $n \geq 0$.

The activation is shown in Fig. 4.11.

The affinity-aware objective. Within a single isobaric transect, most phases never appear anywhere. These phases in these rows do not permit an estimate of dn/dT at saturation, and therefore g and y are undefined for them. We refer to these phases as "untrusted" ($y \notin \mathbb{R}$), and apply a distinct, weaker loss to them. We also reject low-side outlier estimates of dn/dT at saturation (defined as when dn/dT is a factor of $\sqrt{10}$ smaller than the median dn/dT for that phase across all transects). These values are also untrusted. For present phases, $y \in \mathbb{R}$ and is positive. Unlike the affinity-naïve abundance objective, the affinity-aware objective wraps terms for both regression error and binary phase presence/absence into a single loss function with its own weighting scheme.

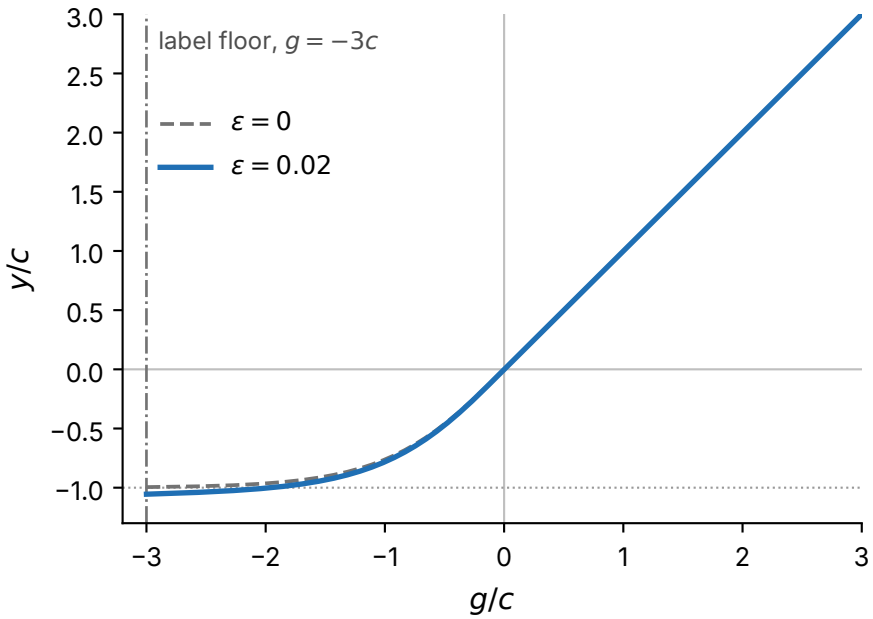


Figure 4.11: The activation of Eq. (4.51) in units of the per-phase scale c, for $\varepsilon = 0.02$ (solid) and $\varepsilon = 0$ (dashed). Above zero $y = g$; below zero the tanh term saturates at $y/c = -1$ (dotted), and the εg term keeps a small slope there. The dash-dotted line marks the floor of $-3c$ applied to the ground-truth g. This saturation behavior ensures that the loss is dominated by saturated or near-saturated phases, which are the cases that matter during inference.

$$\ell = \begin{cases} \frac{(\hat{y}-y)}{c}^2 + s * [w_p * \mathrm{ReLU}\Big(\min(y/c,\ h_p) - \hat{y}/c\Big)^2], & y \in \mathbb{R}, \\ w_n \max(0,\ \hat{y}/c - h_n)^2 + w_\eta\,(\hat{y}/c + 1)^2, & y \notin \mathbb{R}., \end{cases} \tag{4.52}$$

For trusted phases ($y \in \mathbb{R}$), the loss function is a combination of the MSE and a corrective positive hinge loss. The first term is the MSE of the predicted activated abundance $\hat{y}$ against the ground truth y, normalized by the scale factor c. The corrective positive hinge loss penalizes the underprediction of phase abundance for low-abundance phases. This term is a response to the observation that the loss for untrusted phases was leading to underprediction of present phases near saturation, which most obviously manifests in an imbalanced high precision and low recall. A hinge loss uses min or max functions to apply a loss only to one side of a threshold. This hinge loss is applied only to phases which are present in the ground truth ($s = 1$) with abundances less than the threshold $h_p = 0.03$, that are underpredicted by $\hat{y}/c$.

For untrusted phases ($y \notin \mathbb{R}$), the loss function is a combination of a negative

hinge loss and a weak pull towards $-c$. The hinge loss is stronger, but is only applied to predictions that are above the threshold $h_n = -0.2$. The combination of these losses is a compromise that recognizes that untrusted phases are absent and probably would have affinities mapping to decisive undersaturation at $y \approx -c$, however, because this is not known explicitly, only unlikely near-saturation predictions ($\hat{y}$) are substantially penalized.

Table 4.9: Hyperparameters of the affinity-aware loss (Eq. 4.52).

Symbol	Value	Meaning
Trusted phases, $y \in \mathbb{R}$		
h_p	0.03	Predicted abundance threshold for low-abundance, present phases, below which underpredictions are penalized
w_p	3.0	Weight of the underprediction penalty for present phases
Untrusted (absent) phases, $y \notin \mathbb{R}$		
h_n	-0.2	Predicted abundance threshold for untrusted absent phases, above which are penalized
w_n	0.25	Weight of the near-saturation penalty for untrusted absent phases
w_η	0.02	Weight of the weak pull towards $-c$ for untrusted absent phases

Figure 4.12 maps the loss over y/c and $\hat{y}/c$ for the values of Table 4.9.

The HeFESTo emulators within *nGibbs* were trained with the affinity-naïve objective because its transects do not hold entropy constant, so the extrapolation method cannot be applied without rebuilding the training dataset. HeFESTo reports true affinities, and how to scale them into units of "virtual abundance" is left for later work.

The MELTS emulators were trained with the affinity-aware objective because the training dataset was generated as isobaric transects, permitting the estimation of dn/dT at saturation.

Training schedule

All models were trained on an NVIDIA GeForce RTX 5070 Ti GPU with 16 GB of VRAM. Datasets were often too large to fit in memory, and so were streamed from disk in batches of n=10^6.

Quality prediction of equilibrium phase assemblages that conserve mass is a complex task with highly covariate errors. These models are thus trained in

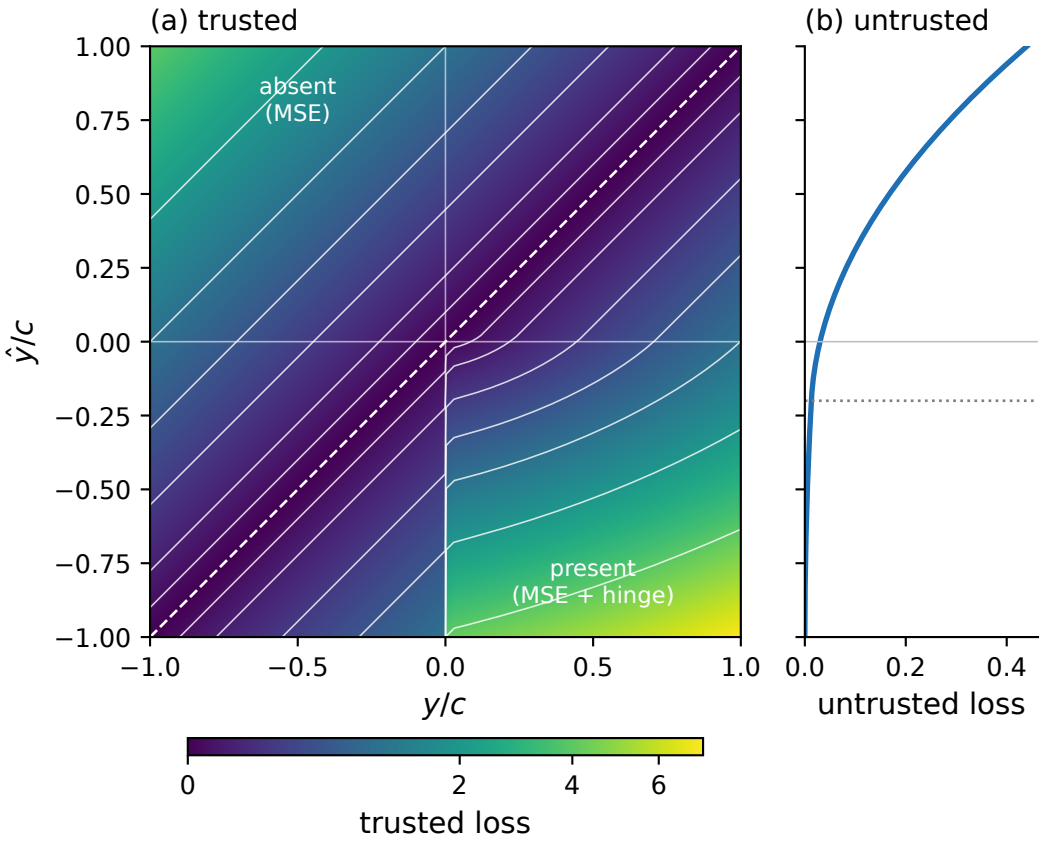


Figure 4.12: The affinity-aware abundance loss (Eq. (4.52)) in units of c, with the values of Table 4.9. (a) Trusted phases, as a function of the label y/c and the prediction $\hat{y}/c$ (square-root color scale; white contours at 0.01, 0.05, 0.2, 0.5, 1, 2 and 4; dashed line $\hat{y} = y$). (b) Untrusted phases, which are absent, have no label, and depend on $\hat{y}/c$ alone. The dotted line marks $\hat{y}/c = h_n = -0.2$, above which the hinge applies. The functional forms follow the *nGibbs* implementation.

episodes that first isolate the distinct abundance and chemistry partitions of the model, training with a high (10^{-3}) maximum learning rate on a single-cycle learning rate scheduler described by L. N. Smith and Topin (2019). Then, the model is trained as an interdependent whole with a more cautious maximum learning rate (10^{-4}) that is periodically annealed to zero then restored following a cosine function. This is effectively a more gentle implementation of the warm restart method described by Loshchilov and Hutter (2017). Initializing the model in this sequential manner ensures that the model does not learn a single task at the expense of the others due to unstable solutions, but optimizes the full unified problem. The model's encoder feeds into both the molar and chemical heads. The encoder must be instantiated in tandem with either the

molar or chemical heads, so that it has an objective to optimize. We choose to instantiate the encoder with the molar heads because they perform the more complex task. Thus, the chemistry heads are frozen and the chemistry loss is zeroed during the first stage of training. During the second stage, the chemistry heads are trained with the chemistry loss, the abundance loss (and phase presence loss for the affinity-naïve model) is zeroed. The weights of both the encoder and molar heads are frozen for this second stage. The encoder feeds into the chemistry heads, but it cannot be optimized for the chemistry loss while the mole heads are frozen, or the abundance predictions will be clobbered as the encoder evolves in isolation from the mole heads that depend on it. It is presumed that an encoded representation of the system state that is useful for predicting phase abundances would also be useful for predicting phase chemistry. Thus, after the second training episode completes, the whole model produces reasonably good abundance and chemistry predictions that both are built upon the same encoder.

Finally, for the third stage, all heads are unfrozen and the model is trained jointly with all losses except the bulk loss. For these two stages, an early stopping patience is implemented that stops training after 10 epochs of no improvement in the model's performance against the validation dataset. For the affinity-naïve model, the smoothing scale $\Upsilon = a\Upsilon_0$ is annealed during this third stage onward, with a linearly decaying from 1 to 0 by the end of training. Bulk loss is the most complex and best captures the systemic error covariation between the abundance and chemistry predictions within the constraint that mass be conserved; it is applied in concert with the other losses in the fourth stage of training, and the adaptive dropout is disabled for stability. Instead, an adaptive input noise is applied as a regularization.

This sequential training schedule is summarized as follows (the losses from an example training history are shown in Fig. 4.13):

1. train the uninitialized encoder and chemistry heads, with the mole heads frozen and $\lambda_n = \lambda_s = 0$;
2. train the uninitialized mole heads, with the encoder and chemistry heads frozen and $\lambda_x = 0$;
3. whole model annealing by training everything jointly with $\lambda_b = 0$.
4. whole model annealing with a mass conservation constraint, by training everything jointly with $\lambda_b > 0$.

The AdamW optimizer is used with weight decay set at 10^{-4} for the encoder and chemistry heads. No weight decay is used on the mole heads.

nGibbs uses an adaptive dropout during training. After each epoch, if the validation loss exceeds 1.05 times the training loss, the dropout rate (outside the mole heads) is raised by a small amount, such as 0.005. If the validation loss is below the training loss, dropout is lowered by an even smaller amount, such as 0.001.

Temperature model for isentropic emulators

The isentropic emulators take pressure and specific entropy as inputs instead of pressure and temperature. Temperature, therefore, is not known *a priori* for an isentropic *nGibbs* emulator, unlike for true free energy minimizers for which temperature emerges as a consequence of the minimization. Because temperature is a required input to calculate material properties (see §4.2), we must retrieve the temperature from the isentropic emulator's output. Thermodynamically, temperature is uniquely defined by the entropy and pressure for an equilibrium assemblage, but recovering it requires an iterative optimization. To avoid this, *nGibbs* implements a separate temperature satellite model for isentropic emulators that predicts the temperature of a state given its pressure, specific entropy and emulated assemblage. This task is divided into two parts: a closed-form reference regression that captures most of the variance, and a fully connected neural network that learns the remaining residual.

The reference regression is an ordinary least-squares fit of temperature to P, P^2, S and S^2, together with a linear and a quadratic term for each element in the bulk composition. Element abundances are expressed as atoms per formula unit, with cations plus oxygen normalized to 24 and total iron counted as a single species. On its own, this smooth polynomial captures the first-order shape of an adiabat and how it shifts with composition. It cannot, however, represent the changes in slope associated with phase transitions. A small neural network corrects for this. It receives the same input as the reference regression, supplemented with the phase abundances and compositions of the equilibrium assemblage predicted by the isentropic emulator. The network then predicts a residual with respect to the reference regression, allowing for more subtle variations in the adiabat's shape to be learned. Because the emulator's predictions are used as inputs during inference, they also serve as the training

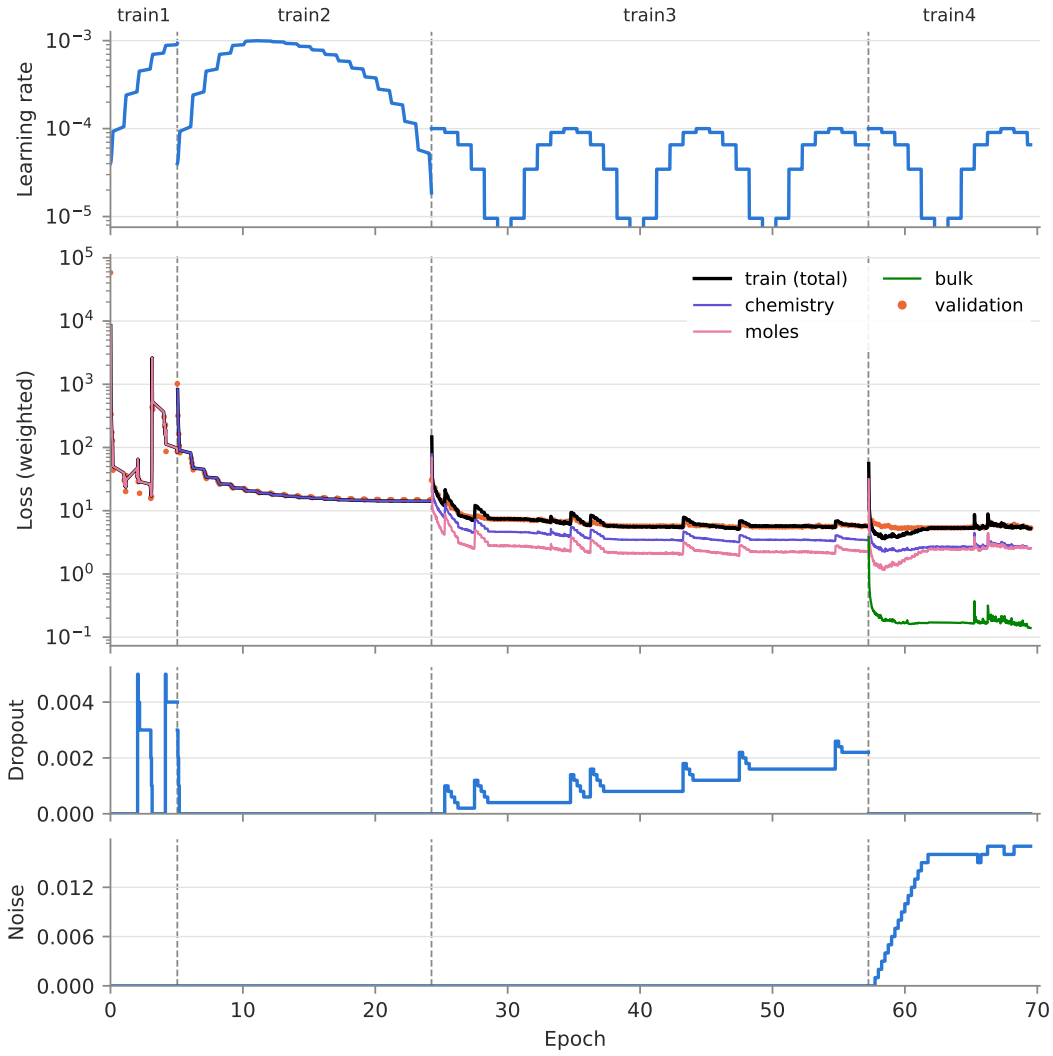


Figure 4.13: Training history of the MELTS 1.0.2 emulator, NoCr variant, across the four stages of the training schedule (dashed lines mark stage boundaries). From top to bottom: learning rate, weighted loss, adaptive dropout rate, and input noise amplitude. The loss panel shows the total training loss and its chemistry, moles, and bulk components, with per-epoch validation loss as points. The bulk loss is zero until the fourth stage, when it is switched on and noise injection begins. The learning rate follows a single-cycle schedule in stages 1 and 2 and cosine warm restarts at a lower maximum in stages 3 and 4. Dropout and noise are adjusted adaptively from the validation-to-training loss ratio.

data. The network has three hidden layers of 256, 128 and 64 units with leaky-ReLU activations. It is trained on a mean-squared-error loss using min–max normalized residuals, with adaptive dropout and early stopping enforced by performance metrics against the validation dataset. The learned residuals span roughly ±200 K. Both the regression and the network are fit only to states with $S \geq 1.75\,\mathrm{J\,g^{-1}\,K^{-1}}$, which encompasses the range spanned by the mantle adiabats the models are designed for, without including temperature conditions at the conductive surface or core-mantle boundary (CMB).

Figure 4.14 compares both stages with HeFESTo for the Mars model, using 10,892 states with $S \geq 1.75\,\mathrm{J\,g^{-1}\,K^{-1}}$ taken from the validation dataset ($T \approx$ 860 K to 2440 K). On its own, the regression reproduces HeFESTo temperatures with an RMSE of 27.5 K ($R^2 = 0.995$), but its errors occasionally exceed 100 K. Adding the network lowers the RMSE to 1.7 K and the mean absolute error to 1.0 K. Ninety-nine percent of states fall within 6.3 K, and the largest error is 47 K.

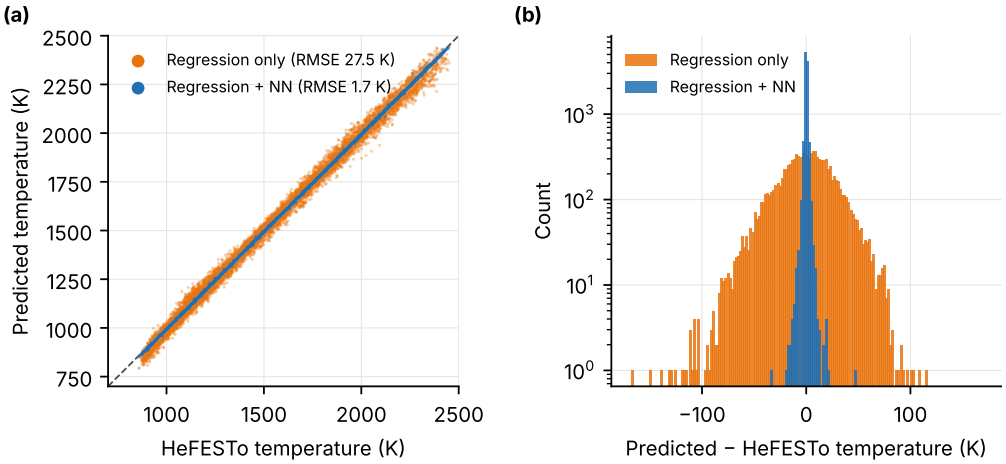


Figure 4.14: Temperature model for the HeFESTo Mars isentropic emulator, evaluated for a subset of the Mars validation dataset. (a) Predicted versus true temperature, for the reference regression alone (orange) and for the regression plus neural-network residual correction (blue); the dashed line is 1:1. (b) Distributions of prediction error (log scale).

Material Properties from Phase Equilibria

To support its use in geodynamic or other physical simulations, *nGibbs* implements the equations of state used by MELTS and HeFESTo to determine the material properties of phases and of the bulk assemblage. These calculations

are cheap relative to the free energy minimizations performed by MELTS and HeFESTo, so there is little benefit to emulating them with a neural network. Furthermore, because these properties are computed from the predicted phase assemblage, they are self-consistent, thermodynamically grounded, and flexible to adapt to changes with the parameters of the reference thermodynamic models. These are implemented as vectorized Python (NumPy/PyTorch) translations of the MELTS models taken from the open source MAGMA C library (commit `705a0fb`), and the HeFESTo models from the open source HeFESTo Fortran repository (commit `9566177`). Parameters for the equations of state are taken from the MAGMA C library for MELTS except for the apatite reference volume, and for HeFESTo from the library (commit `ff99dd0`). The MELTS emulators in *nGibbs* use 15.8228 $\mathrm{J\,bar^{-1}}$ for the apatite reference volume, instead of the 16.4025 $\mathrm{J\,bar^{-1}}$ given in the MAGMA C library so that the volume and density calculations would be consistent with alphaMELTS 2.3.1 output.

Given a phase assemblage, its phase compositions, and a pressure and temperature, *nGibbs* returns density, G, S, V, C_p, for both MELTS and HeFESTo assemblages, H, $\partial V/\partial T$, $\partial V/\partial P$, $\partial C_p/\partial T$ for MELTS only, and V_s, V_p, K_S, K_T, α (referring to the shear wave velocity, compressional wave velocity, isentropic and isothermal bulk moduli, and the thermal expansivity respectively) for HeFESTo only. These properties are computed for each phase and are aggregated to bulk properties. The equations differ from the source codes only in how they are organized and the algorithms used to execute them.

For the HeFESTo properties, the emulator is able to vectorize the isomorphic terms most effectively, whereas the metamorphic terms are more computationally expensive, because they require solving the molar pressure and temperature derivatives $(dn/dp)_T$ and $(dn/dT)_P$. Because the metamorphic terms are only necessary for accurate calculation of the bulk C_p, K_S, K_T, and α (Fig. 4.22), these terms are only evaluated by nGibbs when the user requests properties that need them (Listing 4.1). Thus, it is significantly cheaper to provide S, ρ, V_p, and V_s without the other terms, if they are not necessary.

Listing 4.1: Computing bulk HeFESTo properties for an *nGibbs* assemblage, requesting either the isomorphic terms only or the metamorphic terms as well (pseudocode).

```
from ngibbs.engine.models import HeFESToEmulatorCPU
```

```
input_array = pd.read_csv('bulk_composition.csv')
PROPERTY_VARIANTS = {
    'isomorphic':  ('rho', 'Vp', 'Vs', 'S'),
    'metamorphic': ('rho', 'Vp', 'Vs', 'S', 'cptot', 'KStot'),
    '+fast':       ('rho', 'Vp_fast', 'Vs', 'S', 'cptot', 'KStot'),
}
output = HeFESToEmulatorCPU.ForwardMB(input_array, outputs=['
    component_moles'])
properties = HeFESToEmulatorCPU._compute_bulk_EOS_properties(
    output['component_moles'],
    PT=torch.tensor(input_array[['P(GPa)(System_main)', 'T(K)(System_main)'
    ]]),
    property_names=PROPERTY_VARIANTS['isomorphic'])
```

MELTS and HeFESTo evaluate these equations of state sequentially for one phase at one $(T, P, \mathbf{X})$ state per call, whereas *nGibbs* operates on thousands of states in parallel. The translation, therefore, is a reorganization of the source code to operate on arrays.

These equations of state were empirically validated through comparison against property outputs from MELTS and HeFESTo. Against alphaMELTS 2.3.1 output on the MELTS 1.2.0 reference calculations, bulk S, V, H and C_p computed from the true assemblage agree to within 0.07% (mean relative error). There remains an undiagnosed systematic disagreement in the *nGibbs* implementation for plagioclase H and S. Figs. 4.20–4.23 compare the bulk properties reported by each reference code with those computed by *nGibbs* from the reference code's own assemblage and from the emulator's predicted assemblage.

Performance benchmarking

Both HeFESTo and MELTS are parallelized by running independent simulations in separate processes, so their wall time depends on the number of workers as well as on the number of assemblages N. Each was timed on two machines: at 1, 4 and 8 workers on a 4-core, 8-thread Intel i7-11370H laptop CPU with 24 GB of RAM, and at 1, 4, 8, 16 and 20 workers on the 20-core Intel Core i7-14700H desktop CPU with 32 GB of RAM.

MELTS 1.0.2 was timed through PetThermoTools (`multi_path`) on isobaric cooling paths at 2 kbar from 1200 to 1000 °C in 1 °C steps (201 equilibria per path), each with its own MORB-like bulk composition (every oxide of a

typical MORB multiplied by $\mathcal{N}(1, 0.1)$ noise and renormalized). Every worker count was run with 1 to 5 paths per worker. A PetThermoTools worker process can solve only one MELTS path, so p paths per worker are run as p consecutive calls with one path per worker, and wall time includes process start-up and alphaMELTS library initialization for every path. Equilibria that PetThermoTools failed to return are not counted toward N. Beyond about 10^3 assemblages the wall time is roughly linear in N, so the per-assemblage rate measured for the largest call at each worker count is used to extrapolate to larger N.

HeFESTo was timed on whole-mantle adiabats prepared with the same structure as the training data (§4.2): 141 states each from 0 to 140 GPa. A total of 3 adiabats per worker are measured, run through GNU parallel (Tange, 2021). Preparing the simulation directories was not timed. Both reference codes are timed along paths, where each state is seeded from the last. This is their most efficient mode of use, so the comparison does not favor the emulators. The relative speedup of *nGibbs* is particularly impressive when applied to independent states.

The emulators were timed on a single `ForwardMB` call per batch after an untimed warm-up call. The MELTS 1.0.2 emulator was given the same cooling paths as PetThermoTools, truncated to the same batch sizes as the HeFESTo emulator, $N = 16$–$262{,}144$ rows. The HeFESTo Earth isentropic emulator was given a 0–140 GPa pressure sweep at $S = 2.5\,\mathrm{J\,g^{-1}\,K^{-1}}$, including conservation of mass and temperature prediction. Material properties are then added on top of each emulator. For HeFESTo this is the *nGibbs* equations of state with every term evaluated (§4.2). For MELTS it is the PetThermoTools property calculator (`calc_phase_props_MELTS_parallel`), which calls the alphaMELTS supplemental calculator for the predicted assemblage without minimization, in parallel over PetThermoTools' default worker count. This route is faster than the vectorized translation for large batches (Fig. 4.25). Some rare low-temperature liquids develop negative atoms after the mass conservation correction. For these, the liquid properties cannot be evaluated by alphaMELTS and are removed before timing. MELTS emulator times are the median of three repeats; all other times are single wall-clock measurements.

Table 4.10: Prediction and recall statistics of the closed-system isothermal MELTS 1.0.2 and 1.2.0 emulators, with (Cr) and without (no Cr) chromium, evaluated against their validation datasets. The number of samples (n) in the training dataset is given immediately below the model name. Freq.: fraction of test assemblages in which the phase is present in the ground truth. Prec./Rec.: precision and recall of the predicted phase presence, shaded from white (1.00) to red (0.80 and below). – : phase not in the model.

	1.0.2, Cr $n = 78{,}040{,}752$			1.0.2, no Cr $n = 75{,}621{,}940$			1.2.0, Cr $n = 122{,}908{,}703$			1.2.0, no Cr $n = 122{,}195{,}502$		
Phase	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.
liquid	0.89	1.00	0.99	0.89	1.00	0.99	0.95	1.00	1.00	0.95	1.00	1.00
clinopyroxene	0.42	0.99	0.99	0.47	0.99	0.99	0.26	0.99	0.98	0.27	0.97	0.94
plagioclase	0.41	0.99	0.99	0.46	0.99	0.99	0.26	0.98	0.99	0.28	0.96	0.98
orthopyroxene	0.39	0.97	0.98	0.43	0.97	0.97	0.26	0.97	0.95	0.27	0.93	0.95
olivine	0.37	0.99	0.97	0.40	0.98	0.97	0.26	0.98	0.96	0.24	0.95	0.94
spinel	0.44	0.98	0.97	0.20	0.96	0.96	0.34	0.97	0.95	0.18	0.96	0.80
fluid	0.05	0.98	0.98	0.06	0.97	0.96	0.42	1.00	0.99	0.47	0.99	0.99
rhm-oxide	0.15	0.97	0.97	0.18	0.97	0.96	0.11	0.96	0.96	0.12	0.96	0.95
quartz	0.13	0.98	0.99	0.14	0.98	0.99	0.11	0.98	0.99	0.12	0.97	0.99
apatite	0.13	0.98	0.98	0.14	0.98	0.96	0.10	0.95	0.97	0.11	0.84	0.98
garnet	0.12	0.99	0.98	0.14	0.99	0.97	0.10	0.97	0.97	0.11	0.94	0.96
whitlockite	0.13	0.96	0.99	0.15	0.97	0.98	0.05	0.97	0.97	0.05	0.89	0.93
K-feldspar	0.07	0.98	0.97	0.08	0.98	0.98	0.05	0.97	0.99	0.06	0.95	0.96
tridymite	0.05	0.94	0.97	0.06	0.96	0.96	0.06	0.95	0.96	0.07	0.92	0.95
biotite	0.03	0.94	0.95	0.03	0.95	0.95	0.03	0.95	0.95	0.03	0.93	0.96
graphite	–	–	–	–	–	–	0.02	0.77	0.88	0.02	0.80	0.86
leucite	0.01	0.83	0.93	0.01	0.83	0.85	0.01	0.92	0.88	0.01	0.91	0.88
hornblende	–	–	–	–	–	–	0.01	0.92	0.96	0.01	0.96	0.92
alloy-solid	–	–	–	0.01	0.88	0.68	0.00	0.72	0.83	0.00	0.73	0.88
muscovite	–	–	–	–	–	–	0.00	0.95	0.98	0.00	0.92	0.94
calcite	–	–	–	–	–	–	0.00	0.92	0.92	0.00	0.91	0.84
nepheline	–	–	–	–	–	–	–	–	–	0.00	0.67	0.86

Table 4.11: Prediction and recall statistics of the isothermal (P, T) and isentropic (P, S) HeFESTo Earth and Mars emulators evaluated on their validation datasets. The number of samples (n) in the training dataset is given immediately below the model name. Freq.: fraction of test assemblages in which the phase is present in the ground truth. Prec./Rec.: precision and recall of the predicted phase presence, shaded from white (1.00) to red (0.80 and below). – : phase not in the model.

	Earth (P, T) $n = 26{,}065{,}657$			Earth (P, S) $n = 26{,}065{,}657$			Mars (P, T) $n = 10{,}259{,}324$			Mars (P, S) $n = 10{,}259{,}324$		
Phase	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.
garnet	0.37	0.99	0.97	0.37	0.99	0.98	0.85	0.99	0.99	0.85	0.99	0.99
ferropericlase	0.64	0.99	0.99	0.64	0.99	0.99	0.35	0.97	0.95	0.35	0.96	0.95
clinopyroxene	0.26	0.99	0.98	0.26	0.99	0.98	0.66	0.99	0.99	0.66	0.99	0.99
Ca-perovskite	0.65	1.00	0.99	0.65	1.00	0.99	0.17	0.99	0.95	0.17	0.99	0.95
olivine	0.23	0.99	1.00	0.23	1.00	0.99	0.49	0.99	0.99	0.49	1.00	0.99
bridgmanite	0.52	0.99	0.99	0.52	0.99	0.99	0.09	0.97	0.98	0.09	0.98	0.99
Ca-ferrite	0.40	0.99	0.98	0.40	0.98	0.98	0.14	0.96	0.95	0.14	0.98	0.92
γ-iron	0.28	0.98	0.90	0.28	0.95	0.97	0.24	0.93	0.95	0.24	0.98	0.87
spinel	0.16	0.98	0.98	0.16	0.99	0.98	0.34	0.97	0.99	0.34	0.99	0.97
orthopyroxene	0.13	0.99	0.98	0.13	0.99	0.98	0.28	0.98	0.99	0.28	0.99	0.99
post-perovskite	0.18	0.98	0.99	0.18	0.98	0.99	–	–	–	–	–	–
ringwoodite	0.08	0.97	0.97	0.08	0.97	0.98	0.27	0.98	0.99	0.27	0.99	0.99
stishovite	0.11	0.97	0.97	0.11	0.97	0.97	0.20	0.96	0.98	0.20	0.98	0.98
kyanite	0.04	0.99	0.99	0.04	1.00	0.99	0.13	0.98	0.96	0.13	0.98	0.96
ε-iron	0.16	0.90	0.99	0.16	0.92	0.99	0.01	0.96	0.77	0.01	0.95	0.85
HP clinopyroxene	0.03	0.93	0.97	0.03	0.94	0.97	0.11	0.95	1.00	0.11	0.97	1.00
plagioclase	0.05	0.97	0.97	0.05	0.98	0.96	0.08	0.95	0.96	0.08	0.95	0.95
wadsleyite	0.06	0.96	0.95	0.06	0.97	0.98	0.07	0.94	0.98	0.07	0.97	0.96
akimotoite	0.02	0.86	0.81	0.02	0.86	0.83	0.08	0.77	0.88	0.08	0.79	0.87
quartz	0.02	0.97	0.95	0.02	1.00	0.96	0.07	0.96	0.97	0.07	0.98	0.96
nepheline	0.02	0.96	0.96	0.02	0.98	0.90	0.06	0.98	0.95	0.06	0.98	0.95
α-iron	0.02	0.87	0.86	0.02	0.86	0.90	0.05	0.95	0.82	0.05	0.96	0.77
coesite	0.01	0.96	0.92	0.01	0.97	0.93	0.04	0.96	0.99	0.04	0.98	0.98
α-PbO_2-SiO_2	0.02	0.96	0.97	0.02	1.00	0.99	–	–	–	–	–	–
NAL phase	–	–	–	–	–	–	0.01	0.98	0.88	0.01	0.99	0.83

4.3 Results

The released emulators were evaluated against validation test subsets of 15,000 assemblages per model, and along reference-composition paths computed directly with alphaMELTS and HeFESTo. The training sets contain 10^7–10^8 assemblages per model. We report the results in the order in which a user meets them: which phases are predicted, whether the predicted assemblage conserves mass, how closely phase proportions and physical properties follow the parent model along representative paths, and how long the calculation takes.

Phase presence

Tables 4.10 and 4.11 give the precision and recall of the predicted phase presence. For the MELTS emulators, liquid is predicted with precision and recall of 0.99–1.00, and the common silicate phases (clinopyroxene, orthopyroxene, plagioclase and olivine) with 0.93–0.99. The HeFESTo emulators reach 0.95 or better for the silicate and oxide phases present in more than about 10% of test assemblages, including garnet, the $(Mg,Fe)_2SiO_4$ polymorphs, bridgmanite, post-perovskite, ferropericlase and Ca-perovskite. The exceptions are the Ca-ferrite phase in the Mars isentropic emulator (recall 0.92) and γ-iron, which is present in 24–28% of test assemblages but has a recall as low as 0.87–0.90. The isothermal and isentropic HeFESTo emulators perform similarly.

Classification quality declines for phases that are rare in the training data and stable only over narrow ranges of pressure, temperature or composition. In the MELTS emulators these are graphite (precision 0.77–0.80), Fe–Ni alloy (0.68–0.88 in either metric), leucite (0.83–0.93) and nepheline (0.67 precision in MELTS 1.2.0 without Cr), each present in 2% of test assemblages or fewer. Spinel recall falls to 0.80 in the MELTS 1.2.0 emulator without Cr, where spinel is less frequent than in the Cr-bearing model. In the HeFESTo emulators the weakest phases are akimotoite (0.77–0.88), α-iron (0.77–0.96), ε-iron in the Mars models (recall 0.77–0.85, frequency 0.01) and the NAL phase in the Mars models (recall 0.83–0.88).

Mass conservation

Without correction, the elemental bulk composition reconstructed from the raw network output differs from the requested composition by a median of 0.8–2.3% per element, depending on the emulator, and 45–68% of element

abundances are in error by more than 1% (Fig. 4.4). The largest raw errors are in minor and trace components, such as Cr and Fe^{3+}, where the network assigns small amounts of an element to assemblages whose bulk composition contains little of it. After the correction of §4.2, the median relative error falls to about 5×10^{-6}% for every emulator, the 99th percentile to 0.1% or less, and fewer than 0.4% of element abundances differ from the request by more than 1%.

Phase proportions along reference paths

Figures 4.15 and 4.16 compare phase proportions along isentropes of reference compositions computed with HeFESTo and with the isentropic emulators, given pressure, entropy and bulk composition. For basalt, depleted MORB mantle and Si-enriched pyrolite on Earth, the emulator reproduces the sequence of phase assemblages from the surface to 140 GPa, including the olivine–wadsleyite–ringwoodite transitions, the post-spinel transition near 23–25 GPa and the post-perovskite transition between about 120 and 130 GPa, with phase proportions that are visually indistinguishable over most of each path. The visible differences are confined to thin fields, such as the narrow akimotoite field in basalt near 25–30 GPa, and to small ripples in otherwise smooth phase proportions. The Mars emulator reproduces the basalt, harzburgite and shergottite paths to 25 GPa with the same fidelity, including the stishovite, kyanite and γ-iron fields of the enriched compositions. The largest differences on the Mars paths are small shifts of the clinopyroxene–garnet boundary in the shergottite between about 15 and 18 GPa.

Figures 4.17 and 4.18 show isobaric cooling of the Bishop Tuff, a lherzolite and a MORB with the closed-system isothermal MELTS emulators. Both emulators reproduce the liquidus and the order and temperatures of phase appearance for all three compositions. The lherzolite paths, which span 700–1800 °C in MELTS 1.0.2 and 1330–1800 °C in MELTS 1.2.0, are matched closely throughout, including the clinopyroxene-out reaction near 1380 °C in MELTS 1.0.2. Where alphaMELTS changes abruptly, the emulator smooths the change. The MORB solidus near 1030 °C, at which the alphaMELTS liquid fraction falls from several percent to zero over a few degrees, is spread by the emulator over a few tens of degrees, and a small, sharp step in alkali-feldspar abundance near 780 °C in the MELTS 1.0.2 Bishop Tuff path is absent from the emulator's.

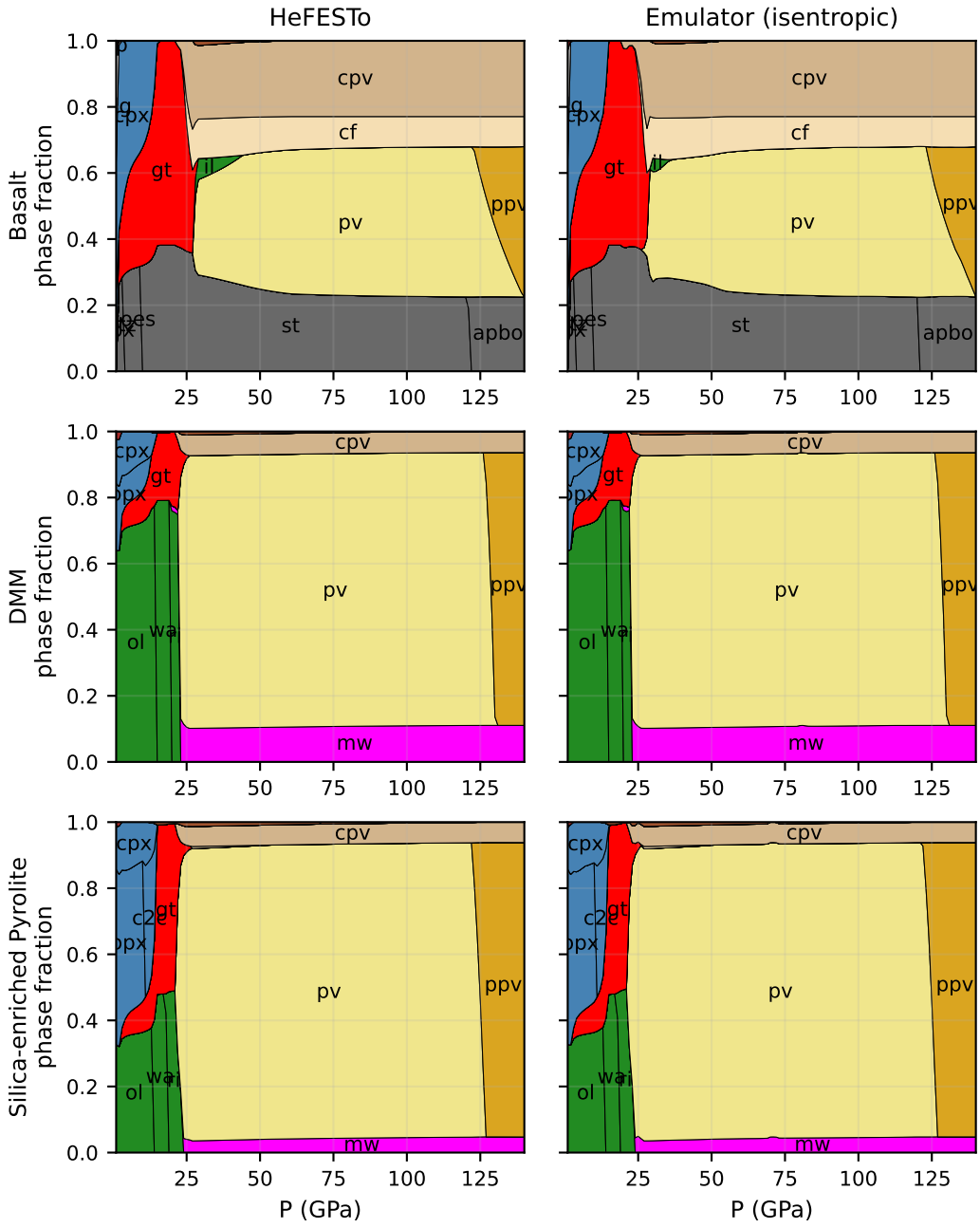


Figure 4.15: Phase abundances for isentropes of reference compositions for the HeFESTo Earth model. Reference compositions are basalt and Depleted MORB Mantle from Stixrude and Lithgow-Bertelloni (2024) after Workman and Hart (2005), and Si-enriched pyrolite (after McDonough and Sun, 1995) reference compositions. Phases: plg, plagioclase; sp, spinel; opx, orthopyroxene; cpx, clinopyroxene; c2c, high-pressure clinopyroxene; gt, garnet; ol, olivine; wa, wadsleyite; ri, ringwoodite; il, akimotoite; pv, bridgmanite; ppv, post-perovskite; cpv, Ca-perovskite; cf, Ca-ferrite-structured phase; nal, NAL phase; mw, ferropericlase; qtz, quartz; coes, coesite; st, stishovite; apbo, α-PbO_2-structured SiO_2; ky, kyanite; feg, γ-iron.

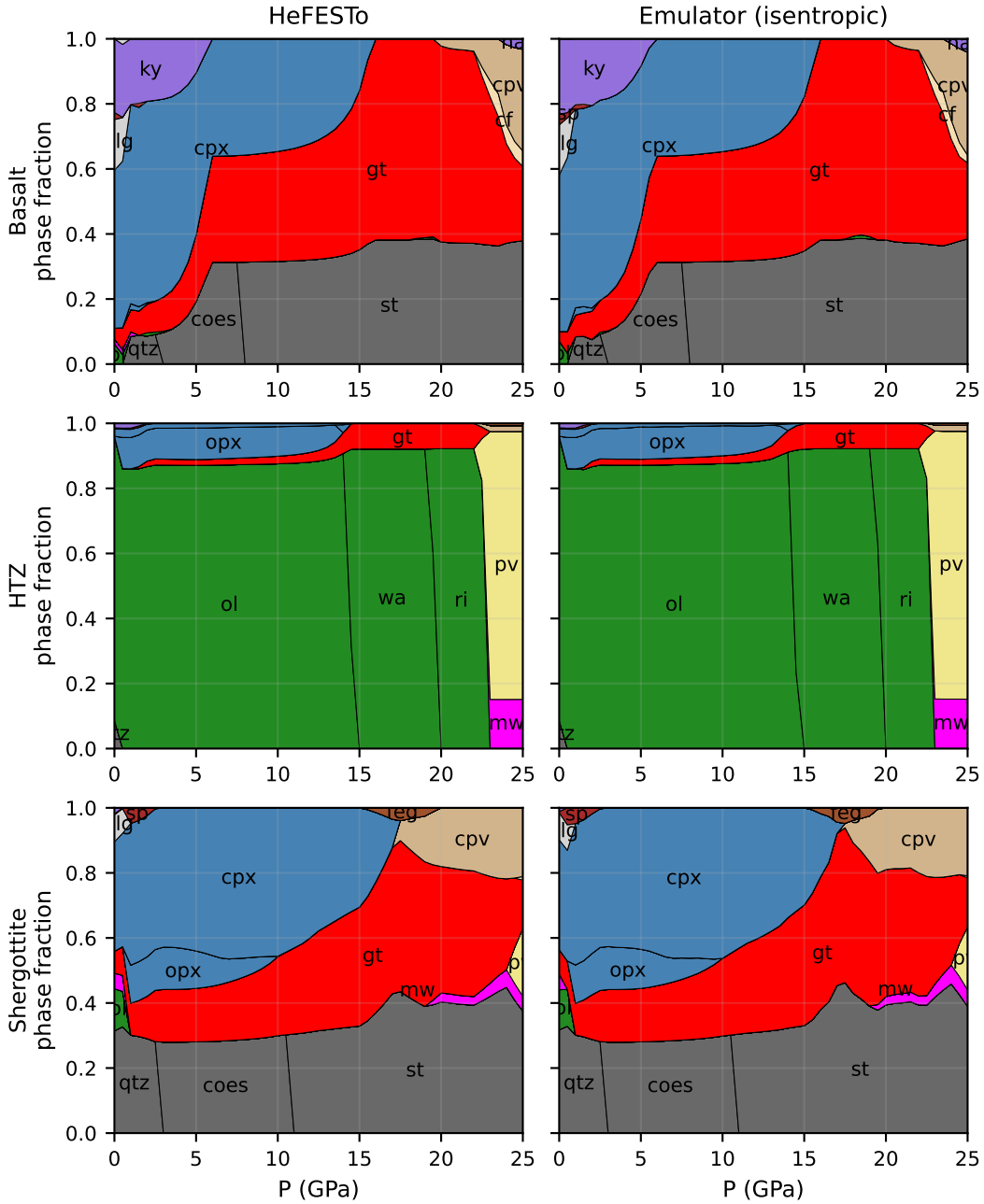


Figure 4.16: As Fig. 4.15, for the HeFESTo Mars model and the basalt, harzburgite (Stixrude and Lithgow-Bertelloni, 2024, after Workman and Hart, 2005), and shergottite (Stolper and McSween, 1979) reference compositions. Phase abbreviations as in Fig. 4.15.

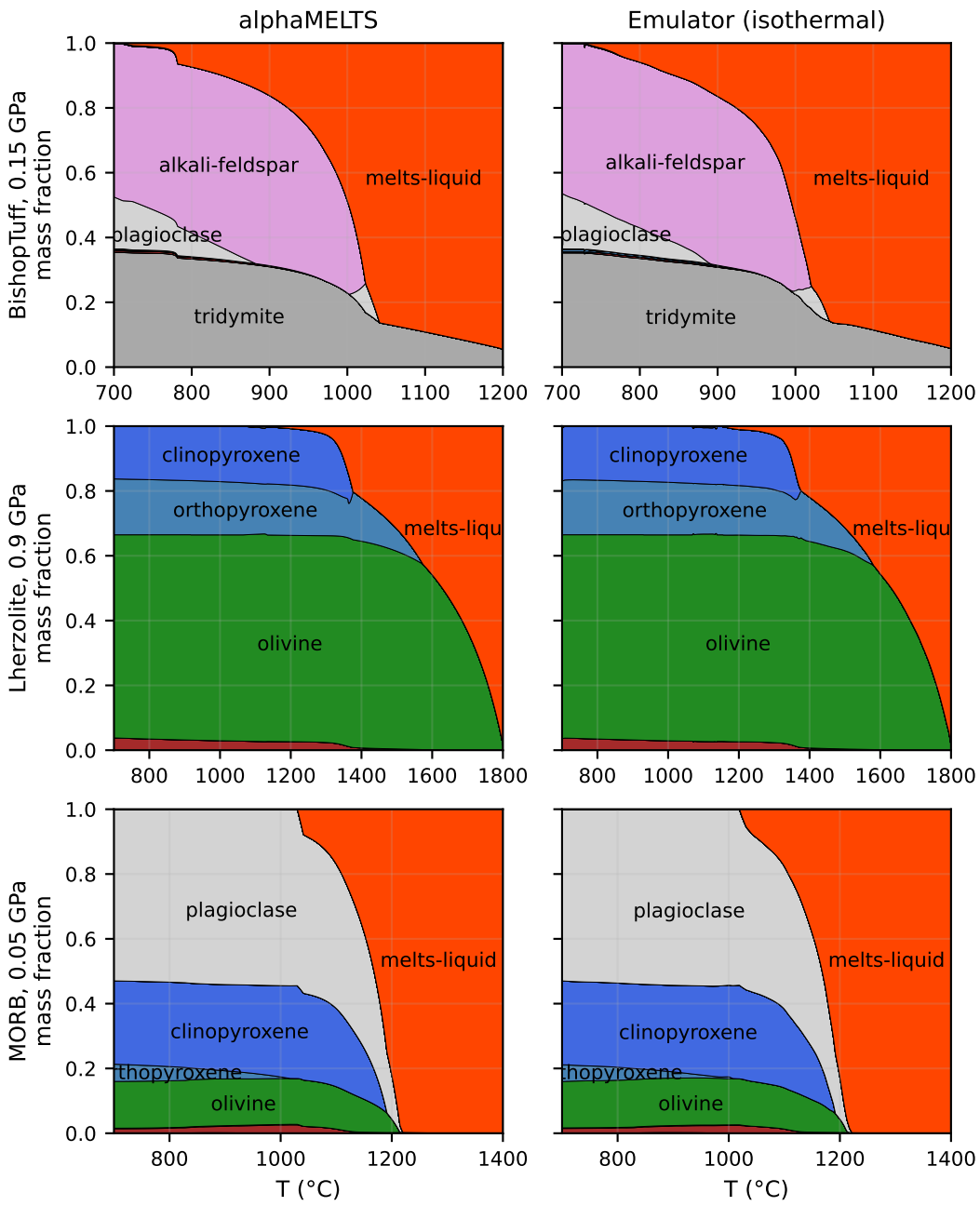


Figure 4.17: Phase mass fractions during isobaric cooling of the MELTS 1.0.2 reference compositions (Cr variant): Bishop Tuff (Hildreth, 1979), lherzolite (Dasgupta and Hirschmann, 2007), and MORB (Allan et al., 1989), each at the pressure given in its panel title. alphaMELTS (left) and the closed-system isothermal emulator, given pressure, temperature and bulk composition (right).

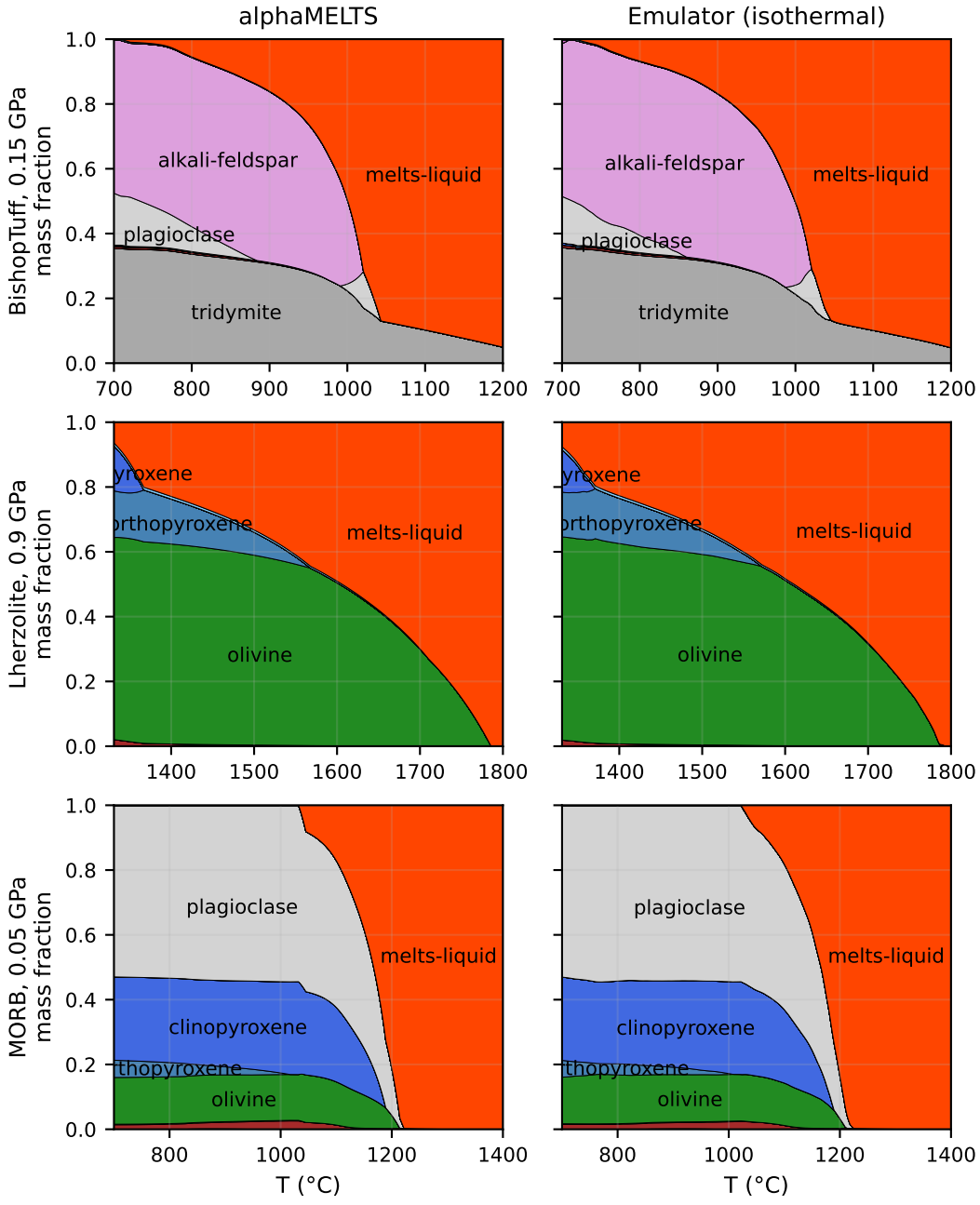


Figure 4.18: As Fig. 4.17, for the MELTS 1.2.0 reference compositions and the closed-system MELTS 1.2.0 isothermal emulator (Cr variant).

Liquid lines of descent

Figure 4.19 compares the liquid compositions along the lherzolite and MORB cooling paths. The emulator follows the alphaMELTS liquid lines of descent closely for MgO, FeO, Fe_2O_3, Al_2O_3 and Cr_2O_3, including the inflections at which phases enter or leave the assemblage. The differences grow as the liquid fraction becomes small. For the most evolved lherzolite liquids ($SiO_2 > 52$ wt%), the emulator underestimates Na_2O and K_2O and overestimates CaO, and its path extends to about 56 wt% SiO_2 where alphaMELTS stops near 54.5 wt%. For the last MORB liquids, H_2O and P_2O_5 diverge from alphaMELTS by up to about 0.4 and 0.15 wt%, respectively. These are the states in which the liquid holds a small fraction of the system's mass, so an error in the liquid composition has little effect on mass balance and is weakly constrained by it.

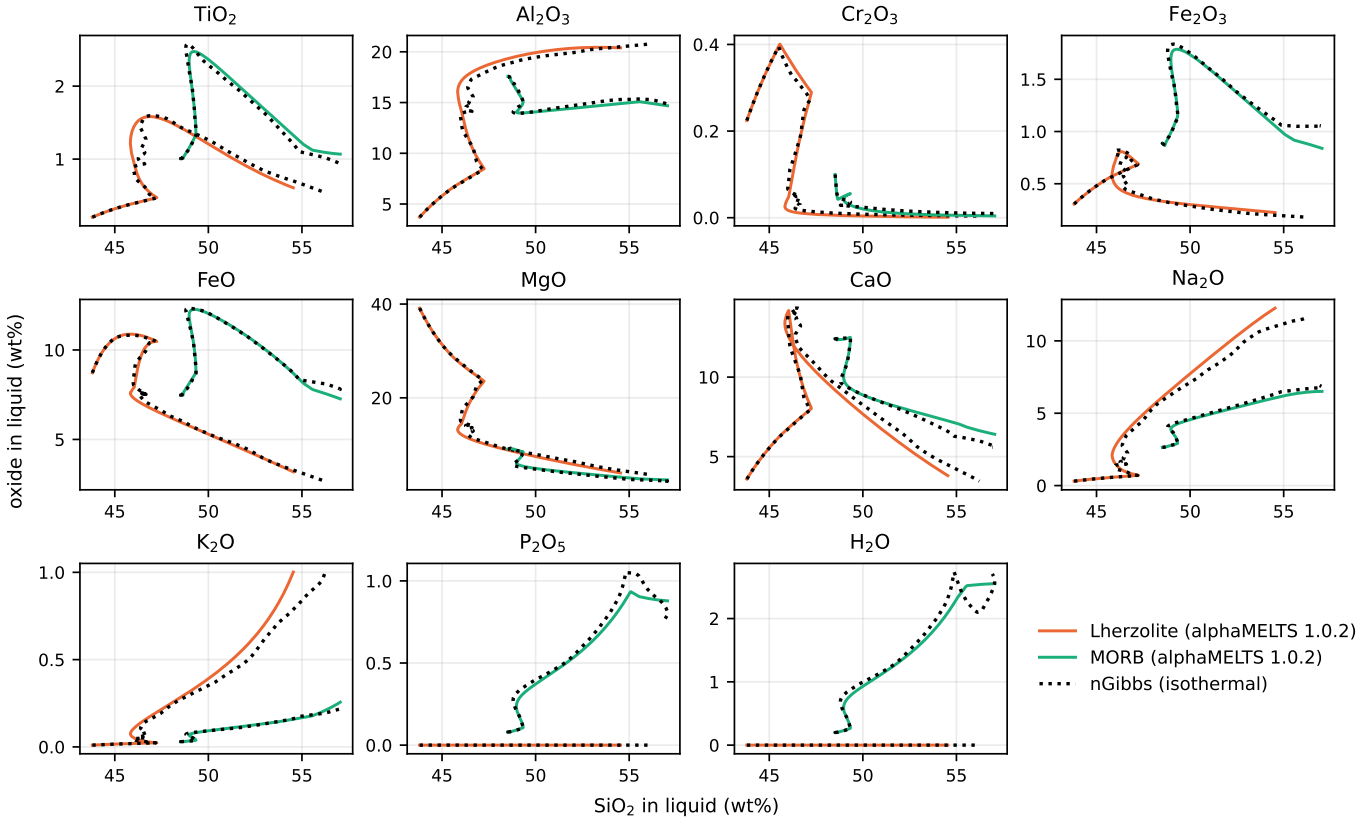


Figure 4.19: Liquid lines of descent during isobaric cooling of the lherzolite and MORB MELTS 1.0.2 reference compositions (Cr variant; Fig. 4.17): Harker diagrams of liquid SiO_2 against each other liquid oxide, in wt%. Solid, colored by rock: alphaMELTS; black dotted: the closed-system MELTS 1.0.2 isothermal emulator, given pressure, temperature and bulk composition. Only states with at least 1% liquid by mass (as predicted by alphaMELTS) are shown.

Material properties and temperature

Along the HeFESTo Earth isentropes (Fig. 4.20), density, V_P, V_S and temperature computed from the emulator's assemblage and predicted temperature cannot be distinguished from those of HeFESTo at the scale of the figure. The emulator also places the peaks in C_p, K_S and α at the phase transitions in the correct positions and with the correct amplitudes. The entropy calculated from the predicted assemblage and temperature stays within about $0.007\,\mathrm{J\,g^{-1}\,K^{-1}}$, or 0.3%, of the entropy imposed on the isentrope, and its mean error on the DMM path is 0.01%. The temperature model that supplies this temperature is evaluated in Fig. 4.14. The HeFESTo Mars isothermal emulator (Fig. 4.21) similarly reproduces the HeFESTo properties along the reference paths to 25 GPa, including the peaks in C_p, K_S and α at the phase transitions for the basalt and harzburgite reference compositions. The shergottite composition is the most exotic to the training domain of the Mars model, and thus the C_p, K_S and α predictions, which are dependent on more temperamental metamorphic terms, are far less accurate. Even for the shergottite composition, the density, V_P, V_S and temperature are indistinguishable from HeFESTo at the scale of the figure.

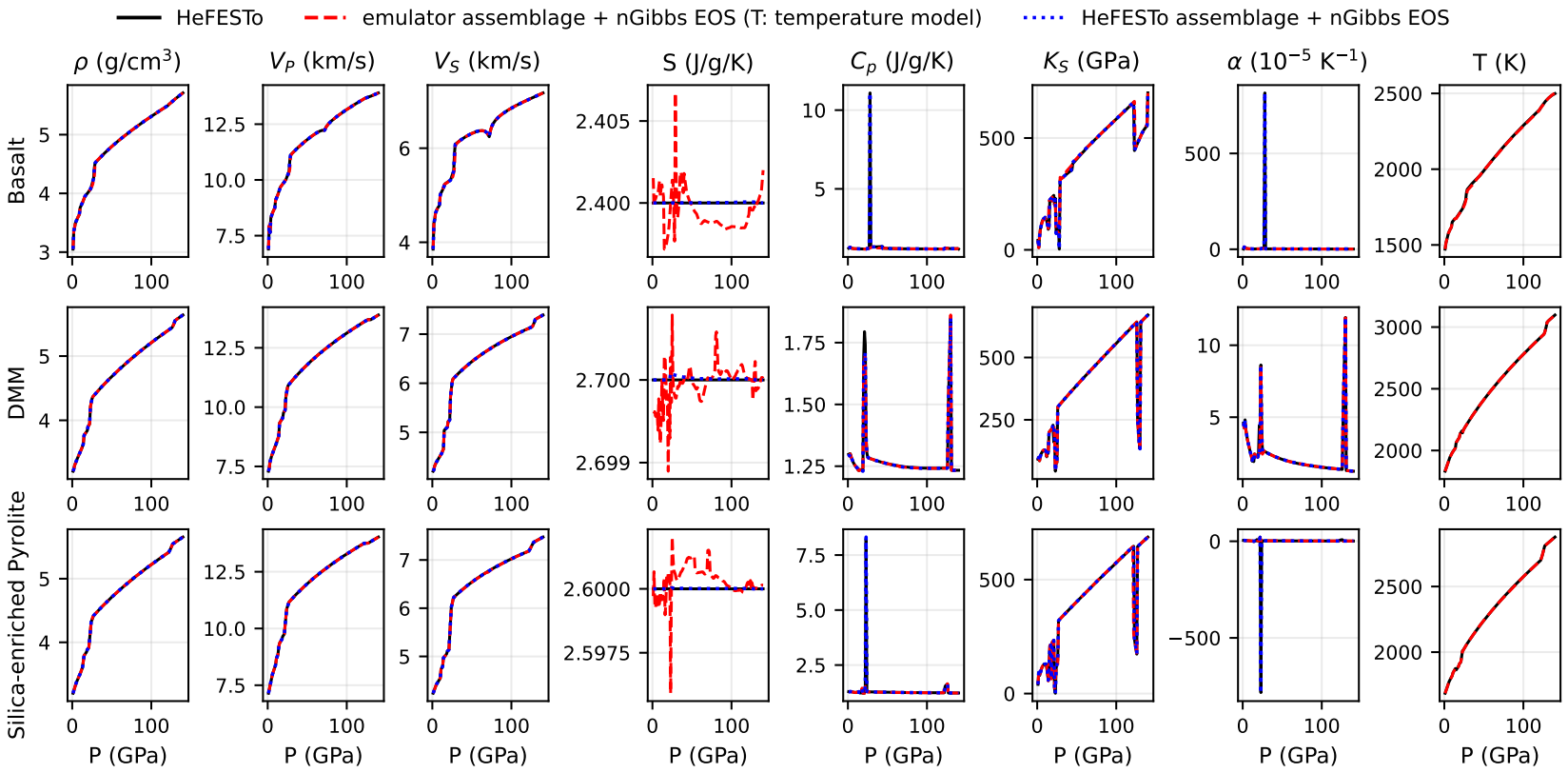


Figure 4.20: Bulk properties along the HeFESTo Earth reference composition paths of Fig. 4.15 (one row per reference composition): density, V_P, V_S, S, C_p, K_S and α against pressure. Black: HeFESTo; red dashed: the isentropic emulator's predicted assemblage, at its predicted temperature, evaluated with the *nGibbs* equations of state (§4.2); blue dotted: the HeFESTo assemblage evaluated with the same equations of state. The right-hand column compares the true temperature with that of the temperature model (Fig. 4.14).

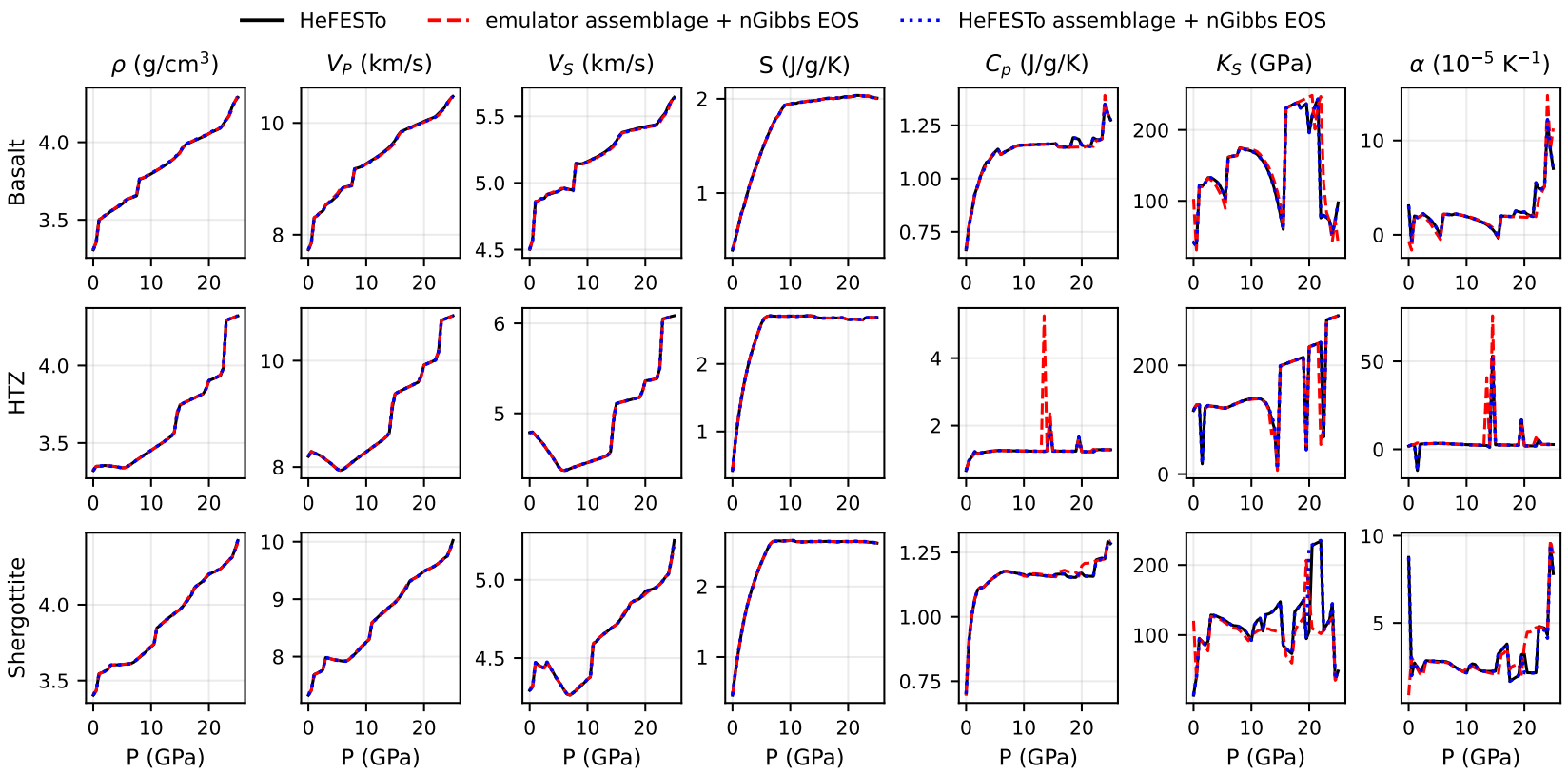


Figure 4.21: As Fig. 4.20, for the HeFESTo Mars model and the reference composition paths of Fig. 4.16, with the isothermal emulator.

Figure 4.22 separates the error of the emulator from that of the equations of state for the DMM isentrope. With every term included, the mean relative errors of the emulator's properties against HeFESTo are 0.02% for ρ, V_P and V_S, 0.01% for S, 0.13% for C_p, 0.63% for K_S and 0.86% for α. The same equations of state applied to the HeFESTo assemblage give 0.11%, 0.53% and 0.69% for C_p, K_S and α, so most of the remaining error comes from the equation-of-state translation rather than from the emulated assemblage. The metamorphic terms are necessary for these three properties: without them, the errors are 2.1% in C_p, 18.8% in K_S and 7.1% in α, because the peaks at the phase transitions are missing. The order–disorder ("fast") term reduces the error in V_P from 0.15% to 0.02% and affects no other property.

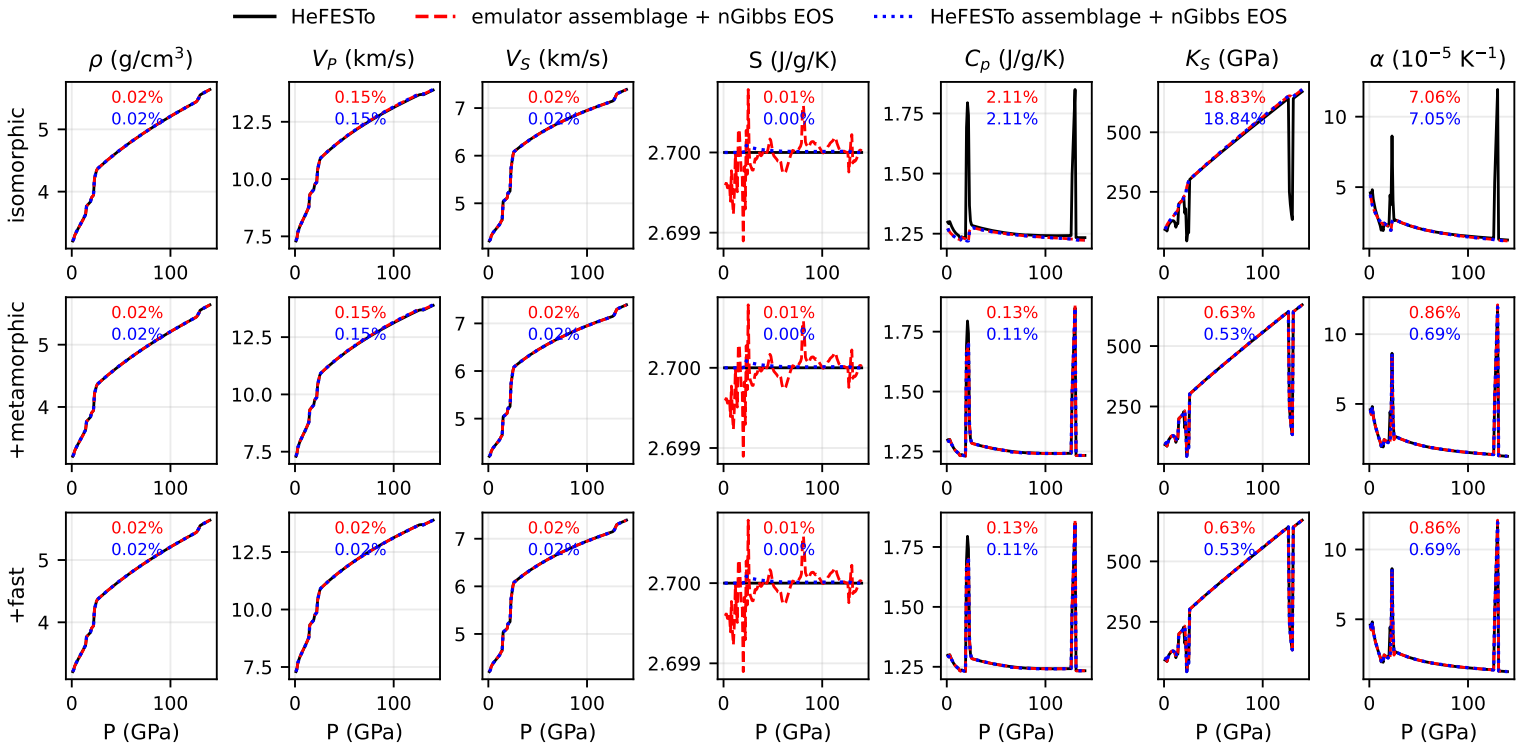


Figure 4.22: Bulk properties along the HeFESTo Earth DMM adiabat of Fig. 4.20, evaluated with successively more terms of the *nGibbs* HeFESTo equations of state (one row per level). Isomorphic: the properties of each phase at fixed composition. +metamorphic: adds the change of the equilibrium assemblage with temperature and pressure to C_p, K_S and α. +fast: also adds the intra-phase order-disorder relaxation to V_P, which completes the set of terms in HeFESTo's reported properties; ρ, V_S and S have no phase-change term and are the same at every level. Line styles as in Fig. 4.20, with the emulator's assemblage at the temperature-model temperature; the numbers in each panel are the mean relative errors against HeFESTo of the emulator assemblage (red) and of the HeFESTo assemblage (blue). Timings for each level are in Table 4.12.

For MELTS 1.0.2 (Fig. 4.23), the properties of the emulated lherzolite and MORB assemblages overlie those of alphaMELTS, including the abrupt changes in density and volume at the MORB solidus. The Bishop Tuff shows the largest differences, in density, volume and enthalpy below about 900 °C, which coincide with the alkali-feldspar step described above.

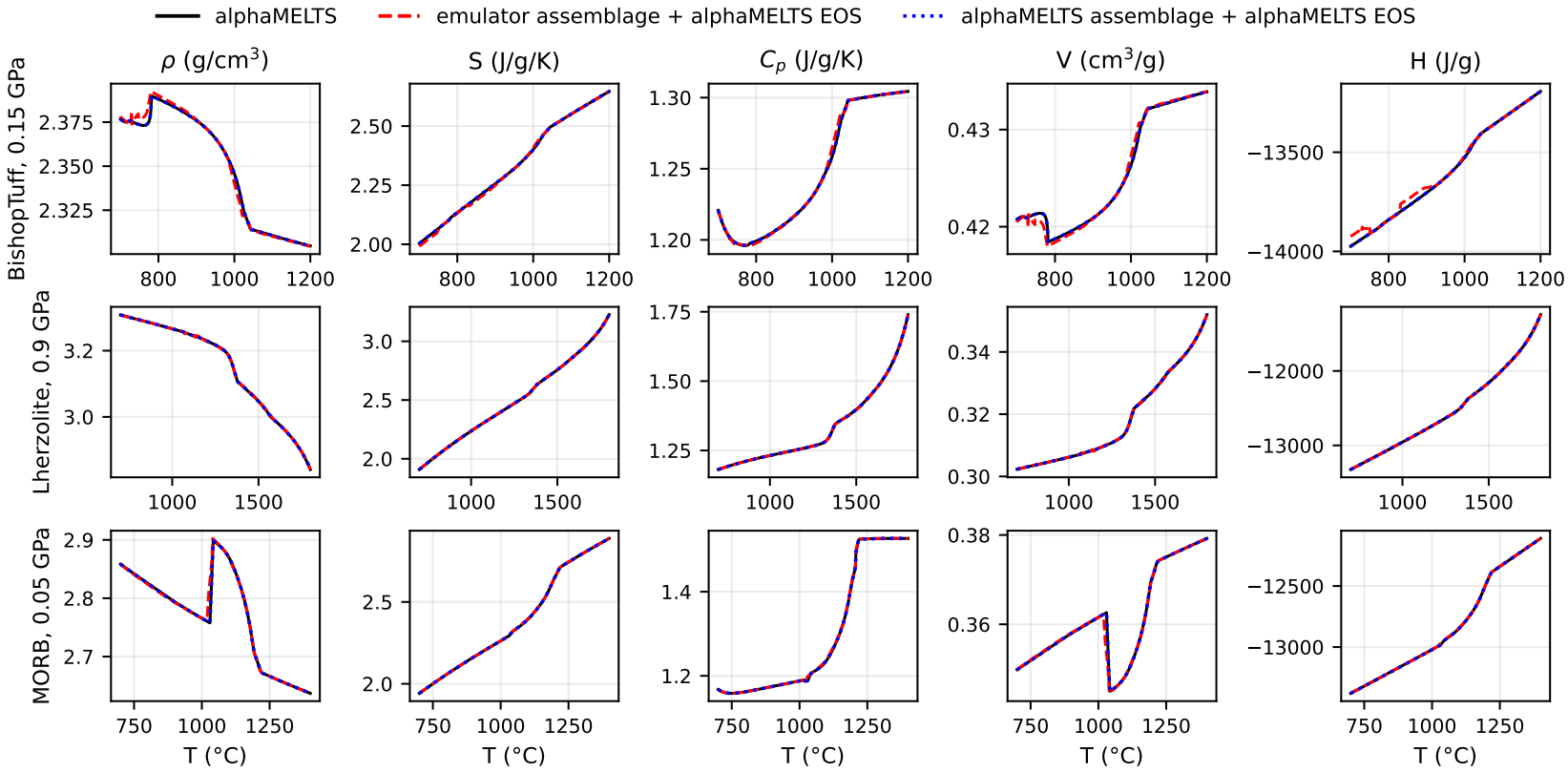


Figure 4.23: Bulk properties during isobaric cooling of the MELTS 1.0.2 reference compositions of Fig. 4.17 (one row per rock): density, S, C_p, V and H against temperature. Black: alphaMELTS; red dashed: the isothermal emulator's predicted assemblage evaluated with the multi-threaded PetThermoTools property solver (§4.2); blue dotted: the alphaMELTS assemblage evaluated similarly.

Table 4.12: Wall time (s) to evaluate $N = 2^{15}$ (32,768) assemblages along the HeFESTo Earth isentrope of §4.2 on the laptop (4-core Intel i7-11370H CPU and NVIDIA GeForce RTX 3060 GPU with 6 GB VRAM) and the desktop (20-core Intel i7-14700H CPU and NVIDIA GeForce RTX 5070 Ti GPU with 16 GB of VRAM). The first column shows the time for the isentropic emulator alone (phase equilibria, mass balance and temperature). Each further column adds the *nGibbs* equations of state with successively more terms (Fig. 4.22): the isomorphic properties only, then the metamorphic (phase-change) terms, then the order-disorder ("fast") relaxation of V_P. The HeFESTo time is extrapolated linearly from the fastest measured run on each machine (8 parallel workers on the laptop, from 3,384 assemblages in 3,479 s; 20 parallel workers on the desktop, from 8,460 assemblages in 1,074 s).

	Emulator	+ isomorphic	+ metamorphic	+ fast	HeFESTo
Laptop CPU	1.52	12.9	22.4	29.6	33,690
Laptop GPU	0.44	2.23	13.7	21.1	–
Desktop CPU	1.38	5.47	8.52	10.6	4,160
Desktop GPU	0.17	0.81	4.03	6.00	–

Computational performance

Table 4.12 and Fig. 4.24 compare the wall time of the emulators and EOS implementations of *nGibbs*, with the source parent models. For 2^{15} (32,768) assemblages on the HeFESTo Earth isentrope, HeFESTo running on all available workers is extrapolated to need about 33,700 s on the 4-core laptop and 4,200 s on the 20-core desktop. The emulator alone, including mass balance and temperature, takes 1.4–1.5 s on either CPU, a speed-up of about 3,000 (desktop) to 22,000 (laptop), and 0.17–0.44 s on a GPU. Adding the full equations of state brings the totals to 10.6 s (desktop CPU) and 29.6 s (laptop CPU), still 390–1,100 times faster than HeFESTo. Most of the property cost lies in the metamorphic and order–disorder terms; with only the isomorphic terms, the GPU totals are 0.81 s (desktop) and 2.2 s (laptop).

The speed-up of *nGibbs* over MELTS is smaller than for HeFESTo because alphaMELTS is itself much faster than HeFESTo, and is timed favorably here along cooling paths on which each equilibrium starts from the previous one. On the laptop, alphaMELTS through PetThermoTools reached at most about 67 equilibria per second (8 workers), against about 8,500 assemblages per second for the emulator on a single CPU call of 262,144 assemblages, a factor of about 130. On the desktop the corresponding rates are about 800

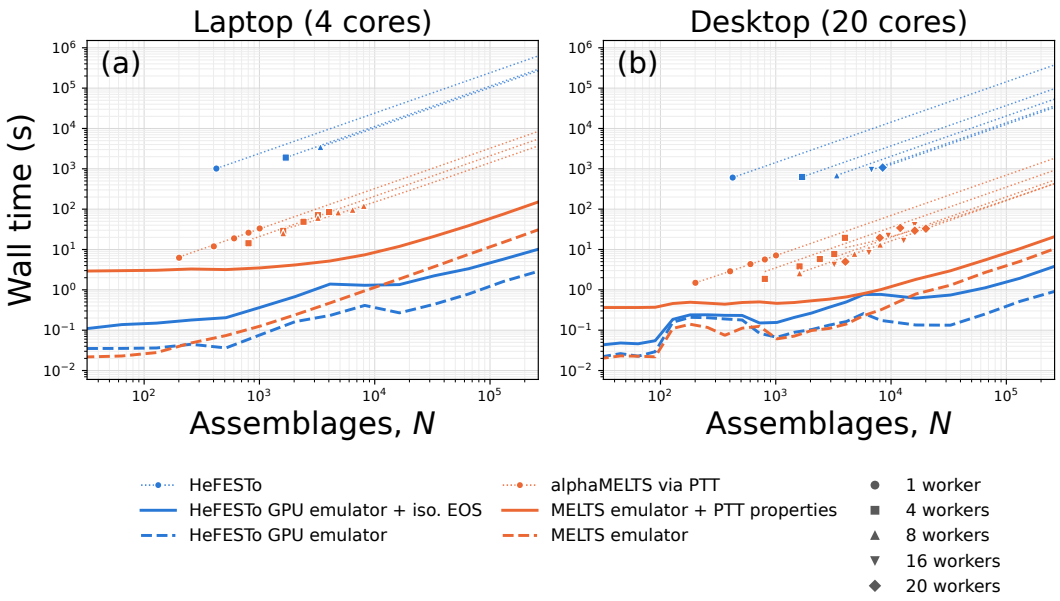


Figure 4.24: Wall time against the number of assemblages N for the reference codes and the emulators (§4.2) on (a) a 4-core Intel i7-11370H laptop and (b) a 20-core Intel i7-14700H desktop, on shared axes. Markers are measured HeFESTo and alphaMELTS (via PetThermoTools) runs, with marker shape giving the worker count; each dotted line is a straight-line fit through the runs of one worker count, or through the origin for a single run, extrapolated to the largest N. The *nGibbs* HeFESTo emulator performance was measured on the CPU for $N < 8192$ and on the GPU for $N \geq 8192$. This cross-over is managed automatically by the *nGibbs* `ForwardMB` function. All other emulator curves are CPU-only timings.

equilibria per second (20 workers) and 24,000 assemblages per second, a factor of about 30. When properties are added through the PetThermoTools property calculator, which dominates the cost for large batches, the factors fall to about 26 (laptop) and 15 (desktop). The emulator's cost is nearly constant below about 10^3 assemblages, where fixed overheads dominate, and grows roughly linearly above about 10^4 (Fig. 4.24). The crossover points between the vectorized and sequential property routes for MELTS are shown in Fig. 4.25.

For the MELTS property calculation benchmark, the sequential approach is more efficient for evaluating fewer states because previous calculations can be reused and previous states become good initial guesses for optimization problems, but a vectorized approach becomes faster when many states are evaluated at once. A multi-threaded approach, however, is able to leverage benefits afforded to both sequential and parallel workflows. For MELTS properties, the

vectorized translation is faster than the single-threaded sequential workflow, but slower than the multi-threaded workflow leveraged by the PetThermoTools `calc_phase_props_MELTS_parallel()` function (M. L. M. Gleeson et al., 2025) for more than about 2000 states (Fig. 4.25). The single-threaded sequential workflow avoids the overhead cost of deploying multiple threads, and is fastest for fewer than about 700 states. The vectorized approach is the fastest between 700 and 2000 states evaluated at once, but because this speedup is marginal and *nGibbs* is meant for applications at much larger batch sizes, we generally recommend that users calculate properties using the multi-threaded implementation in PetThermoTools.

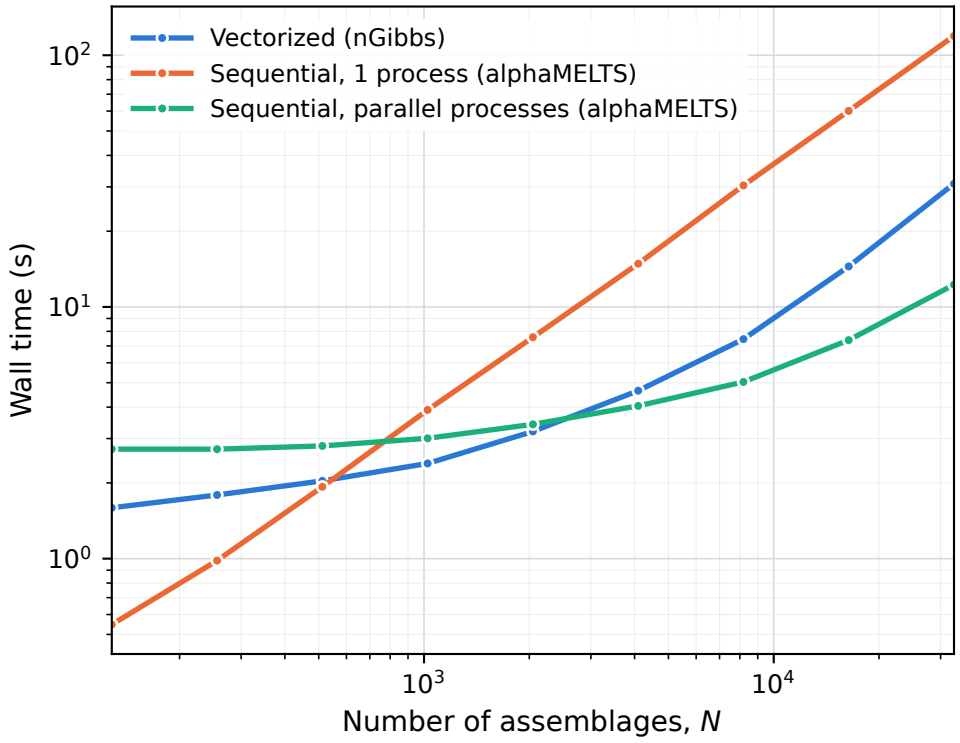


Figure 4.25: Wall time to compute MELTS 1.0.2 phase and bulk properties for an assemblage table predicted by the MELTS 1.0.2 emulator, against the number of assemblages, benchmarked on the 4-core Intel i7-11370H laptop. Vectorized: the *nGibbs* array translation of the MELTS equations of state (§4.2), taken as the difference between an emulator call with and without properties. Sequential: the alphaMELTS supplemental calculator called once per state and phase through PetThermoTools, in one process or split across PetThermoTools' default number of worker processes. No calculation includes free-energy minimization. Inputs are a fixed MORB composition on a P–T grid over 2–10 kbar and 900–1400 °C.

4.4 Application

Availability and installation

nGibbs is open source and available at https://github.com/benjthyer/nGibbs, and the release used in this chapter (v0.1.0b4) is archived on Zenodo (Thyer, 2026). It can be installed from PyPI with

```
pip install ngibbs
```

The MELTS emulators are also available through PetThermoTools (`pip install petthermotools`) with the `"nMELTS"` model keywords described in §4.4. Running the emulators on a GPU requires a CUDA-enabled installation of PyTorch.

PetThermoTools Integration

PetThermoTools (M. L. M. Gleeson et al., 2025) is a Python package that runs alphaMELTS for Python (Antoshechkina and Ghiorso, 2018) and MAGEMin (Riel et al., 2022) through one interface. It serves as a front end interface that allows users to interact with different thermodynamic models through a single, consistent API, which accesses different underlying models through a `Model` keyword argument. Its outputs are pandas DataFrames keyed by phase in the alphaMELTS naming convention, and it runs independent calculations in separate child processes so that a failed MELTS calculation does not halt a batch (M. L. M. Gleeson et al., 2025). The MELTS emulators of *nGibbs* are now also integrated within PetThermoTools, accessible with the `Model` keyword of a MELTS model preceded by the letter `n` (e.g. `"nMELTSv1.0.2"`), and their predicted assemblages are returned in the same structure as an alphaMELTS result. This serves two purposes. First, it makes *nGibbs* accessible to users who already work in PetThermoTools and who need not learn a separate interface. Second, it makes the emulators straightforward to benchmark, because the same function call can be run with the emulator and with the MELTS model it replaces, and the results compared directly. M. L. M. Gleeson et al. (2025) argued that a common interface would encourage systematic comparison of thermodynamic models across P–T–X space, and the same reasoning applies to comparing an emulator with its parent.

By default, the emulator output is passed to the PetThermoTools property calculator, which calls the alphaMELTS supplemental calculator for each predicted phase without performing a minimization, returning material proper-

ties for each phase and for the bulk assemblage. For applications where these properties are not required, this step can be bypassed by appending `NoProp` to the `Model` keyword (e.g. `"nMELTSv1.0.2NoProp"`), resulting in significantly faster performance. At this time, *nGibbs* through PetThermoTools is only supported on the CPU. Users interested in running nGibbs on their CUDA-enabled GPU may do so through the *nGibbs* interface directly.

Phase diagrams

Listing 4.2 computes the same phase diagram for a Mauna Loa primary melt twice, once with rhyolite-MELTS 1.0.2 and once with the *nGibbs* emulator, changing only the `Model` keyword. The emulator needs no initial guess and no path from a neighboring state, so it evaluates the grid as one batch. In the run shown, the emulator returned the diagram in about a tenth of the wall time of rhyolite-MELTS, although the two calls are not identical in workload because PetThermoTools also applies grid refinement to the MELTS calculation (M. L. M. Gleeson et al., 2025). The phase fields produced by the two calculations are compared in Fig. 4.26.

Listing 4.2: Phase diagram of a Mauna Loa primary melt computed with rhyolite-MELTS 1.0.2 and with *nGibbs* through PetThermoTools.

```
import multiprocessing
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from time import time

# Mauna Loa estimated primary melt composition from Wieser et al. 2025
ML_starting = {
    'SiO2_Liq': 48.5858,
    'TiO2_Liq': 1.4381,
    'Al2O3_Liq': 9.73,
    'FeOt_Liq': 10.144,
    'MgO_Liq': 20.0,
    'CaO_Liq': 7.2186,
    'Na2O_Liq': 1.7327,
    'K2O_Liq': 0.3227,
    'H2O_Liq': 0.2,
    'Fe3Fet_Liq': 0.12
}

```

```
# Guard required for multiprocessing on Windows: each PTT worker re-imports this
# file, and without it would rerun phaseDiagram_calc (and reload nGibbs) itself.
if __name__ == "__main__":
    multiprocessing.freeze_support()
    import petthermotools as ptt

    ptt.nGibbs_bridge._ensure_ngibbs_loaded()

    start_time = time()
    PD_ML = ptt.phaseDiagram_calc(Model = "MELTSv1.0.2",
                                  bulk = ML_starting,
                                  T_C = np.linspace(1100,1400,20),
                                  P_bar = np.linspace(250,5000,20),
                                  refine = 2,
                                  i_max = 3)
    print(f"rhyolite-MELTS Phase diagram calculation time: {time() - start_time:.2f} seconds")

    start_time = time()
    nPD_ML = ptt.phaseDiagram_calc(Model = "nMELTSv1.0.2NoProp", # No properties needed for this analysis
                                  bulk = ML_starting,
                                  T_C = np.linspace(1100,1400,20),
                                  P_bar = np.linspace(250,5000,20))

    print(f"nGibbs Phase diagram calculation time: {time() - start_time:.2f} seconds")

    ptt.plot_phaseDiagram_multi({'nGibbs': nPD_ML, 'rhyolite-MELTS v1.0.2': PD_ML})
```

Output:

```
rhyolite-MELTS Phase diagram calculation time: 52.30 seconds
nGibbs Phase diagram calculation time: 1.23 seconds
```

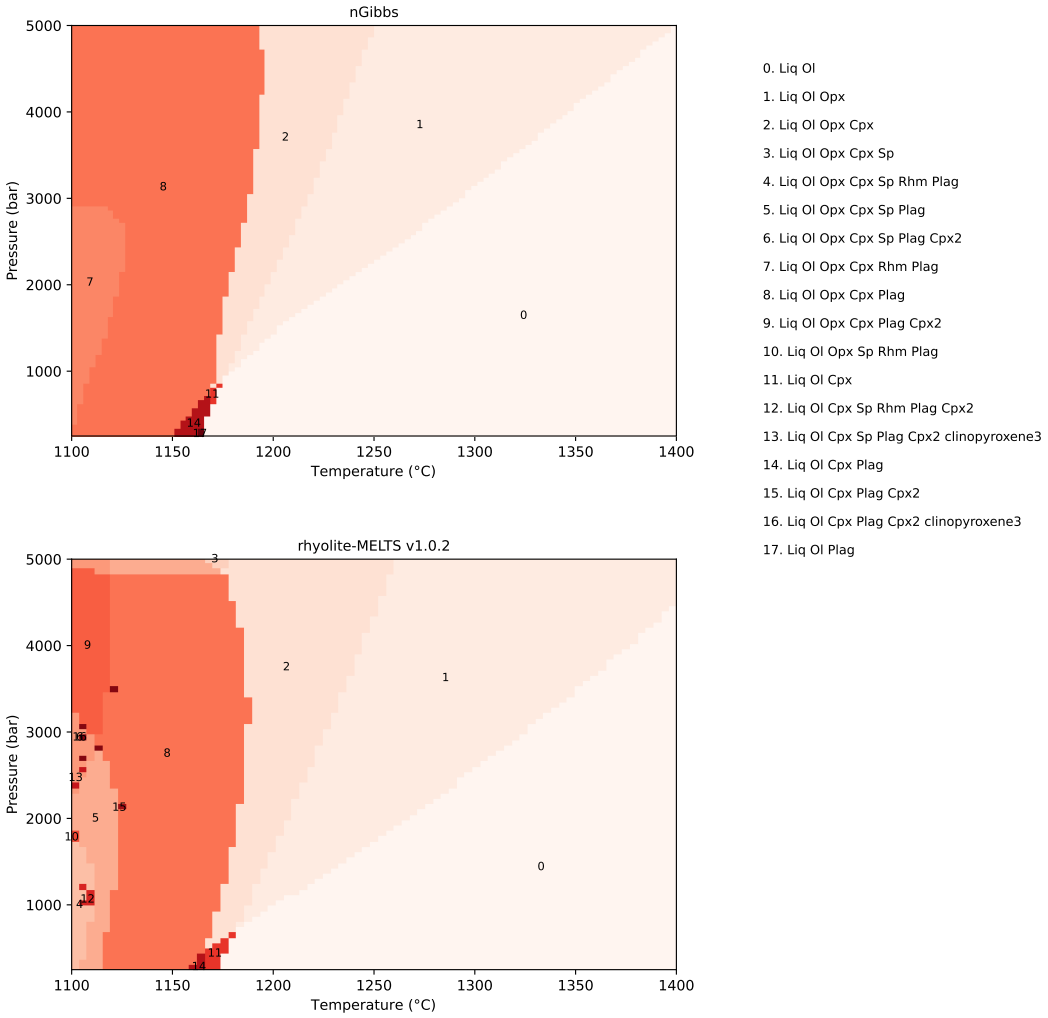


Figure 4.26: Stable phase assemblages for the Mauna Loa primary melt (Wieser, Shi, et al., 2025; Wieser, M. L. Gleeson, et al., 2025) over 1100–1400 °C and 250–5000 bar, from *nGibbs* (top) and RHYOLITE-MELTS 1.0.2 (bottom), produced by Listing 4.2.

Seismic inversions of the thermochemical structure of planetary mantles

Seismic data constrain the thermal and compositional structure of planetary mantles (Stixrude and Lithgow-Bertelloni, 2012; Jackson and Thomas, 2021; Khan et al., 2018; Wood and Rubie, 1996). A thermochemical inversion of seismic data begins with a forward model: bulk composition and thermal profiles are converted to density and seismic velocities that follow from a phase assemblage predicted through free energy minimization. These seismic and density profiles then are compared with seismic and physical observations (Khan and Connolly, 2008; Khan et al., 2021). A Markov chain Monte Carlo

(MCMC) sampler requires many thousands of such likelihood evaluations, which in turn each require many free energy minimizations. Every additional compositional parameter enlarges the volume the sampler must explore. This is why compositional dimensions are often fixed, as in the InSight inversion of Khan et al. (2021), which assumed a single bulk composition derived from the martian meteorites.

The HeFESTo emulators are well-suited to this problem. Listing 4.3 builds 201 whole-mantle profiles which span Earth's interior from 0–135 GPa, as an example workflow for this problem. Each profile is a conductive lid over an isentropic mantle over a conductive thermal boundary layer at the core–mantle boundary; the lid and isentrope follow the geophysical parameterization of Khan et al. (2021). Across the set, adiabat entropy, pressure at the adiabatic-conductive interfaces, temperature of the mantle at the core–mantle boundary, and composition of the mantle are varied. The isentropic segments are evaluated first in parallel as a single batch with the isentropic emulator, which also predicts their temperatures. Both conductive layers are then evaluated as a second batch with the isothermal emulator, and the vectorized HeFESTo equations of state return density, V_P, V_S and bulk modulus using isomorphic terms only for every node (Figs. 4.27 and 4.28). In the run shown, the full set of 35,175 assemblages completes in under 5 seconds on a GPU, whereas the true HeFESTo calculation would take about 10 hours. At that rate an inversion can sample bulk composition alongside the thermal parameters, rather than fixing it in advance, enabling future studies to use seismic data to constrain composition with a level of detail that has never before been possible. The Mars emulators can be used in the same way: they are trained to 25.5 GPa, which spans the pressure at the martian core–mantle boundary of ∼20 GPa estimated by Khan and Connolly (2008). We are already exploring this avenue in ongoing work.

On Earth, where seismic data are abundant, density and velocity profiles are more tightly constrained and carry more chemical information. A joint inversion for temperature and composition beneath plumes and subduction zones could separate the thermal contribution to discontinuity topography and sharpness from the compositional one, and could test whether the transition zone is compositionally stratified relative to the mantle above and below it. The 2024 HeFESTo database, which the Earth emulator reproduces, includes

ferric iron, the high-spin to low-spin transition in ferropericlase, and native iron phases (Stixrude and Lithgow-Bertelloni, 2024). Those additions make the emulator a candidate forward model for the iron-rich compositions proposed for ULVZs and LLVPs (Dobson and Brodholt, 2005; Jackson and Thomas, 2021; Q. Yuan et al., 2023). Some of these candidates, however, such as banded iron formation with up to 85 wt% iron oxide (Dobson and Brodholt, 2005), lie far outside the composition range of the current training data and would require targeted training data before the emulator can be applied to them.

Listing 4.3: Seismic and phase profiles of 201 whole-mantle columns (a conductive lid, an adiabatic mantle and a conductive boundary layer at the core–mantle boundary) of basalt mixed with silica-rich pyrolite, computed in parallel with the HeFESTo Earth isentropic and isothermal emulators and the vectorized HeFESTo equations of state.

```
import numpy as np
import torch
from time import time
from ngibbs.utils.math_utils import mix_compositions, IDX_2D_Split_3Part
from ngibbs.engine.models import HeFESToEmulatorCPU, HeFESToEmulatorGPU

# Each profile is a whole-mantle column: a conductive lid over an
# adiabatic (isentropic) mantle over a conductive thermal boundary layer
# (TBL) at the core-mantle boundary (CMB). All profiles are evaluated in
# one batch, on the GPU when one is available.
HeFESToEmulator = HeFESToEmulatorGPU or HeFESToEmulatorCPU
device = HeFESToEmulator.device
print(f"Using the {str(device).upper()} emulator")

n_pressures = 175  # pressure nodes per profile
n_profiles = 201   # profiles evaluated in parallel
P_CMB = 135        # GPa
T_surface = 300    # K
pressures = np.linspace(0, P_CMB, n_pressures)

# Profile parameters, swept together from the first profile to the last
X_basalt = np.linspace(1, 0, n_profiles)     # basalt fraction
S_ad = np.linspace(2.3, 2.8, n_profiles)     # isentrope entropy (J/g/K)
P_lid = np.linspace(8, 1, n_profiles)        # base of the lid (GPa)
P_TBL = np.linspace(132, 125, n_profiles)    # top of the TBL (GPa)
T_CMB = np.linspace(3500, 5000, n_profiles)  # CMB temperature (K)

```

```
# Snap both transitions to the nearest pressure node. The isentrope
# covers nodes lid_idx to TBL_idx; the lid lies above it, the TBL below.
lid_idx = np.abs(pressures - P_lid[:, None]).argmin(axis=1)
TBL_idx = np.abs(pressures - P_TBL[:, None]).argmin(axis=1)
P_lid, P_TBL = pressures[lid_idx], pressures[TBL_idx]

# (row, column) index pairs into the (profile x pressure) grid for each
# regime. The lid and the TBL are both conductive, so they are evaluated
# together.
TBL_IDX, isentropic_IDX, lid_IDX = IDX_2D_Split_3Part(
    lid_idx, TBL_idx + 1, n_pressures)
conductive_IDX = tuple(np.concatenate(pair)
                       for pair in zip(lid_IDX, TBL_IDX))

# End-member bulk compositions in HeFESTo element moles, from the
# SILICAadiabat and BASALTadiabat standards ('O' sets the ferric iron)
SILICA = {'Si': 4.35196, 'Mg': 4.35196, 'Fe': 0.51993, 'Ca': 0.29376,
          'Al': 0.40504, 'Na': 0.05391, 'Cr': 0.02345, 'O': 14.54576}
BASALT = {'Si': 4.61116, 'Mg': 1.33197, 'Fe': 0.61701, 'Ca': 1.22855,
          'Al': 1.81090, 'Na': 0.39533, 'Cr': 0.00508, 'O': 15.35841}
mixes = [mix_compositions([SILICA, BASALT], [1 - x, x])
         for x in X_basalt]
comp_headers = list(mixes[0])
comp_array = np.array([list(mix.values()) for mix in mixes])

# Results on the (profile x pressure) grid, filled one regime at a time
properties = {prop: np.zeros((n_profiles, n_pressures))
              for prop in ['temperature', 'S', 'rho', 'Vp', 'Vs', 'Kh']}
indexer = HeFESToEmulator.isentropic_emulator.ml_indexer
mass_phasedict = indexer.mass_phasedict  # phase name -> column
phase_fractions = np.zeros((n_profiles, n_pressures,
                            len(mass_phasedict)))

def emulate(IDX, condition, condition_header, outputs):
    """Emulate the stable assemblage at each node in IDX from its
    pressure, its S or T, and its composition."""
    inputs = np.column_stack(
        [pressures[IDX[1]], condition, comp_array[IDX[0]]])
    headers = ['P(GPa)(System_main)', condition_header] + comp_headers
    with torch.no_grad():
        return HeFESToEmulator.ForwardMB(inputs, headers=headers,
                                         outputs=outputs)

```

```
def store(IDX, output, T):
    """Fill the grid at IDX with the phase mass fractions, and with bulk
    properties from the vectorized HeFESTo equations of state evaluated
    on the emulated assemblages."""
    properties['temperature'][IDX] = T
    mass_percent = output['phase_tables'][1].cpu().numpy()
    phase_fractions[IDX] = mass_percent / 100
    PT = torch.tensor(np.column_stack([pressures[IDX[1]], T]),
                      dtype=torch.float64, device=device)
    eos = HeFESToEmulator._compute_bulk_EOS_properties(
        output['component_moles'].to(device), PT,
        property_names=['S', 'rho', 'Vp', 'Vs', 'Kh'])
    for name, values in eos.items():
        properties[name][IDX] = values

# --- Step 1: the adiabatic mantle (isentropic emulator) ---------------
# Run first, because both conductive layers start from the isentrope
# temperatures, which the isentropic model predicts at each (P, S).
t0 = time()
output = emulate(isentropic_IDX, S_ad[isentropic_IDX[0]],
                 'S(J/g/K)(System_main)',
                 ['component_moles', 'phase_tables', 'temperature'])
T_isentrope = output['temperature'].cpu().numpy().reshape(-1)
store(isentropic_IDX, output, T_isentrope)
print(f'Adiabatic mantle:  {len(isentropic_IDX[0])} assemblages '
      f'in {time() - t0:.2f} s')

# --- Step 2: the lid and the TBL (isothermal emulator) ----------------
# Temperature is linear in pressure: in the lid from T_surface to the
# isentrope at P_lid, and in the TBL from the isentrope at P_TBL to
# T_CMB. Each linspace drops the node it shares with the isentrope.
T = properties['temperature']
T_lid = [np.linspace(T_surface, T[i, j], j + 1)[:-1]
         for i, j in enumerate(lid_idx)]
T_TBL = [np.linspace(T[i, j], T_CMB[i], n_pressures - j)[1:]
         for i, j in enumerate(TBL_idx)]
T_conductive = np.concatenate(T_lid + T_TBL)

t0 = time()
output = emulate(conductive_IDX, T_conductive, 'T(K)(System_main)',
                 ['component_moles', 'phase_tables'])
store(conductive_IDX, output, T_conductive)
print(f'Conductive layers: {len(conductive_IDX[0])} assemblages '
```

```
114        f'in {time() - t0:.2f} s')
```

Output:

```
Using the CUDA emulator
Adiabatic mantle:  32327 assemblages in 3.33 s
Conductive layers: 2848 assemblages in 1.14 s
```

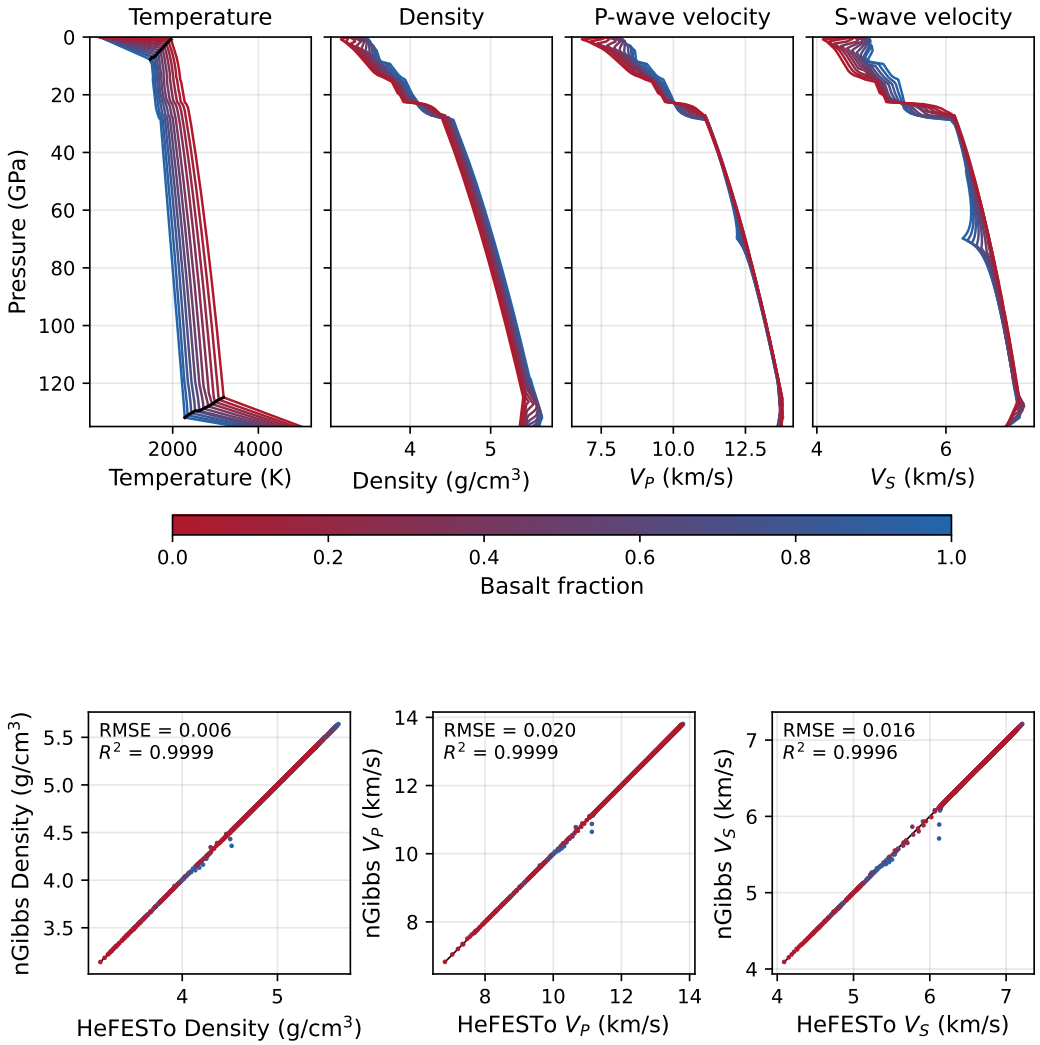


Figure 4.27: Temperature, density, V_P and V_S against pressure for every 20th of the 201 profiles produced by Listing 4.3, colored by basalt fraction. Temperature is linear in pressure in the conductive lid (from 300 K at the surface) and in the core–mantle boundary layer, and follows the isentrope between them; black lines join the transitions of the plotted profiles. The end members are the silica-rich pyrolite (after McDonough and Sun, 1995) and basalt reference compositions from Stixrude and Lithgow-Bertelloni (2024) after Workman and Hart (2005).

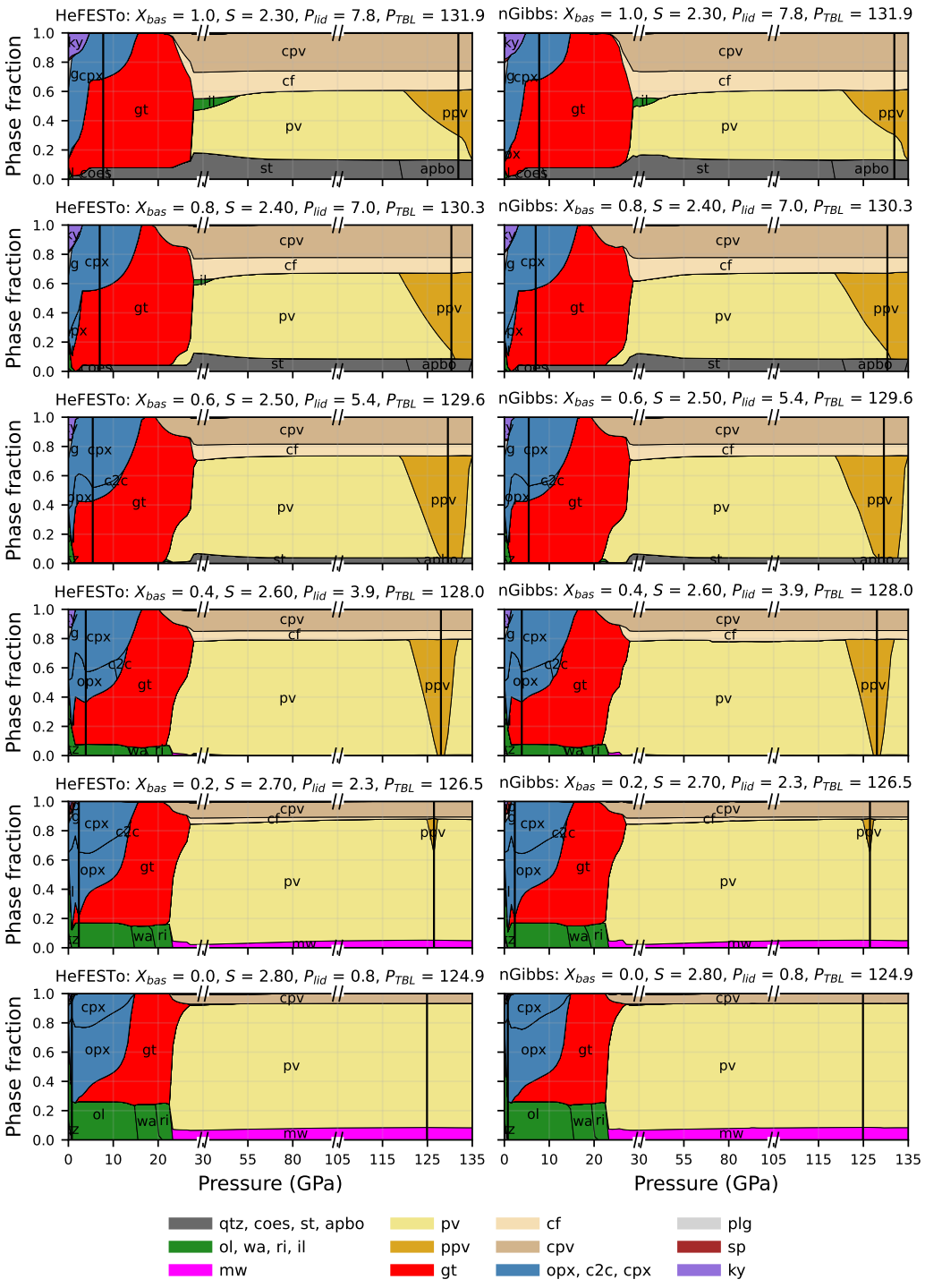


Figure 4.28; caption on the following page.

Figure 4.28: Phase mass fractions along six of the profiles of Listing 4.3, with basalt fractions evenly spaced between 1.0 and 0.0, computed with HeFESTo (left) and with the *nGibbs* emulators (right). Both profiles are computed with the same workflow: the isentrope is evaluated first between the base of the conductive lid and the top of the core–mantle conductive boundary layer, then the conductive layers are evaluated along linear paths between the surface temperature and the CMB temperature and the isentropic temperature at the top and bottom of the isentrope. Each panel gives the basalt fraction X_{bas}, the isentrope entropy S ($\mathrm{J\,g^{-1}\,K^{-1}}$), the pressure at the base of the lid P_{lid} and the pressure at the top of the core–mantle boundary layer P_{TBL} (GPa), which are also marked by solid black lines. The pressure axis condenses the relatively uninteresting middle mantle between 30 and 105 GPa to represent a disproportionately small part of the plot in order to emphasize the portions of the mantle where mineralogy strongly evolves as a function of depth. The 6 profiles evaluated by HeFESTo took 29.8 minutes on the 4-core Intel i7-11370H laptop, while all 201 profiles evaluated by the *nGibbs* emulators are completed in under 5 seconds on the same machine.

Mantle thermochemistry beneath spreading centers

An emulator of pHMELTS is planned. pHMELTS uses the pMELTS calibration for mantle melting and it approximates the storage of water in nominally anhydrous minerals by partitioning H_2O into a hidden reservoir (Asimow et al., 2004; P. M. Smith and Asimow, 2005). This model shows that trace amounts of water can have a significant effect on mantle melting (Asimow and Langmuir, 2003; Asimow et al., 2004). A fast emulator of polybaric melting followed by crystallization may allow individual mid-ocean ridge basalts to be inverted for the potential temperature, source composition and water content of the mantle that produced them, and repeating the inversion along ridge segments would build a map of mantle thermochemistry beneath spreading centers.

Existing methods for recovering primary magmas show why such an inversion is difficult. PRIMELT2 restores a lava to its primary magma by adding or subtracting olivine alone and infers potential temperature from the primary magma's MgO content (Herzberg and Asimow, 2008). It cannot be applied to lavas that have also fractionated clinopyroxene, which usually happens once the liquid has evolved below 7–10 wt% MgO, and applying olivine addition to such lavas yields primary magmas with too much MgO and source temperatures that are too high (Herzberg and Asimow, 2008). PRIMARSMELT, the martian

member of the same family, has the same restriction: liquids that have fractionated olivine, clinopyroxene and plagioclase cannot be easily inverted by this method without a significant increase in computational cost and uncertainty (Hernández-Montenegro et al., 2024). This restriction effectively excludes every lava that has evolved beyond olivine-only fractionation.

Multiple saturation is also ambiguous evidence of origin. Asimow and Longhi (2004) showed that aggregate liquids from polybaric near-fractional melting can display near-intersections of liquidus surfaces that cannot be distinguished experimentally from true multiple saturation points, so a multiply saturated liquid need not be a single batch that equilibrated with its residue. They did show, however, that such multiple saturation points preserve information about the average pressure at which the constituent increments of liquid were extracted (Asimow and Longhi, 2004). Lavas with complex melting and differentiation history likely encode some of this information in their geochemistry, but attempting to recover it requires a fast and flexible forward model that properly treats melting and multi-phase fractionation together. A pHMELTS emulator would provide such a forward model. Whether the inversion is well posed for a given suite of lavas, given the trade-offs among potential temperature, source fertility and water content, remains to be tested.

4.5 Discussion

Strengths

The central result of this chapter is that a single emulator per parent model can reproduce that model's phase equilibria across a broad range of composition, pressure, and temperature conditions and needn't be restricted to a single compositional path or magma series. The *nGibbs* HeFESTo Earth emulator follows basaltic, peridotitic and Si-enriched compositions from the surface to 140 GPa, and the MELTS emulators follow compositions from lherzolite to rhyolite, with the same network in each case (§4.3). The surrogates reviewed in §4.1 were each trained on a narrower slice: arc magmas with fixed water content (Candioti et al., 2026), or compositions near the basalt–harzburgite join (Hartmeier and Lanari, 2026; Q. Yuan et al., 2025).

The emulators predict the same quantities as their parent models, the abundance of every phase and the composition of every solution phase, so physical properties are computed with the parent model's own equations of state rather

than learned as separate targets. These differences are far smaller than the uncertainty of the thermodynamic databases themselves (Ghiorso and Sack, 1995; Ghiorso et al., 2002; Stixrude and Lithgow-Bertelloni, 2005; Riel et al., 2022; Holland and Powell, 1998). For most applications, therefore, the choice of database matters more than the choice between the emulator and its parent.

As in the parent models, mass balance is enforced exactly rather than encouraged. This is a requirement for reactive transport and for any calculation that removes or adds material step by step, such as fractional crystallization, where a small bias per step would accumulate. Hartmeier and Lanari (2026) penalize mass-balance violations during training and report small residual deviations, while the surrogate of Candioti et al. (2026) closes mass balance only for the bulk solid, by difference.

The emulators are fast without a GPU. On a 4-core laptop CPU, the HeFESTo Earth emulator is four orders of magnitude faster than HeFESTo on the same machine, and a GPU adds a further factor of 3–8 (§4.3). Because *nGibbs* reaches its largest CPU speed-up on a 4-core laptop, it is usable on ordinary workstations. Through PetThermoTools, it is also usable without learning a new interface (§4.4).

Limitations

An emulator can be no more accurate than its parent model. The MELTS emulators inherit the known limitations of MELTS, including its calibration toward low-pressure, nearly anhydrous experiments (Ghiorso and Sack, 1995) and its poor treatment of hydrous minerals such as amphibole (Candioti et al., 2026). The HeFESTo emulators inherit a database that has no silicate liquid. The emulators are also reliable only inside their training coverage (Figs. 4.5–4.8). Compositions far from that coverage, such as banded iron formation or carbonatite, need targeted training data and validation before use.

Three systematic weaknesses appear in the results. First, phases that are rare and stable over narrow ranges are predicted less reliably (§4.3). Akimotoite is the most consequential of these for mantle applications, because its stability near the 660-km discontinuity is one of the composition-sensitive reactions that a transition-zone inversion would aim to resolve (Hartmeier and Lanari, 2026); oversampling its stability field is a priority before that application. Second, the emulator smooths sharp features in the parent model, such as the MORB

solidus in MELTS, over a few tens of degrees. This follows from the continuous architecture (§4.2), which is correct for continuous phase-out reactions but blurs changes that are nearly discontinuous, as at this solidus. Third, properties that depend on derivatives of the phase proportions (C_p, K_S and α) and liquid compositions at small melt fractions carry larger errors than phase proportions do (§4.3 and 4.3) and require higher fidelity predictions. Applications that rely on these quantities, such as latent-heat effects at phase transitions or the composition of the last liquids, should check them against the parent model.

The speed advantage also has limits. Fixed overheads make the emulators no faster than the parent model for very small batches, and the advantage over alphaMELTS on a multicore machine is about one to two orders of magnitude, not four, especially once properties are computed through PetThermoTools (§4.3). The emulators pay off most when many states are evaluated together, as in the inversions and coupled models of §4.4.

Finally, the release is deliberately partial. Only the closed-system isothermal MELTS emulators and the HeFESTo emulators are released; the open-system MELTS emulators await further testing. Emulator error is not yet propagated into downstream calculations, which an inversion would need in its likelihood. Planned work includes a pHMELTS emulator (§4.4), targeted training for rare phases and unusual compositions, and a systematic map of the depth, width and velocity contrast of the transition-zone discontinuities as functions of composition.

4.6 Conclusions

This chapter introduced *nGibbs*, a set of deep-learning emulators of the MELTS 1.0.2 and 1.2.0 and HeFESTo thermodynamic models. Its goal is to stand in for its parent models in essentially any calculation they would be used for, rather than for one composition range or one set of outputs. Four design choices support that goal:

1. Each emulator predicts the full equilibrium assemblage, so thermodynamically consistent physical properties follow from the parent model's own equations of state.

2. Phase abundances are predicted as rectified affinities, which keeps them continuous across phase boundaries like the parent model they emulate.

3. Mass is conserved exactly at inference by a constrained correction that itself is constrained to produce legal (non-negative) atomic abundances.

4. The training compositions are drawn from whole-rock databases of igneous rocks that span the breadth of compositional diversity of igneous rock compositions known on Earth. MELTS 1.2.0 is additionally trained on a database of sedimentary rock compositions, and HeFESTo models are strongly perturbed to cover compositions that are rare or unknown at Earth's surface. The validation data are drawn from distinct bulk compositions to ensure that the emulators generalize to new compositions.

The released emulators reproduce their parent models closely within their training coverage. The major silicate and oxide phases are predicted with precision and recall of 0.93 or better. Element abundances are conserved to a median relative error of about $5 \times 10^{-6}\%$. Phase proportions, densities and seismic velocities along reference paths are nearly indistinguishable from those of alphaMELTS and HeFESTo. Subtle behaviors observed in liquid lines of descent are learned and reproduced by the emulators. Errors are largest for rare phases with narrow stability fields, such as akimotoite and the native iron phases. They are also larger where sharp changes occur in the parent model, for material properties such as heat capacity, bulk modulus and thermal expansivity that depend on molar derivatives; and for liquid compositions at small melt fractions. These differences remain smaller than the uncertainties of the underlying thermodynamic databases for most applications. They do, however, mark where targeted training data are needed before specific applications, such as resolving akimotoite stability near the 660-km discontinuity.

The emulators are fast enough to change what is computationally practical. On a 4-core laptop CPU, the HeFESTo Earth emulator evaluates 2^{15} assemblages about 22,000 times faster than HeFESTo, and about 1,100 times faster with the full equations of state. A set of 201 whole-mantle profiles of phase equilibria, density, and sound velocity, comprising about 35,000 assemblages, takes under 5 s on a GPU, compared with about 10 hours for HeFESTo. The MELTS emulators run two orders of magnitude faster than alphaMELTS and, through PetThermoTools, can replace it in existing workflows by changing a single keyword.

Several applications follow directly from this speed and generality. In seismic

inversions of the Earth's and Mars' mantles, bulk composition can be sampled alongside the thermal parameters rather than fixed in advance. Composition-dependent transition-zone structure and candidate compositions for ultralow-velocity zones can be explored systematically. Geodynamic models can carry a chemically heterogeneous mantle, or the cumulates of a solidifying magma ocean, without reducing composition to a single mixing variable. Planned extensions include an emulator of pHMELTS for inverting mid-ocean ridge basalts for the thermochemical state of the mantle beneath spreading centers, and spatially resolved models of reactive porous flow in crystal mushes.

Code and Data Availability

The *nGibbs* source code is available at `https://github.com/benjthyer/nGibbs` and can be installed with `pip install ngibbs` (§4.4). The release used in this chapter (v0.1.0b4) is archived on Zenodo at `https://doi.org/10.5281/zenodo.21345830` (Thyer, 2026). Subsets (n = 15,000) of the validation datasets are shipped with every PyPI installation of nGibbs. Subsets of the training datasets are available **[TODO: on zenodo]** and the full training datasets are available on request from the author.

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